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Sécurité Wi-Fi, une expérience d'établissement



Le service informatique

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Les stagiaires

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Puis les deux stagiaires qui ont passé du temps sur un sujet difficile avec peu (ou pas) de documentation disponible.

Enfin nos collègues des services techniques de l'I.U.T. qui ont tiré des kilomètres de câbles, installé les équipements Wi-Fi et réalisé les branchements dans des endroits parfois difficile d'accès.

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Avant-propos

Le Creusot se situe en Bourgogne, en Saône-et-Loire pour être précis. L'Institut Universitaire de Technologie est né en 1975 autour des départements Génie Electrique et Informatique Industrielle et Génie Mécanique et Productique. Ensuite sont arrivés les départements de Mesures Physiques et de Techniques de Commercialisation. L'I.U.T. Héberge six licences professionnelles depuis plus de deux ans. L'informatique est présente dans l'établissement depuis ses débuts et à beaucoup évolué ces dernières années.

Le problème d'un établissement délocalisé comme le nôtre est d'être attractif pour les étudiants. C'est pour cela que depuis quelques années, le service informatique essaie de développer des services pour les étudiants, pour proposer un petit plus par rapport aux autres établissements. C'est ainsi qu'est né en 2004 le concept d'E-work zone.

E-Work zone, c'est un ensemble de services fournis aux étudiants de l'I.U.T. au cours de leurs années d'études dans notre établissement:

- Un Environnement Numérique de Travail (ENT): UNIV-R, basé sur le projet EPPUN de l'université de Strasbourg,
- Le réseau des anciens étudiants de l'établissement,
- Une plateforme de streaming vidéo IP pour la recherche et la pédagogie,
- Une plateforme de montage vidéo pour la recherche et la pédagogie,
- Le réseau Wi-Fi accessible sur tout le campus.

D'autres expériences ont vu le jour pour présenter des activités attractives pour les étudiants comme par exemple technovision: « <http://www.espace-technovision.com> ». Toutes ces activités poussent l'I.U.T. à assurer une veille technologique permanente sur les techniques les plus récentes.

L'équipe informatique de l'I.U.T. est composée de cinq personnes, trois ingénieurs d'études, un assistant-ingénieur et une emploi-jeune. Nous gérons plus de 1000 ordinateurs, dont 800 reliés à l'Internet. Nous ne sommes pas des spécialistes du Wi-Fi, mais nous y avons passé beaucoup de temps tout en continuant à gérer le quotidien (mail en panne, machines à installer, virus divers et variés, connexion Internet impossible, serveurs hors services, imprimante déprimée, etc...).

On a souvent reproché à l'équipe informatique de l'I.U.T. de ne pas assez communiquer sur ce qu'elle faisait. Partant de cette remarque, ce manuel est l'aboutissement de notre sens de la communication.

1. Introduction

Les réseaux sans fil sont de plus en plus utilisés. On trouve dans de nombreux lieux publics ou privés des réseaux accessibles gratuitement ou bien par abonnement. Le principal problème posé par ces réseaux est leur sécurisation.

La technologie actuelle rend l'installation d'un réseau sans fil très facile. Il suffit de placer un point d'accès à un endroit d'un bâtiment, de configurer quelques paramètres souvent entrés avec un simple navigateur pour permettre aux utilisateurs de se connecter.

Les réseaux sans fil ont été conçus à l'origine pour être entièrement *plug-and-play* (c'est-à-dire on branche et ça marche). Aussi il n'est pas rare d'allumer son ordinateur et de constater qu'il propose de lui-même de se connecter à un réseau sans fil sans qu'on ne lui ait rien demandé dans un contexte où l'on ne supposait même pas l'existence d'un tel réseau: place publique d'un

village, hall d'un hôtel, salle d'attente d'une gare...

Cette facilité d'utilisation s'est faite au détriment de la sécurité. En effet, il faut aujourd'hui garantir à l'utilisateur un niveau de sécurité aussi important sur le réseau sans fil que sur le réseau filaire avec du cryptage des données transmises, une authentification, des autorisations, du suivi de compte et une qualité de service optimale.

Ce manuel traite des différentes solutions de sécurisation testées et retenues par le service informatique de l'I.U.T. du Creusot afin d'assurer la sécurité de son réseau sans fil. Il s'adresse à un large public ayant tout de même un minimum de connaissances sur l'utilisation du système d'exploitation libre et ouvert Linux.

Le projet Wi-Fi de l'I.U.T. est l'aboutissement de deux ans de travail et d'une volonté forte de la direction de l'établissement qui a misé sur les nouvelles technologies de l'information et de la communication depuis quelques années en multipliant les demandes de financement auprès de l'université de Bourgogne, de la région Bourgogne ainsi que sur fonds propres. Sans cette politique de développement du Wi-Fi, nous n'aurions pas pu mettre en place le réseau sans fil de l'établissement.

Les marques citées en référence n'ont pas une vocation publicitaire, mais donnent le matériel de notre établissement et les choix qui ont été retenus en fonction de ce matériel. Ainsi les commutateurs et les points d'accès sans fil de l'établissement sont de marque Cisco et le pare-feu matériel de l'établissement est un NETASQ F100.



2. Le réseau Wi-Fi

Le nom Wi-Fi désigne non pas une norme, mais un label défini par de nombreux constructeurs de matériel réseau afin d'assurer une compatibilité matérielle et logicielle des solutions sans fil avec la norme IEEE 802.11. Le site web officiel de la *Wi-Fi Alliance* « <http://www.wi-fi.org> » présente les activités de cet organisme en matière de sécurité des réseaux Wi-Fi. Wi-Fi est l'abréviation de *Wireless Fidelity*.

2.1. Les liaisons radio

Les liaisons radios Wi-Fi utilisent 14 canaux dans la bande de fréquence de 2,412 GHz à 2,477 GHz. En France, l'ART (Autorité de Régulation des Télécommunications, <http://www.art-telecom.fr>) qui régule les télécommunications n'autorise que l'utilisation des canaux 1 à 13 (Les USA n'autorisent que les canaux 1 à 11 et le Japon autorise jusqu'au canal 14).

Le tableau ci-dessous décrit les différents canaux ainsi que la puissance d'émission en intérieur et en extérieur préconisées par l'ART.

Canal	Fréquence (Ghz)	Puissance maxi en intérieur (mW)	Puissance maxi en extérieur (mW)
1	2,412	100	100
2	2,417	100	100
3	2,422	100	100
4	2,427	100	100
5	2,432	100	100
6	2,437	100	100
7	2,442	100	100
8	2,447	100	100
9	2,452	100	100
10	2,457	100	10
11	2,462	100	10
12	2,467	100	10
13	2,472	100	10
14	2,477	Non utilisé en France	Non utilisé en France

Ces canaux situés dans la bande de fréquences de 2,412 à 2,477 GHz à 100 mW possède une propriété physique contraignante, les spectres de fréquences sont recouvrants. La figure 1 illustre cette propriété.

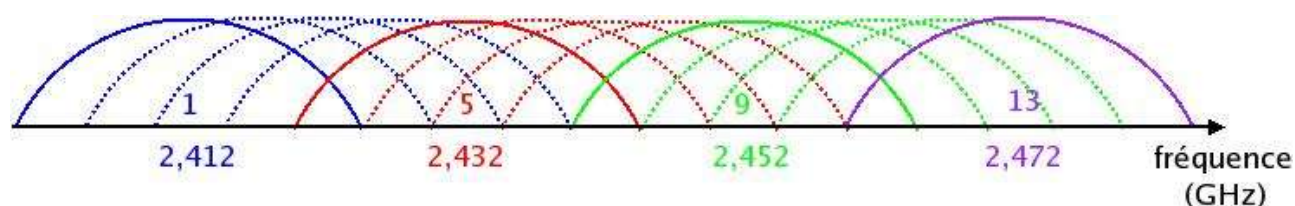


Figure 1: Recouvrement spectral des canaux utilisables en Wi-Fi.

Le problème du recouvrement des canaux est qu'on ne peut pas tous les utiliser. En pratique, il est recommandé d'utiliser:

- Dans le cas idéal: 3 canaux sont sans recouvrement de fréquence: 1, 6 et 11,
- Si on veut un peu plus de souplesse, on peut utiliser 4 canaux: 1, 5, 9, et 13 dont le recouvrement est minimal.

Pour assurer une bonne couverture Wi-Fi, il faut faire en sorte d'occuper l'espace disponible avec plusieurs fréquences non recouvrantes. En effet, deux points d'accès sur le même canal situés à peu de distance l'un de l'autre sont vus comme un seul et même point d'accès, il y a donc partage de la bande passante entre les utilisateurs connectés sur ces points d'accès.

On considère pour simplifier que la couverture d'un point d'accès est sphérique et que les ondes se propagent de la même façon dans toutes les directions (milieu isotrope). Le placement des points d'accès est facilité par la configuration de deux paramètres: le canal et la puissance d'émission. La figure 1 présente un partitionnement de l'espace en canaux Wi-Fi non recouvrants. Dans le cas (a), les puissances d'émission des points d'accès est identique, seule le canal varie. Dans le cas (b), le point d'accès sur le canal 11 émet plus faiblement que les autres points d'accès afin de combler le vide entre les zones de couverture et afin d'éviter le phénomène de partage de bande passante.

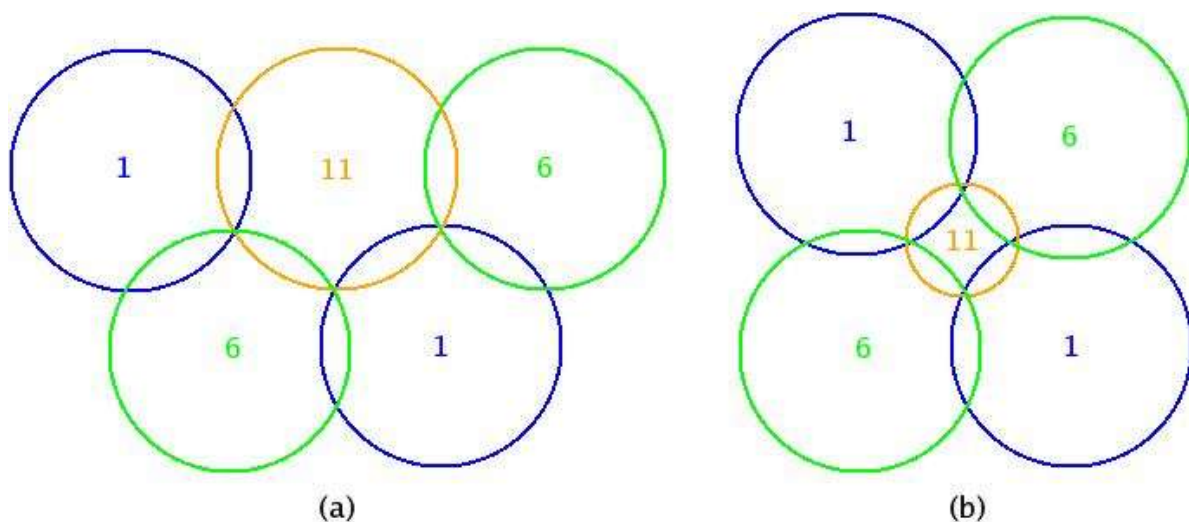


Figure 2: Partitionnement de l'espace en trois canaux Wi-Fi non recouvrants, (a) en pavage décalé (même puissance d'émission) ou (b) en pavage fleur (puissance d'émission plus faible pour le canal 11).

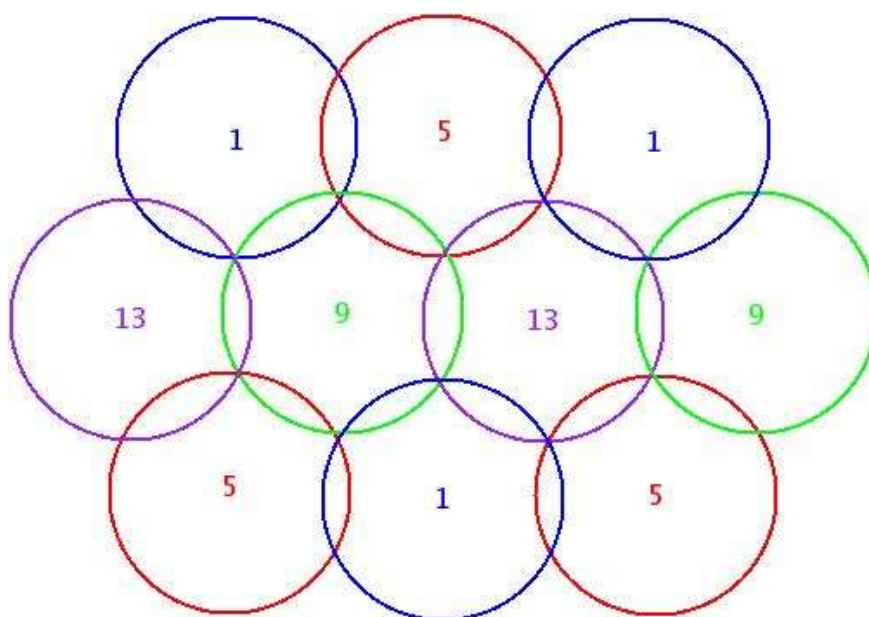


Figure 3: Partitionnement de l'espace en quatre canaux Wi-Fi avec peu de recouvrement.

Ce partitionnement est bien évidemment théorique et on s'aperçoit très rapidement en pratique que les structures des bâtiments et des zones à couvrir ne sont pas sphériques et que la couverture radio n'est pas homogène.

2.2. Les modes d'utilisation du réseau Wi-Fi

2.1.1. Le mode *ad hoc*

Dans ce mode d'utilisation, les ordinateurs sont connectés en réseau local privé. Chaque participant configure sa carte réseau sur un canal particulier. Les ordinateurs émettent sur ce canal et tous les participants font partie intégrante du réseau local privé. Un exemple de réseau *ad hoc* est présenté en figure 4.

Le mode *ad hoc* est utilisé ponctuellement pour utiliser une ressource réseau locale. Par exemple pour imprimer sur une imprimante Wi-Fi ou bien projeter avec un vidéoprojecteur Wi-Fi. Nous avons testé une imprimante et un vidéoprojecteur Wi-Fi qui ne proposaient pas d'EAP. Il faut donc les utiliser en réseau *ad hoc*, ce qui est un problème dans un établissement équipé Wi-Fi. Les prochains modèles de périphériques devraient proposer plusieurs EAP pour intégrer les périphériques Wi-Fi à un réseau (infrastructure) d'établissement. En attendant le réveil des constructeurs, le mode *ad hoc* permet de créer un réseau local en cas de besoin.

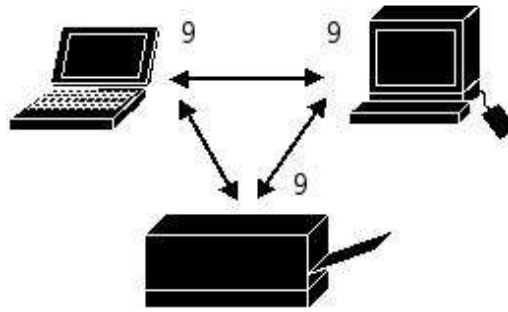


Figure 4: Un réseau ad hoc avec deux ordinateurs et une imprimante sur le canal 9.

L'inconvénient majeur de ce mode est la faible sécurité qu'il procure puisque quelqu'un à portée du réseau peut écouter le trafic et récupérer des données sans aucune difficulté.

2.2.2. Le mode infrastructure

Dans ce mode, les clients Wi-Fi se connectent au réseau à travers un ensemble de points d'accès. Ce sont ces points d'accès, répartis dans l'ensemble de l'établissement qui assurent la connexion des clients Wi-Fi et garantissent la sécurité de l'établissement contre des attaques éventuelles.

Le réseau sans fil minimal est donné sur la figure 1. Il se compose d'un point d'accès, de clients reliés à Internet par le réseau filaire commuté de l'établissement.

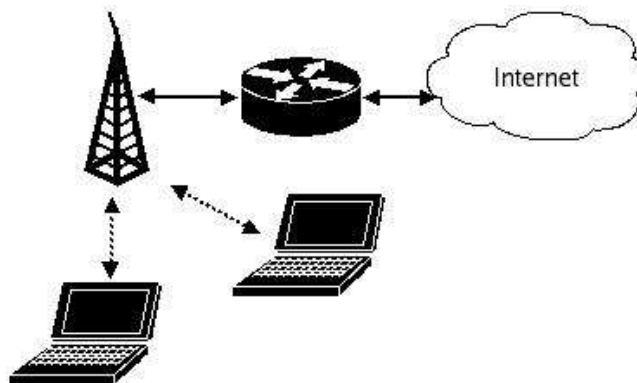


Figure 5: Réseau Wi-Fi minimal.

Un réseau plus complexe dispose de plusieurs points d'accès avec des canaux différents dans l'établissement afin d'optimiser la couverture radio.

2.3. L'ESSID

Dans un réseau Wi-Fi, il est nécessaire de disposer d'un nom de réseau afin d'identifier le réseau. Cet identifiant (l'ESSID) est défini sur les points d'accès et est diffusé ou non pour les clients potentiels.

2.3.1. En mode infrastructure

En mode infrastructure, chaque client se connecte au réseau à travers un point d'accès. L'ensemble formé par le point d'accès et les clients situés dans sa zone de couverture est appelé ensemble de services de base (BSS: *basic service set*). Chaque BSS est identifié par un BSSID, un identifiant de 48 bits (6 octets). En mode *infrastructure*, le BSSID correspond simplement à l'adresse MAC du point d'accès.

Les BSS sont reliés entre eux par un DS (*distribution system*), ce qui permet de former un ensemble de service étendus (ESS: *extended service set*). Cet ensemble de services étendu est identifié par un ESSID (*ESS identifier*). L'ESSID est aussi très souvent appelé SSID (*service set identifier*). Le DS peut être un réseau filaire commuté ou un réseau sans-fil.

Puisque les BSS sont reliés entre eux par le DS, les déplacements de l'utilisateur sont répercutés au sein de tous les points d'accès de l'ESS. Le client se connecte donc toujours au point d'accès dont le signal est le plus fort ou à celui qui est le moins sollicité. Le changement de point d'accès est totalement transparent pour l'utilisateur, c'est le *roaming* (nomadisme).

Tous les points d'accès émettent en permanence des trames balises (*beacon*) dans lesquelles ils fournissent leur BSSID, leurs caractéristiques et éventuellement leur ESSID. Pour des raisons de sécurité, il faut absolument interdire la diffusion de l'ESSID dans les *beacons* afin que seuls les clients disposant de l'ESSID puissent se connecter en Wi-Fi.

Chaque client qui arrive dans un BSS diffuse sur chaque canal une requête de sondage (*probe request*) contenant l'ESSID pour lequel il est configuré. Si aucun ESSID ne lui a été fourni, il écoute les *beacons* pour tenter de s'accrocher à un ESSID diffusé.

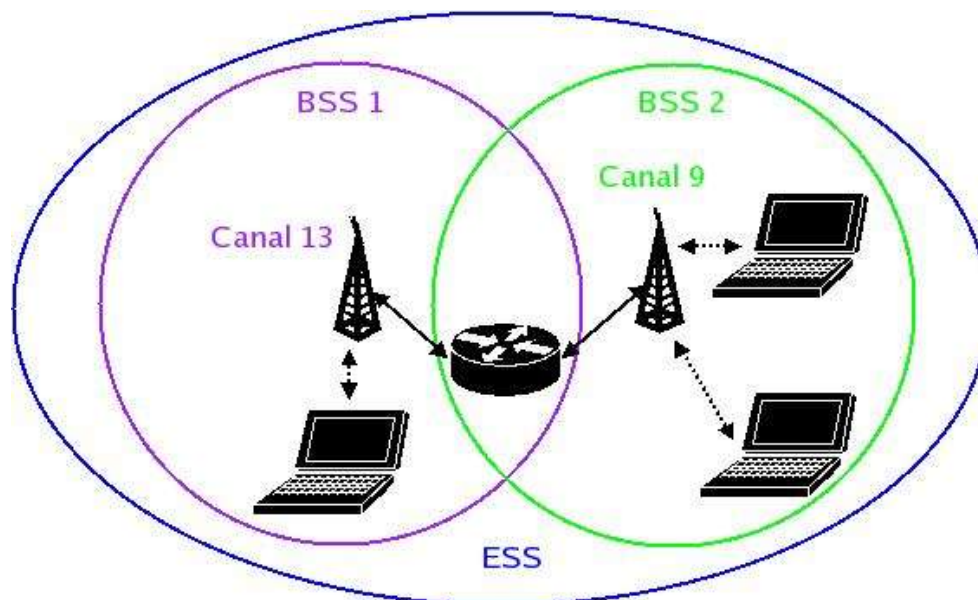


Figure 6: ESS et BSS.

Les points d'accès répondent à une requête de sondage en vérifiant l'ESSID demandé et en renvoyant des informations de débit et de charge au client demandeur. C'est le client qui choisit le point d'accès sur lequel il va s'accrocher en fonction des informations de charge et de débit des points d'accès.

La connexion d'un client à un point d'accès se nomme l'association. Un client Wi-Fi est donc associé à un point d'accès et un seul à un instant t.

2.3.1. En mode ad hoc

Les ordinateurs se connectent entre eux en choisissant tous le même canal Wi-Fi et un ESSID commun. Chaque ordinateur joue à la fois le rôle de client et le rôle de point d'accès. L'ensemble formé par les ordinateurs ainsi configurés est un IBSS (*independent basic service set*). La figure 7 montre un IBSS avec deux ordinateurs sur le canal 5.

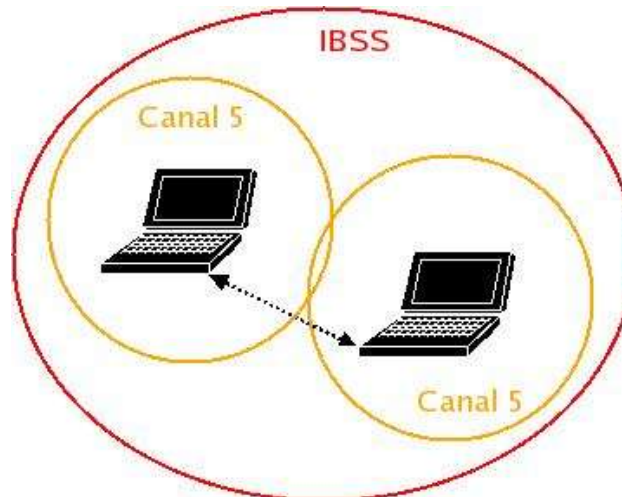


Figure 7: IBSS en mode ad hoc.

L'IBSS est un réseau destiné à l'échange de données entre deux ou trois ordinateurs pendant une durée limitée. Les ordinateurs doivent rester à portée les uns des autres sous peine de perdre la connexion. Il n'y a pas de roaming possible dans un réseau ad hoc. On est à portée ou non.

Ce type de réseau permet par exemple de se servir lors d'une conférence d'un vidéo-projecteur Wi-Fi ou bien d'imprimer sur une imprimante Wi-Fi. Il s'agit d'une utilisation ponctuelle du réseau qui offre une sécurité très faible.

2.4. Le cryptage, les clés WEP statiques

A l'origine, le Wi-Fi a été conçu pour permettre aux utilisateurs de se connecter simplement au réseau sans aucune configuration particulière (*plug-and-play*). Mais cette facilité de connexion s'est faite au détriment de la sécurité.

Afin de garantir la sécurité, des mécanismes de cryptage des données ont été ajoutés au Wi-Fi d'origine. Ainsi, le WEP (Wired Equivalent Privacy) est un algorithme de cryptage des données sur le réseau Wi-Fi. Il fonctionne à l'aide d'une clé symétrique de 64 bits (40+24) ou de 128 bits (104+24) de longueur. Cette clé est entrée au niveau du point d'accès et permet le chiffrement du trafic entre le client et le point d'accès. WEP utilise l'algorithme de cryptage RC4.

La norme 802.11 prévoit jusqu'à 4 clés WEP interchangeables afin de simuler une rotation de clés en changeant la clé WEP statique selon un intervalle de temps programmé. La figure 8 présente l'écran de configuration d'un point d'accès CISCO 1100 où l'on trouve les quatre clés WEP statiques et la rotation programmée toutes les 10 secondes.

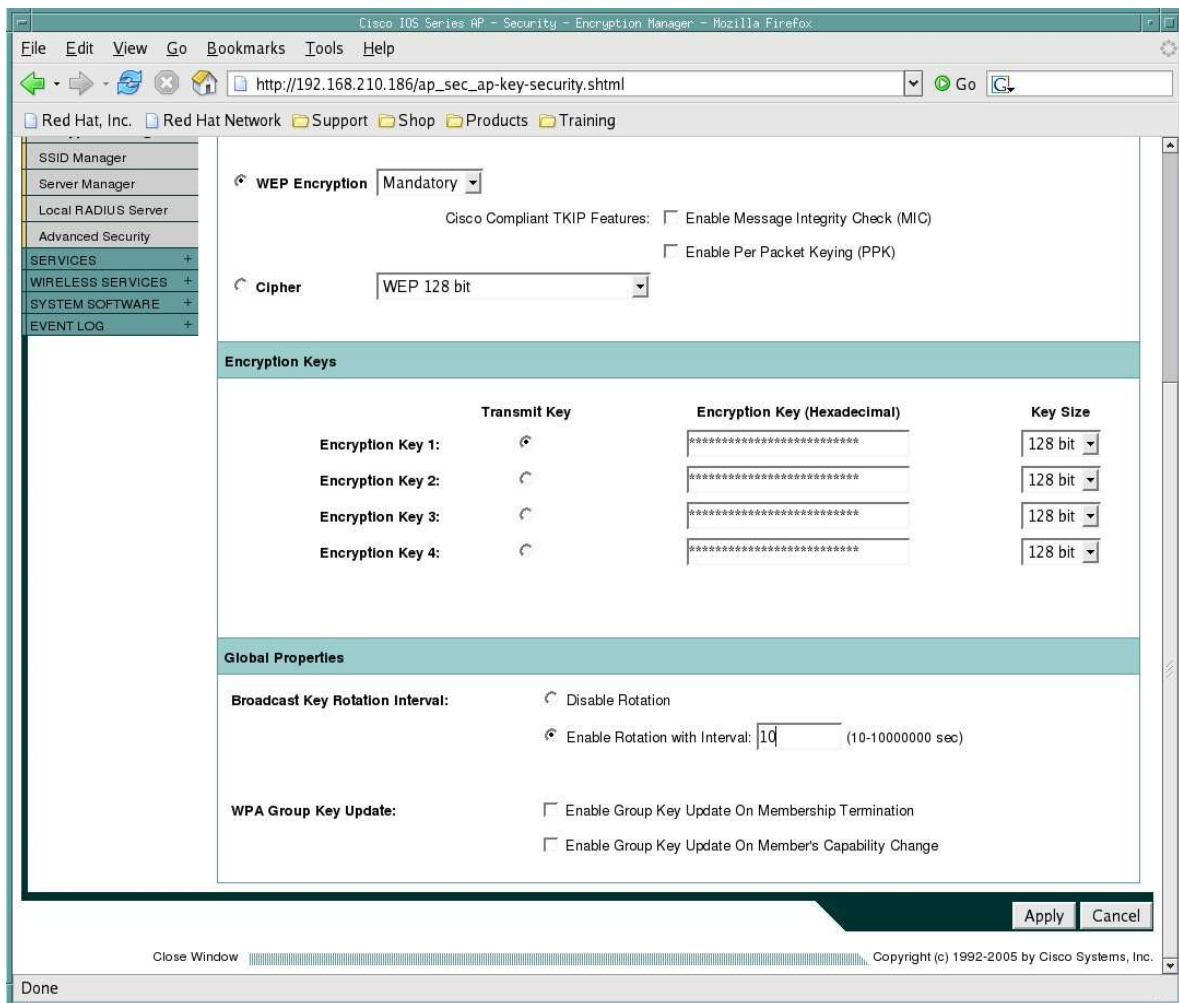


Figure 8: Définition des quatre clés WEP et rotation toutes les 10 secondes pour un pseudo-dynamisme.

Le principal problème du WEP est la facilité avec laquelle on peut intercepter le trafic réseau avec un logiciel d'écoute pour trouver la clé. Par exemple, *airsnort* (figure 9) permet de trouver assez facilement la clé Wi-Fi d'un réseau en écoutant directement le trafic pendant 3 heures environ (si il y a beaucoup de transferts). Un autre exemple est *airodump* qui permet de récupérer un ensemble de paquets Wi-Fi et de les stocker dans un fichier que l'on va exploiter avec *aircrack* (figure 10) hors-ligne pour récupérer la clé.

Bien que WEP ne soit pas une garantie de sécurité suffisante, il peut être utilisé pour sécuriser un réseau ad hoc ponctuel pour la durée d'une conférence par exemple. WEP est souvent, pour l'instant, la seule méthode de sécurisation sur une imprimante ou un vidéo-projecteur Wi-Fi. C'est donc la protection minimum à mettre en place, mais elle doit rester ponctuelle.

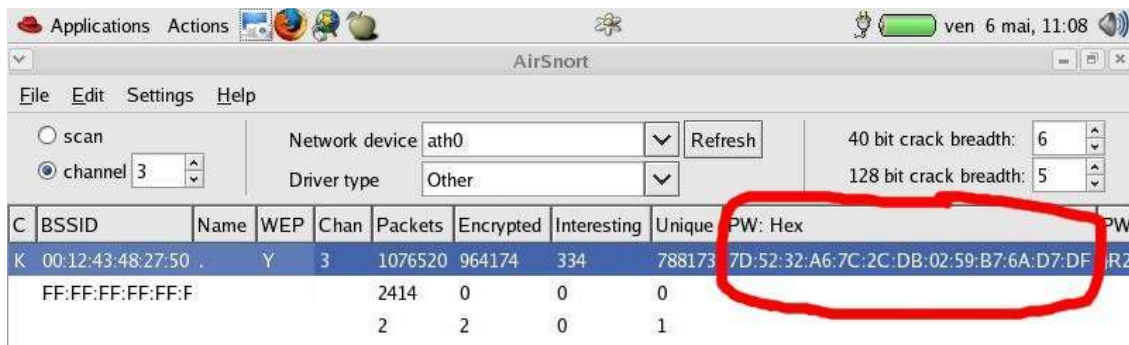


Figure 9: Exemple de récupération de clé WEP avec airtsnort.

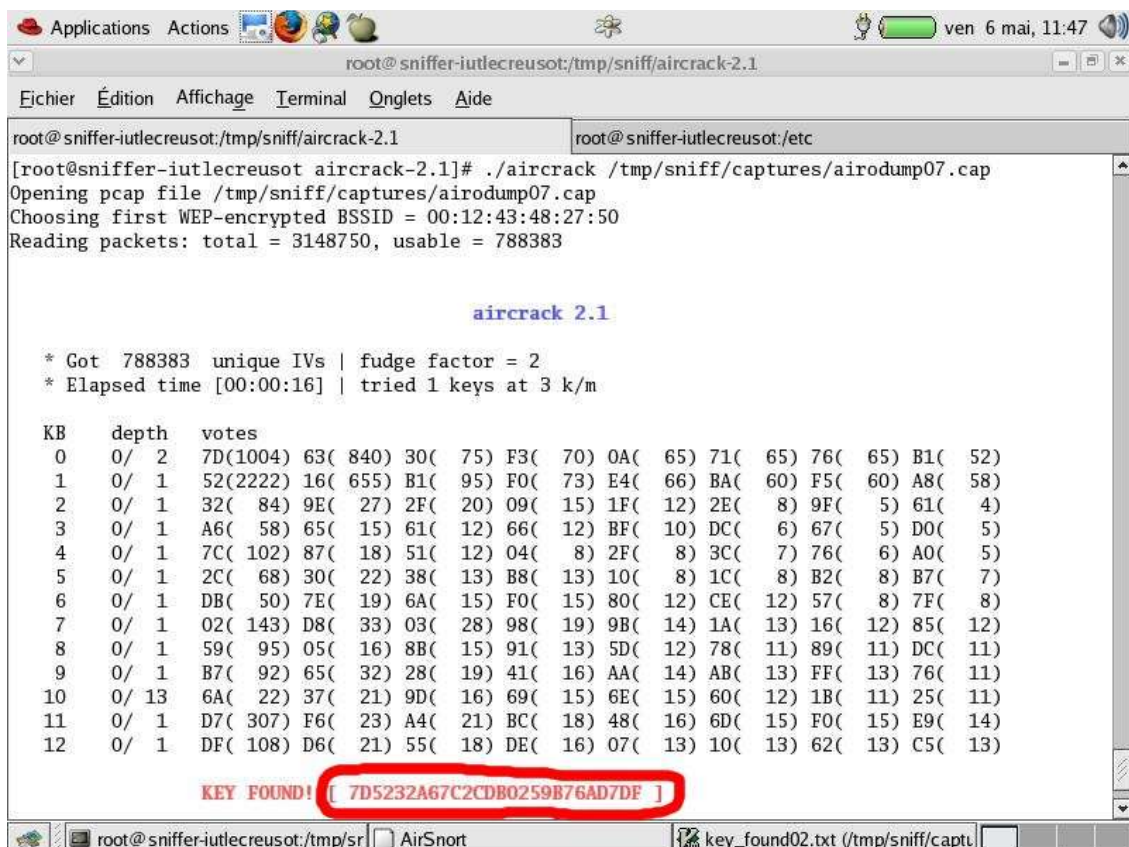


Figure 10: Un exemple de découverte de clé WEP avec aircrack.

2.5. WPA, WPA2, AES, la dynamique des clés

Comme on l'a vu précédemment, le WEP possède de très nombreux défauts en raison de la faiblesse de l'algorithme de cryptage RC4. Ces failles de sécurité ont été résolues par l'émergence d'un nouveau standard: WPA (Wireless Protected Access). En 2003, la Wi-Fi Alliance a introduit WPA pour faire face à la faiblesse du WEP. En 2004, elle a introduit WPA2, la nouvelle génération de la sécurité des réseaux Wi-Fi. WPA et WPA2 assurent une authentification mutuelle entre le client et le serveur d'authentification à travers le point d'accès. WPA et WPA2 font partie de la norme IEEE 802.11i qui assure enfin un cryptage fort et une protection optimale des données sur un réseau Wi-Fi.

WPA utilise l'algorithme de cryptage TKIP (*Temporal Key Integrity Protocol*) avec vérification des messages MIC (*Message Integrity Check*), sa mise en place nécessite seulement une mise à jour logicielle des points d'accès et des pilotes de cartes Wi-Fi pour fonctionner. WPA2 utilise quant à lui l'algorithme de cryptage CCMP (*Counter-Mode/CBC-MAC protocol*) appelé également AES (*Advanced Encryption Standard*) qui nécessite, en raison de sa complexité, une mise à jour matérielle des points d'accès et des adaptateurs Wi-Fi clients.

La Wi-Fi alliance a défini deux utilisations du Wi-Fi:

	WPA	WPA2
Mode entreprise	IEEE 802.1X/EAP cryptage: TKIP/MIC	IEEE 802.1X/EAP cryptage: AES/CCMP
Mode personnel	PSK cryptage: TKIP/MIC	PSK cryptage: AES/CCMP

Le mode personnel utilise en WPA et en WPA2 une clé pré-partagée (PSK: *Pre-Shared Key*). Le cryptage est assuré par une clé partagée entre le client et le serveur d'authentification ce qui fait de cette méthode où la clé est partagée entre tous les clients une solution personnelle non applicable dans un environnement professionnel.

Le mode entreprise offre la garantie d'une sécurité optimale basée sur 802.1X et un EAP alliés aux algorithmes de cryptage TKIP et AES. Ce mode est le plus sûr pour les entreprises et les administrations. L'algorithme AES utilise des clés de 256 et 512 bits, qui nécessitent une architecture matérielle dédiée en raison de la complexité de décryptage.

En WPA et WPA2, les clés sont générées par paquet, par session et par utilisateur, ce qui rend la tâche de piratage quasi impossible.

2.6. La santé

Les questions de santé sont souvent abordées par les utilisateurs lors de l'arrivée d'une nouvelle technologie. Des études américaines ont montré que le réseau sans fil ne nuit pas à la santé. Les points d'accès utilisent une puissance maximale d'émission de 100 mW alors qu'un téléphone portable possède une puissance de 1 W. Un point d'accès Wi-Fi est donc 10 fois moins puissant qu'un téléphone et ne reste pas des heures collé à l'oreille. Le Wi-Fi est donc beaucoup moins nocif que le téléphone portable.

Cependant, par principe de précaution, aucun point d'accès ne devra être placé à moins de 60 centimètres d'un bureau où une personne travaille toute la journée. Cette restriction n'est pas valable pour les lieux de passage où on ne reste pas des heures (armoires réseaux, couloirs, placards à balai, ...).

3. La couverture radio

3.1. La cartographie de l'établissement

Un problème lié au Wi-Fi est de connaître la couverture radio des points d'accès que l'on va installer. Il faut en effet qu'ils inondent l'établissement d'ondes Wi-Fi sans inonder le domaine public hors de l'établissement. C'est un problème difficile à résoudre sur lequel il faut se pencher sérieusement sans quoi la mise en place de la sécurité risque d'échouer.

Le test le plus sûr de sa couverture radio est la promenade avec portable, sport qui peut se pratiquer seul ou en groupe suivant l'étendue que l'on souhaite vérifier. Cette vérification est nécessaire mais sans doute pas suffisante pour avoir une idée de la portée des points d'accès.

Les entreprises de service qui proposent une étude radio sur plan uniquement sont amenées à avoir des surprises lors du déploiement du Wi-Fi. Cette étude peut donner une idée de la couverture à prévoir, du nombre de points d'accès à installer et des fréquences à utiliser, mais ne peut en aucun cas être satisfaisante du point de vue sécurité sans la promenade avec portable.

A l'issue de la promenade, on a une bonne idée de la couverture radio et de la zone d'influence de chaque point d'accès. Il convient de limiter la puissance des points d'accès qui émettent sur le domaine extérieur à l'établissement ou de les déplacer de quelques mètres. Ensuite, on recommence la promenade avec portable pour vérifier. C'est un sport épuisant, mais malheureusement nécessaire.

3.2. L'outil graphique WLSE

Les acteurs des réseaux sans fil proposent de plus en plus d'outils de gestion de la couverture radio. Cisco a développé la solution WLSE (*Wireless LAN Secure Engine*) qui propose un outil de cartographie et d'optimisation du réseau sans fil. A l'aide de bornes dédiées et d'une méthode de balayage des fréquences radio, cet outil est capable de dresser un plan de la couverture (2D pour l'instant, 3D prévu bientôt) afin de visualiser l'ensemble de l'établissement, de définir les puissances d'émission idéales pour chaque point d'accès et de détecter les points d'accès Wi-Fi pirates (*rogue AP*) non autorisés.

WLSE se présente sous la forme d'un ordinateur de type PC au format *rackable* 1U sous Linux. Après une configuration au démarrage (adresse IP, nom...), l'interface d'administration est une interface web sécurisée (https) très simple d'emploi protégée par un login et un mot de passe. La figure 5 donne la fenêtre d'accès à WLSE.



Figure 11: Accès à WLSE.

Le menu de WLSE propose un grand nombre de fonctions sur l'analyse du réseau Wi-Fi: état des points d'accès, fautes, cartographie, détection de points d'accès non autorisés... La figure 11 décrit le menu de WLSE avec les différentes options accessibles. WLSE est capable de proposer une configuration radio automatique en écoutant le réseau avec un point d'accès spécial (qui n'émet pas mais écoute les autres points d'accès), le serveur WDS (*Wireless Domain Services*).

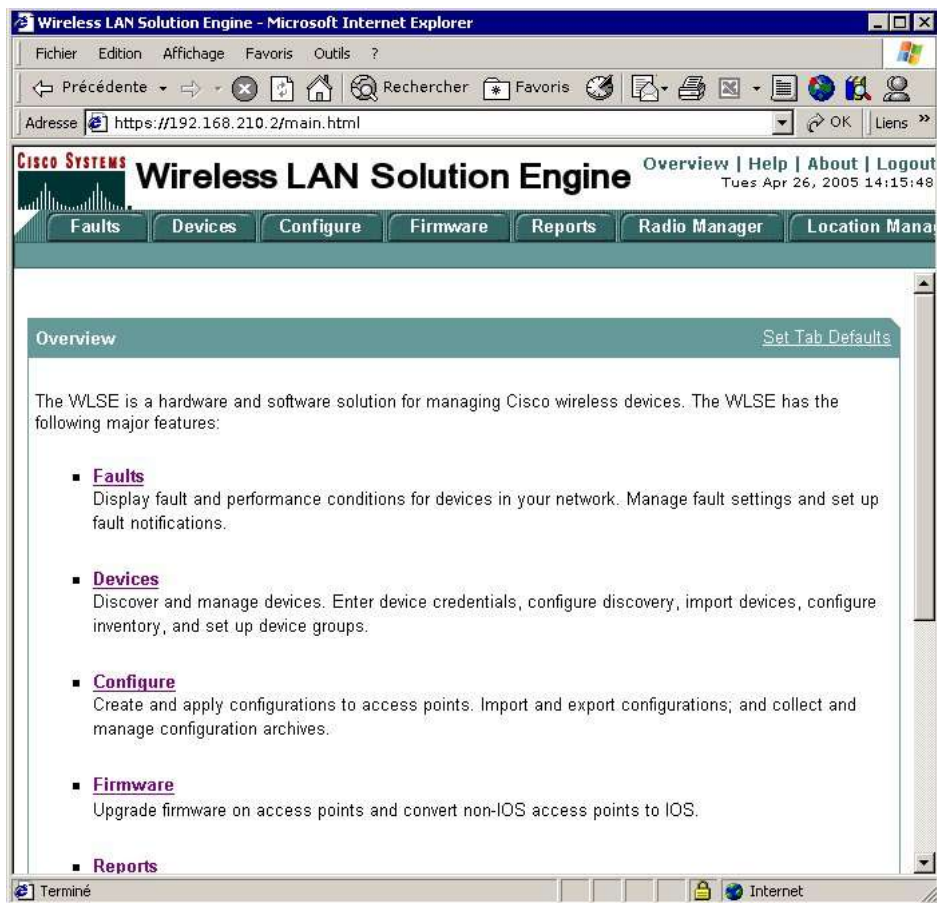


Figure 12: Menu de gestion de WLSE.

WLSE fait partie de l'architecture Cisco SWAN (Structured Wireless-Aware Network) qui a pour but d'intégrer complètement l'architecture filaire et l'architecture Wi-Fi. Cette architecture est matérielle et logicielle et est malheureusement propriétaire Cisco. WLSE intègre des outils de génération de rapports de défaillances sur les points d'accès, mais également sur les clients.

Un outil important de WLSE est l'outil de cartographie qui donne en temps réel l'état du réseau Wi-Fi sur le plan de l'établissement en deux dimensions. Cet outil est présenté en figure 13. cet outil permet de visualiser en temps réel la couverture des points d'accès ainsi que les problèmes du réseau. Sur cet exemple, il y a trois points d'accès (en rouge sur la liste à gauche et sur le plan de droite) qui posent problème.

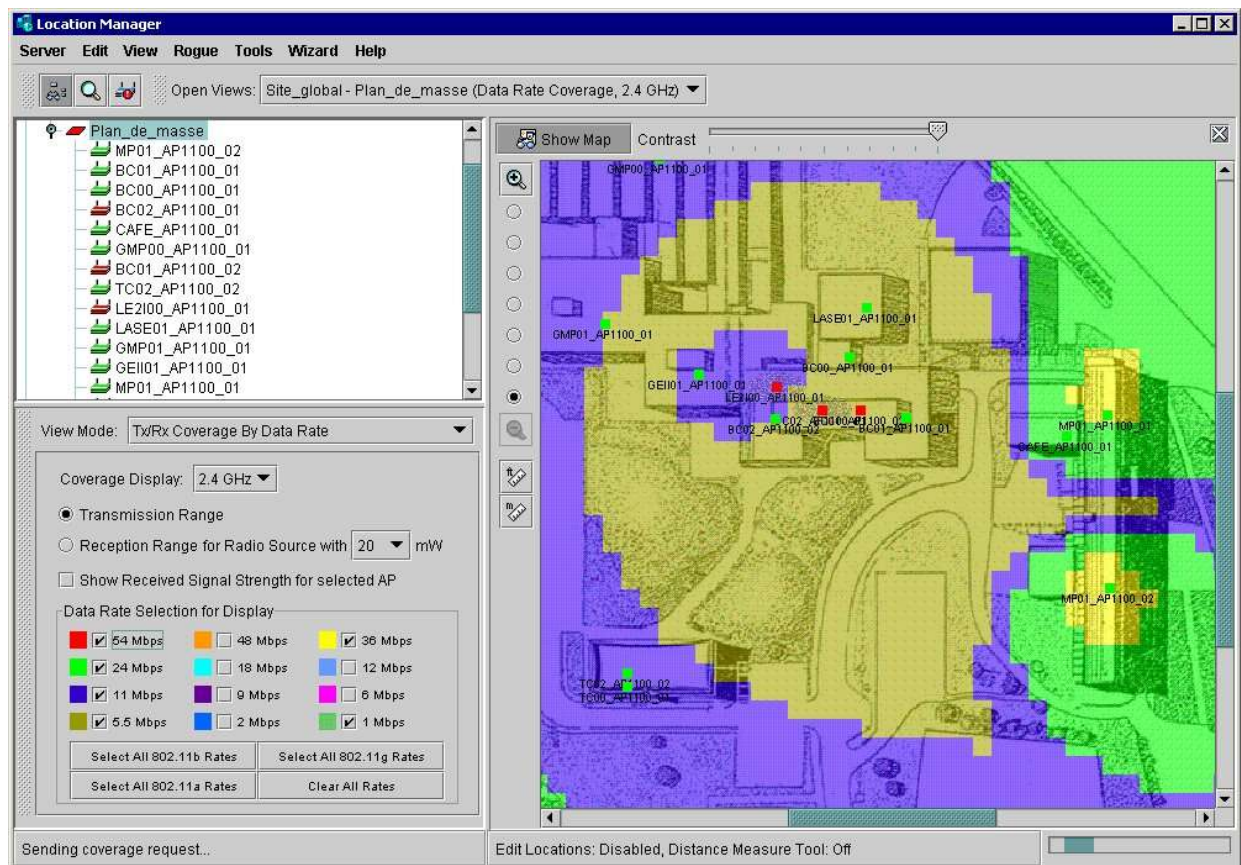


Figure 13: Outil graphique de localisation des points d'accès autorisés ou non.

Une fonctionnalité bien pratique de WLSE est l'optimisation de la couverture radio. Il teste sur chacun des points d'accès tous les canaux et écoute le résultat des interférences entre les canaux. Il détermine ainsi pour chaque point d'accès la puissance d'émission optimale pour éviter les interférences entre canaux. C'est une fonction très utile qui donne une couverture radio optimale de l'établissement. Seule contrainte, ce test qui dure environ 10 minutes coupe le réseau Wi-Fi pendant toute sa durée (en raison des changements de canaux sur chacun des points d'accès).

En résumé, WLSE est un outil complet de gestion de la couverture radio d'un établissement, il propose de nombreux outils indispensables pour avoir une vue globale et locale de son réseau Wi-Fi.

4. Les différentes contraintes du réseau

Un réseau informatique se décompose en quatre « parties » distinctes:

- Le matériel dont le rôle est d'assurer la mise en oeuvre physique (électronique) de la transmission de l'information,
- Le logiciel qui fournit des protocoles de communication et assure la mise en oeuvre logique des transmissions,
- L'administrateur qui s'arrache les cheveux pour essayer de faire fonctionner tout ça ensemble,
- Les utilisateurs qui souvent ne savent pas la quantité de travail nécessaire pour que cela fonctionne, qui considèrent que ça doit marcher et que si ça ne marche pas, c'est la faute de l'administrateur.

L'intégration du réseau Wi-Fi au réseau de l'établissement doit prendre en compte les contraintes techniques existantes sur le réseau déjà en place.

4.1. Les VLANs

Depuis quelques années déjà, le réseau de l'I.U.T. du Creusot sépare les flux entre différentes communautés d'utilisateurs en utilisant des VLANs (*virtual local area network*). Chaque trame réseau contient un identifiant de VLAN (le *tag*). Chaque port d'un commutateur est configuré dans un certain VLAN qui dépend de l'ordinateur qui se connecte sur ce port.

Les flux réseaux utilisent donc le même chemin physique, mais les commutateurs ne distribuent les paquets d'un VLAN qu'aux ports associés à ce VLAN. La mise en place des VLANs est assez difficile car elle nécessite de connaître très exactement les prises murales et les panneaux de brassage de l'établissement. Par contre, cette approche une fois en place simplifie grandement l'administration du réseau et offre un niveau de sécurité important par la séparation des flux.

L'I.U.T. du Creusot comporte trois VLANs principaux:

- le VLAN pédagogie regroupe toutes les salles de cours, les salles informatiques libre accès.
- Le VLAN gestion regroupe les ordinateurs de gestion de la vie étudiante: service scolarité (Apogée, Harpège) et service comptabilité (Nabuco).
- Le VLAN recherche comprend les ordinateurs des enseignants et du personnel administratif et technique de l'établissement.

Afin de permettre d'échanger des données entre enseignants et étudiants ou bien enseignants et scolarité, une machine spéciale a été mise en place avec un serveur ftp qui permet l'échange de fichiers. C'est la seule machine de l'établissement qui se situe dans tous les VLANs. Toutes les autres machines sont uniquement dans un seul VLAN en fonction de leur utilisation.

Ces VLANs sont à prendre en compte lors de la mise en place qu Wi-Fi car il faudra sans doute séparer les flux sans fil en WVLANs (*Wireless VLANs*) grâce au serveur RADIUS qui pourra faire basculer l'utilisateur dans un certain WVLAN juste après la phase d'authentification.

4.2. Les différentes communautés d'utilisateurs

Les utilisateurs n'ont pas tous les mêmes besoins d'accès au réseau. Une grosse partie du travail de l'administrateur est de définir quelles sont les différentes communautés d'utilisateurs en présence et quels sont les besoins de ces différentes communautés.

Le choix des droits des différentes communautés dépend de l'établissement concerné et il n'y a pas de règle générale pour les définir, chacun doit faire en fonction de ses besoins et de ses ressources. L'I.U.T. du Creusot a défini trois communautés Wi-Fi:

- La communauté « personnel » qui comprend les enseignants et le personnel technique et administratif,
- La communauté « étudiant » qui s'adresse aux étudiants de l'établissement,
- La communauté « invité » qui définit les différents visiteurs de l'établissement (chercheurs invités des laboratoires, stagiaires de DEA, conférenciers ponctuels...).

Les droits donnés à chacune des communautés sont résumés ci-dessous:

communauté	droits
personnel	Tous les protocoles réseaux de l'établissement.
étudiant	http, https, ftp, pop, smtp, ssh, rdp, X11
invité	http, https, ftp, pop, smtp, ssh, rdp, X11

En plus de ces services, l'établissement est responsable de l'utilisation des ressources réseaux mises à sa disposition par la charte RENATER. La communauté « étudiant » est donc surveillée de près pour éviter toute utilisation frauduleuse du réseau. Les étudiants signent tous la charte

d'utilisation du Wi-Fi mise en place par l'établissement. De plus, les fichiers de surveillance (*logs*) sont archivés et disponibles pour le directeur de l'établissement en cas de réprimande pénale contre une utilisation abusive du réseau.

Pour les communautés « personnel » et « invité », il y a seulement une mise en garde contre les abus à ne pas commettre et la signature de la charte d'utilisation du réseau Wi-Fi.

Après avoir mis en place le réseau et défini les différents utilisateurs potentiels ainsi que leurs droits et devoirs, il faut passer à la sécurisation proprement dite.

5. Les différentes méthodes d'authentification

Le choix d'une méthode d'authentification doit prendre en compte un certain nombre de contraintes. L'idée générale est de simplifier au maximum les configurations tant au niveau utilisateur qu'au niveau administrateur du réseau Wi-Fi. L'utilisateur veut un Wi-Fi qui marche tout le temps sans avoir à se soucier d'une quelconque configuration matérielle ou logicielle. L'administrateur veut un réseau sécurisé qui marche tout le temps sans devoir passer des heures à configurer les postes clients ou à administrer les points d'accès ou le serveur d'authentification.

5.1. Le portail captif

Le portail captif est une solution d'authentification simple à mettre en oeuvre, qui ne nécessite pas d'installation particulière sur le client. L'authentification est basée sur une page web sécurisée dans laquelle l'utilisateur donne son identifiant (*login*) et son mot de passe. Il est alors authentifié sur le réseau pour une durée définie dans le portail.

La figure 14 donne le schéma de ce mécanisme. Le client Wi-Fi est associé avec le point d'accès et récupère une adresse IP par le serveur DHCP. Lors de la première demande de connexion http vers l'extérieur, l'utilisateur est renvoyé automatiquement vers la page d'authentification du portail captif (en https). Il saisit ensuite son identifiant et son mot de passe. Si le firewall valide cette authentification, l'utilisateur est redirigé vers la page qu'il avait choisi au départ (en http).

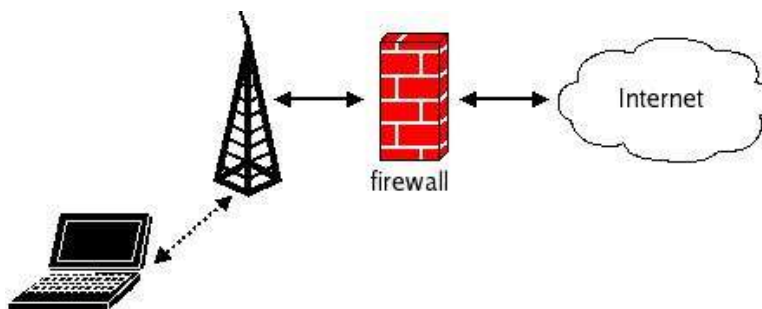


Figure 14: Le portail captif hébergé par le firewall permet (après authentification) de sortir sur Internet.

Notre établissement est pourvu d'un pare-feu (*firewall*) NETASQ F100 possédant un portail captif. Nous avons défini une plage d'adresse privée pour ce réseau sans fil et le pare-feu est serveur DHCP sur cette plage d'adresse. Un ESSID unique est utilisé par une communauté d'utilisateurs afin de se connecter au Wi-Fi. La base des utilisateurs est stockée au niveau du firewall sous la forme d'une base LDAP.

Cette méthode d'authentification a été retenue pour deux communautés d'utilisateurs: les étudiants et les invités de l'établissement. Au niveau des points d'accès, deux ESSID ont été définis: Wi-Fi-etudiant et Wi-Fi-invite. Ces deux ESSID sont définis en mode *Open authentication*

avec quatre clés WEP statiques en rotation toutes les 10 secondes.

Sur les ordinateurs clients, il suffit de définir les paramètres IP en DHCP et d'entrer l'ESSID voulu. Le logiciel client est natif sous Windows 2000 et XP, sous Linux et sous Mac OS X. L'ordinateur est alors prêt à se connecter. Il faut ensuite entrer l'utilisateur et son mot de passe au niveau du firewall. Et c'est fini, l'utilisateur peut immédiatement se connecter.

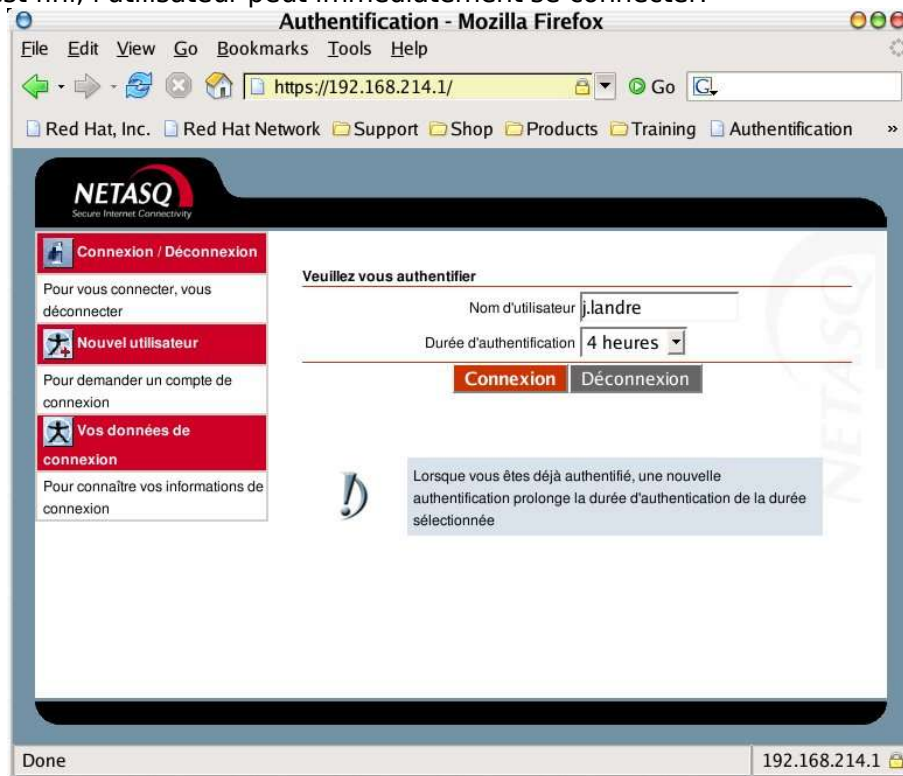


Figure 15: Ecran d'accueil du portail captif (https).

Dès qu'il active la carte Wi-Fi, l'utilisateur reçoit ses paramètres IP du serveur DHCP du firewall. Dès qu'il lance son navigateur sur une page (google.fr par exemple), il est automatiquement redirigé vers la page d'accueil de l'authentification (figure 15). Il entre alors son login (sécurisé par https). Puis il saisit son mot de passe (toujours sécurisé par https), voir figure 16.

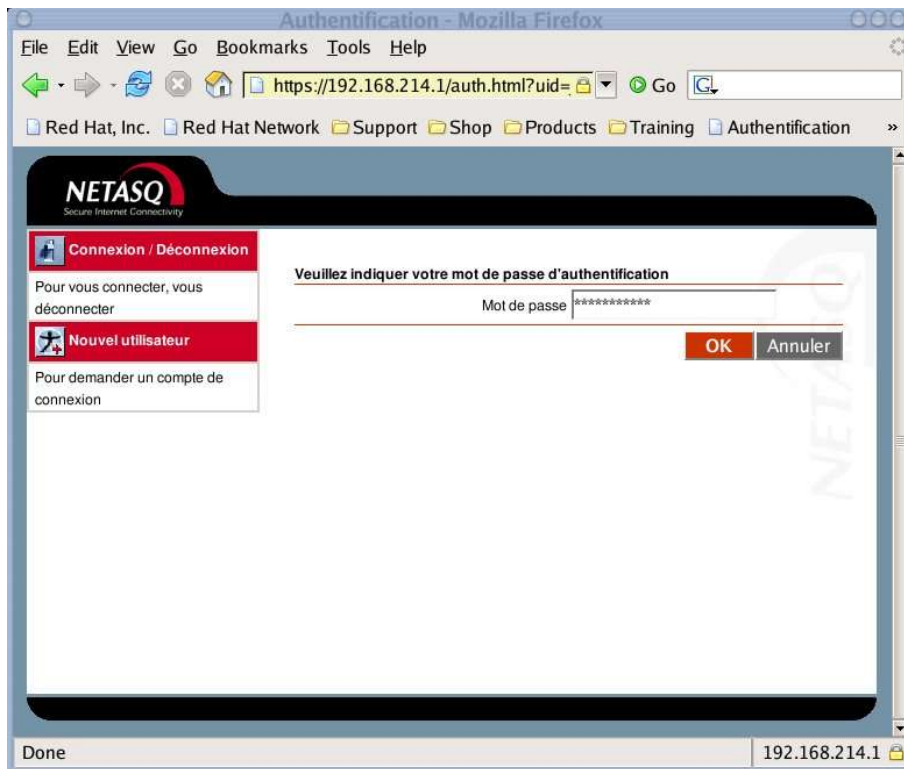


Figure 16: Saisie du mot de passe du portail captif (https).

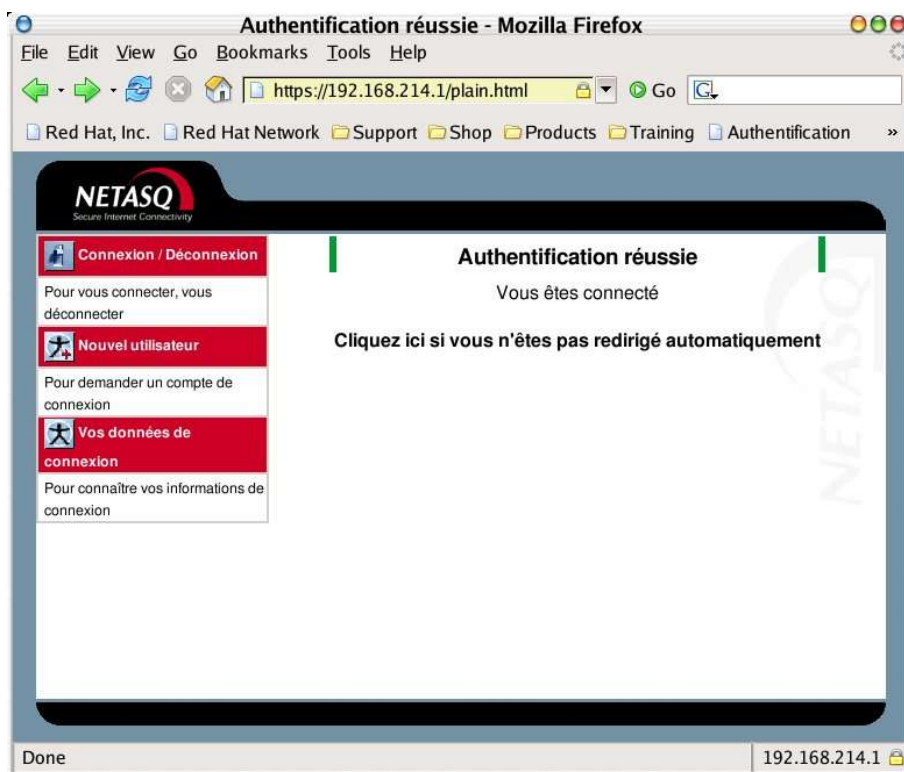


Figure 17: Succès de l'authentification, redirection vers la page demandée (http).

Dès que l'authentification est réussie, l'utilisateur est dirigé automatiquement vers la page qu'il avait demandée au départ.

Chaque utilisateur signe une charte d'utilisation de RENATER et en plus une charte de bonne utilisation du Wi-Fi dans laquelle il est prévenu de cette conservation des traces. Nous avons choisi d'ouvrir aux étudiants et aux invités de l'I.U.T. les services suivants:

- pop, smtp pour le courrier électronique,
- http, https pour la navigation Internet,
- ftp pour le transfert de fichiers,
- ssl et ssh pour des connexions sécurisées,
- RDP et X11 pour l'accès à distance à des ordinateurs.

En ce qui concerne la communauté Wi-Fi-recherche qui regroupe les enseignants et le personnel administratif et technique, la sécurité était beaucoup plus importante que les communautés étudiants et invités. Le choix s'est donc porté vers une authentification forte grâce à 802.1X et un EAP.

5.2. L'accès 802.1X et EAP, RADIUS

IEEE 802.1 est le groupe de travail gestion et interconnexions de réseaux (*LAN/MAN Bridging & Management*) de l'IEEE (*Institute of Electrical and Electronics Engineers*). Il propose des normes de gestion et d'interconnexion sur les réseaux locaux et étendus. Ce groupe a proposé une méthode de sécurisation des réseaux nommée 802.1X (*Port Based Network Access Control*).

802.1X est une norme d'accès au réseau par port, indépendante du support physique (elle marche avec ethernet, Wi-Fi, etc...). Le port est le point d'attache d'un système au réseau physique. Tout système qui tente de se connecter au réseau doit d'abord être identifié avant d'accéder aux ressources. Un port d'accès au réseau peut être soit contrôlé, soit non contrôlé.

En Wi-Fi, on utilise 802.1X avec un EAP (*Extensible Authentication Protocol*) qui effectue l'authentification des systèmes sur un serveur d'authentification. Le processus d'authentification comprend quatre étapes, il est décrit en figure 18.

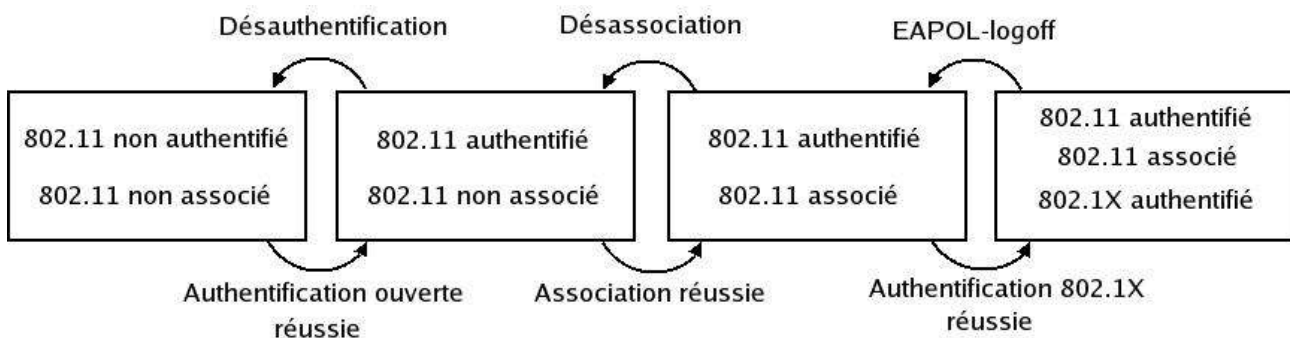


Figure 18: Les quatre étapes d'une authentification 802.1X.

La phase d'authentification est réalisée par un serveur d'authentification. Freeradius est un serveur AAA (*Authentication Authorization Accounting*) basé sur le protocole RADIUS (*Remote Authentication Dial In User Service*). Ce protocole effectue une authentification à distance du client et lui donne certains droits d'accès aux ressources réseaux. Il est défini dans le RFC 2865. Le suivi du compte de l'utilisateur (*accounting*) est une extension de RADIUS définie dans le RFC 2866.

Voici quelques définitions nécessaires pour comprendre les mécanismes d'authentification 802.1X.

- Le client (*supplicant*) est un ordinateur qui demande une ressource réseau Wi-Fi.

- L'authentificateur (*authenticator*) est un point d'accès au ressource Wi-Fi qui va recevoir et traiter la demande du client.
- Le serveur AAA (*authentication, authorization and accounting server*) est le serveur qui va recevoir la demande d'authentification de l'authentificateur et qui va répondre à la requête d'accès au client.

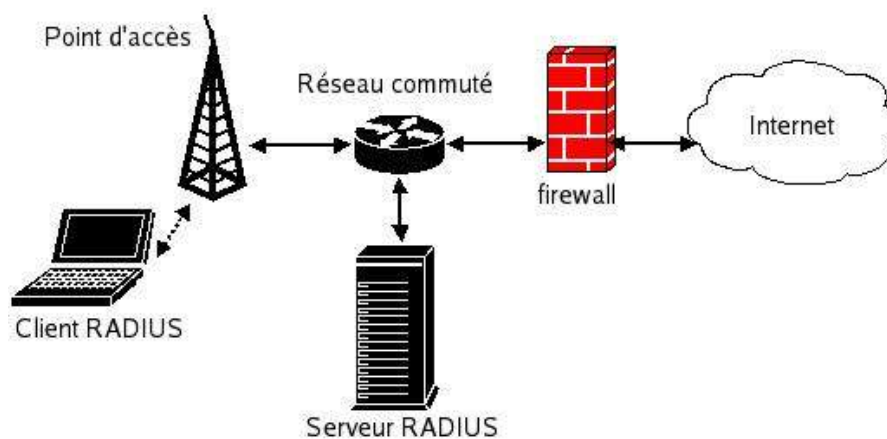


Figure 19: Architecture d'authentification RADIUS.

Le serveur AAA est RADIUS (*Remote Authentication Dial-In User Service*). Il est chargé de trois tâches principales:

- 1) Vérifier l'identité du client (*Authentication*),
- 2) Définir les droits du client (*Authorization*),
- 3) Archiver l'utilisation des ressources réseaux du client (*Accounting*).

Freeradius fonctionne par un échange de messages entre les points d'accès et le serveur. Le serveur utilise le port 1812 (udp) pour l'authentification et le port 1813 (udp) pour la gestion de comptes (*accounting*).

Les messages RADIUS sont définis ci-dessous:

- Access-Request: demande d'accès du client,
- Access-Accept: acceptation du client,
- Access-Reject: rejet du client,
- Access-Challenge: échange entre client et serveur,
- Accounting-Request: demande de gestion du compte client,
- Accounting-Response: réponse de gestion de compte client,
- Attributes: Attributs nécessaires à RADIUS.

Il y a une authentification mutuelle entre le serveur et les points d'accès grâce à un mot de passe partagé qui n'est jamais envoyé sur le réseau. Ce mot de passe commun sert à crypter les échanges entre le serveur freeradius et les points d'accès.

Freeradius est gratuit et libre, il supporte de nombreux EAP, souvent plus que certains produits concurrents payants. Il permet d'authentifier les utilisateurs selon plusieurs moyens: mots de passe UNIX, fichier texte, base de données SQL (MySQL, PostgreSQL...) ou annuaire LDAP (OpenLDAP, Active Directory). Dans la suite de ce manuel, nous utiliserons une base d'utilisateurs locale (fichier texte) afin de ne pas compliquer les procédures.

L'utilisation de MySQL, OpenLDAP et Active Directory sera expliquée plus tard pour ne pas compliquer les choses. Nous allons maintenant aborder l'installation de tous les outils nécessaires au déploiement de freeradius et des EAP.

5.2.1. Installation de Linux

Le but de ce manuel n'est pas de décrire l'installation de Linux. Linux est un système d'exploitation libre de plus en plus facile à installer. Il faut installer une version minimale de Linux: le noyau, les outils et bibliothèques de base, une interface graphique (toujours plus agréable mais pas indispensable) et ne surtout pas oublier le compilateur C gcc avec les outils de développement standard.

La distribution qui a servi à l'installation de freeradius est une « FEDORA Core 3 » qui est basée sur le noyau 2.6.9 avec de nombreux outils de configuration du système. Notre choix s'est porté sur une installation à partir des sources (compilation) plutôt qu'un paquet « .rpm » standard afin d'obtenir la dernière version de freeradius et d'openssl (nécessaire pour générer les certificats de sécurité).

5.2.2. Installation de FreeRADIUS

Il faut récupérer les sources de la dernière version sur le site web <http://www.freeradiusd.org> en tar.gz avec votre navigateur favori. Il faut être super-utilisateur (root) de la machine Linux afin d'avoir les droits suffisants. Nous avons installé les sources dans « /tmp » et freeradius dans « /usr/local/freeradius-1.0.2 ».

a) Décompression de l'archive compressée « .tar.gz »:

```
[root@ordi ~]# cd /tmp

[root@ordi tmp]# ll
total 2168
-rwxr-xr-x  1 root root 2208884 avr 16 05:22 freeradius-1.0.2.tar.gz

[root@ordi tmp]# tar xvzf freeradius-1.0.2.tar.gz
freeradius-1.0.2/
freeradius-1.0.2/debian/
freeradius-1.0.2/debian/README.Debian
freeradius-1.0.2/debian/TODO
freeradius-1.0.2/debian/changelog
freeradius-1.0.2/debian/compat
freeradius-1.0.2/debian/control
freeradius-1.0.2/debian/copyright
freeradius-1.0.2/debian/freeradius-dialupadmin.README.Debian
freeradius-1.0.2/debian/freeradius-iodbc.postinst
freeradius-1.0.2/debian/freeradius-krb5.postinst
freeradius-1.0.2/debian/freeradius-ldap.postinst
freeradius-1.0.2/debian/freeradius-mysql.postinst
freeradius-1.0.2/debian/freeradius-postgresql.postinst
freeradius-1.0.2/debian/freeradius.dirs
freeradius-1.0.2/debian/freeradius.examples
freeradius-1.0.2/debian/freeradius.init
freeradius-1.0.2/debian/freeradius.logrotate
freeradius-1.0.2/debian/freeradius.postinst
freeradius-1.0.2/debian/freeradius.postrm
freeradius-1.0.2/debian/freeradius.prerm
```

```

freeradius-1.0.2/debian/freeradius.radiusd.pam
freeradius-1.0.2/debian/freeradius.undocumented
freeradius-1.0.2/debian/rules
freeradius-1.0.2/COPYRIGHT
freeradius-1.0.2/CREDITS
[...]
freeradius-1.0.2/src/tests/hmac-md5-01/digest1.txt
freeradius-1.0.2/src/tests/hmac-sha1-01/
freeradius-1.0.2/src/tests/hmac-sha1-01/digest1.txt
freeradius-1.0.2/suse/
freeradius-1.0.2/suse/freeradius.spec
freeradius-1.0.2/suse/radiusd-logrotate
freeradius-1.0.2/suse/radiusd-pam
freeradius-1.0.2/suse/rcradiusd
freeradius-1.0.2/todo/
freeradius-1.0.2/todo/TODO
freeradius-1.0.2/todo/proposed-new-users
freeradius-1.0.2/todo/serverside-ip-pools

[root@ordi tmp]# ll
total 2176
drwxr-xr-x 15 1166 1166 4096 fév 16 20:36 freeradius-1.0.2
-rwxr-xr-x 1 root root 2208884 avr 16 05:22 freeradius-1.0.2.tar.gz

```

b) Déplacement dans le répertoire freeRADIUS.

```

[root@ordi tmp]# cd freeradius-1.0.2

[root@ordi freeradius-1.0.2]# ll
total 1008
-rw-r--r-- 1 1166 1166 3369 mai 17 2004 acconfig.h
-rw-r--r-- 1 1166 1166 133030 fév 13 02:03 aclocal.m4
-rwxr-xr-x 1 1166 1166 43609 jui 17 2004 config.guess
-rw-r--r-- 1 1166 1166 457 fév 13 02:40 config.h.in
-rwxr-xr-x 1 1166 1166 31160 jui 17 2004 config.sub
-rwxr-xr-x 1 1166 1166 274535 fév 13 02:40 configure
-rw-r--r-- 1 1166 1166 26268 fév 7 20:52 configure.in
-rw-r--r-- 1 1166 1166 1106 jui 31 2002 COPYRIGHT
-rw-r--r-- 1 1166 1166 1383 oct 2 2002 CREDITS
drwxr-xr-x 2 1166 1166 4096 fév 16 20:36 debian
drwxr-xr-x 9 1166 1166 4096 fév 16 20:36 dialup_admin
drwxr-xr-x 4 1166 1166 4096 fév 16 20:36 doc
-rw-r--r-- 1 1166 1166 5842 jan 31 20:01 INSTALL
-rwxr-xr-x 1 1166 1166 5598 fév 22 2000 install-sh
drwxr-xr-x 2 1166 1166 4096 fév 16 20:36 libltdl
-rw-r--r-- 1 1166 1166 18083 déc 28 2000 LICENSE
-rwxr-xr-x 1 1166 1166 96333 jun 24 2003 ltconfig
-rwxr-xr-x 1 1166 1166 139120 jun 24 2003 ltmain.sh
-rw-r--r-- 1 1166 1166 3074 jun 9 2004 Makefile
-rw-r--r-- 1 1166 1166 1674 jan 23 2004 Make.inc.in
drwxr-xr-x 5 1166 1166 4096 fév 16 20:36 man
drwxr-xr-x 2 1166 1166 4096 fév 16 20:36 mibs
-rwxr-xr-x 1 1166 1166 10270 jun 16 2003 missing
drwxr-xr-x 3 1166 1166 4096 fév 16 20:36 raddb
-rw-r--r-- 1 1166 1166 4070 avr 11 2003 README
drwxr-xr-x 2 1166 1166 4096 fév 16 20:36 redhat
drwxr-xr-x 2 1166 1166 4096 fév 16 20:36 scripts
drwxr-xr-x 2 1166 1166 4096 fév 16 20:36 share
drwxr-xr-x 8 1166 1166 4096 fév 16 20:36 src
drwxr-xr-x 2 1166 1166 4096 fév 16 20:36 suse

```

drwxr-xr-x 2 1166 1166 4096 fév 16 20:36 todo

c) Configuration de l'installation dans « /usr/local/freeradius-1.0.2 ».

```
[root@ordi freeradius-1.0.2]# ./configure --prefix=/usr/local/freeradius-1.0.2
creating cache ./config.cache
checking for gcc... gcc
checking whether the C compiler (gcc ) works... yes
checking whether the C compiler (gcc ) is a cross-compiler... no
checking whether we are using GNU C... yes
checking whether gcc accepts -g... yes
checking how to run the C preprocessor... gcc -E
checking for AIX... no
checking whether gcc needs -traditional... no
checking whether we are using SUNPro C... no
checking for ranlib... ranlib
checking whether byte ordering is bigendian... no
checking for gmake... yes
checking for gmake... /usr/bin/gmake
checking for lt_dlinit in -lldl... yes
checking for Cygwin environment... no
[...]
checking for gcc... (cached) gcc
checking whether the C compiler (gcc -g -O2 -D_REENTRANT
-D_POSIX_PTHREAD_SEMANTICS -DOPENSSL_NO_KRB5 -Wall -D_GNU_SOURCE -DDEBUG )
works... yes
checking whether the C compiler (gcc -g -O2 -D_REENTRANT
-D_POSIX_PTHREAD_SEMANTICS -DOPENSSL_NO_KRB5 -Wall -D_GNU_SOURCE -DDEBUG ) is
a cross-compiler... no
checking whether we are using GNU C... (cached) yes
checking whether gcc accepts -g... (cached) yes
checking how to run the C preprocessor... (cached) gcc -E
checking for regex.h... (cached) yes
creating ./config.status
creating Makefile
creating config.h
```

d) Lancement de la compilation.

```
[root@ordi freeradius-1.0.2]# make
gmake[1]: Entering directory `/tmp/freeradius-1.0.2'
Making all in src...
gmake[2]: Entering directory `/tmp/freeradius-1.0.2/src'
gmake[3]: Entering directory `/tmp/freeradius-1.0.2/src'
Making all in include...
gmake[4]: Entering directory `/tmp/freeradius-1.0.2/src/include'
gmake[4]: Rien à faire pour « all ».
gmake[4]: Leaving directory `/tmp/freeradius-1.0.2/src/include'
Making all in lib...
gmake[4]: Entering directory `/tmp/freeradius-1.0.2/src/lib'
gcc -g -O2 -D_REENTRANT -D_POSIX_PTHREAD_SEMANTICS -DOPENSSL_NO_KRB5 -Wall
-D_GNU_SOURCE -DDEBUG -D_LIBRADIUS -I../include -DHMAC_SHA1_DATA_PROBLEMS -c
dict.c -o dict.o
gcc -g -O2 -D_REENTRANT -D_POSIX_PTHREAD_SEMANTICS -DOPENSSL_NO_KRB5 -Wall
-D_GNU_SOURCE -DDEBUG -D_LIBRADIUS -I../include -DHMAC_SHA1_DATA_PROBLEMS -c
print.c -o print.o
gcc -g -O2 -D_REENTRANT -D_POSIX_PTHREAD_SEMANTICS -DOPENSSL_NO_KRB5 -Wall
-D_GNU_SOURCE -DDEBUG -D_LIBRADIUS -I../include -DHMAC_SHA1_DATA_PROBLEMS -c
```

```
radius.c -o radius.o
[...]
gcc -g -O2 -D_REENTRANT -D_POSIX_PTHREAD_SEMANTICS -DOPENSSL_NO_KRB5 -Wall
-D_GNU_SOURCE -DDEBUG -I../include -DHOSTINFO="" -DRADIUSD_VERSION=
"1.0.2" -L../lib -o radrelay radrelay.o mainconfig.o util.o nas.o client.o
log.o conffile.o files.o xlat.o -lnsl -lresolv -lpthread -lcrypto -lssl
-lradius
gmake[4]: Leaving directory `/tmp/freeradius-1.0.2/src/main'
gmake[3]: Leaving directory `/tmp/freeradius-1.0.2/src'
gmake[2]: Leaving directory `/tmp/freeradius-1.0.2/src'
Making all in raddb...
gmake[2]: Entering directory `/tmp/freeradius-1.0.2/raddb'
gmake[2]: Rien à faire pour « all ».
gmake[2]: Leaving directory `/tmp/freeradius-1.0.2/raddb'
Making all in scripts...
gmake[2]: Entering directory `/tmp/freeradius-1.0.2/scripts'
gmake[2]: Rien à faire pour « all ».
gmake[2]: Leaving directory `/tmp/freeradius-1.0.2/scripts'
Making all in doc...
gmake[2]: Entering directory `/tmp/freeradius-1.0.2/doc'
gmake[3]: Entering directory `/tmp/freeradius-1.0.2/doc'
Making all in rfc...
gmake[4]: Entering directory `/tmp/freeradius-1.0.2/doc/rfc'
gmake[4]: Rien à faire pour « all ».
gmake[4]: Leaving directory `/tmp/freeradius-1.0.2/doc/rfc'
gmake[3]: Leaving directory `/tmp/freeradius-1.0.2/doc'
gmake[2]: Leaving directory `/tmp/freeradius-1.0.2/doc'
gmake[1]: Leaving directory `/tmp/freeradius-1.0.2'
```

e) Installation des exécutables.

```
[root@ordi freeradius-1.0.2]# make install
/tmp/freeradius-1.0.2/install-sh -c -d -m 755 /usr/local/freeradius-1.0.2/sbin
/tmp/freeradius-1.0.2/install-sh -c -d -m 755 /usr/local/freeradius-1.0.2/bin
/tmp/freeradius-1.0.2/install-sh -c -d -m 755 /usr/local/freeradius-
1.0.2/etc/raddb
[...]
gmake[1]: Leaving directory `/tmp/freeradius-1.0.2'
Installing dictionary files in /usr/local/freeradius-1.0.2/share/freeradius
/tmp/freeradius-1.0.2/libtool --finish /usr/local/freeradius-1.0.2/lib
PATH="$PATH:/sbin" ldconfig -n /usr/local/freeradius-1.0.2/lib
-----
Libraries have been installed in:
/usr/local/freeradius-1.0.2/lib
```

If you ever happen to want to link against installed libraries in a given directory, LIBDIR, you must either use libtool, and specify the full pathname of the library, or use the `-LLIBDIR` flag during linking and do at least one of the following:

- add LIBDIR to the `'LD_LIBRARY_PATH'` environment variable during execution
- add LIBDIR to the `'LD_RUN_PATH'` environment variable during linking
- use the `'-Wl,--rpath -Wl,LIBDIR'` linker flag
- have your system administrator add LIBDIR to `'/etc/ld.so.conf'`

See any operating system documentation about shared libraries for more information, such as the `ld(1)` and `ld.so(8)` manual pages.

f) Validation de la recherche des librairies dans ce nouveau répertoire.

```
[root@ordi freeradius-1.0.2]# vi /etc/ld.so.conf
include ld.so.conf.d/*.conf
/usr/lib/opencv
/usr/lib/ipp
/usr/lib/ipp/linux32
/usr/lib/mysql
/usr/X11R6/lib
/home/tools/intel/opencv/lib
/usr/java/jdk1.5.0_01/jre/lib/i386
/usr/local/openssl-certgen/lib
/usr/local/freeradius-1.0.2/lib
```

g) Reconfiguration du chargeur de librairies dynamiques.

```
[root@ordi freeradius-1.0.2]# ldconfig -v | grep radius
/usr/local/freeradius-1.0.2/lib:
    libradius-1.0.2.so -> libradius.so
```

h) Vérification du bon déroulement de l'installation.

```
[root@ordi freeradius-1.0.2]# cd /usr/local/freeradius-1.0.2/
[root@ordi freeradius-1.0.2]# ll
total 64
drwxr-xr-x  2 root root  4096 mai  2 08:53 bin
drwxr-xr-x  3 root root  4096 mai  2 08:52 etc
drwxr-xr-x  2 root root 12288 mai  2 08:52 lib
drwxr-xr-x  5 root root  4096 mai  2 08:52 man
drwxr-xr-x  2 root root  4096 mai  2 08:53 sbin
drwx----- 4 root root  4096 mai  2 08:53 share
drwxr-xr-x  4 root root  4096 mai  2 08:52 var
```

i) Déplacement dans le répertoire de configuration.

```
[root@ordi freeradius-1.0.2]# cd etc
[root@ordi etc]# ll
total 8
drwxr-xr-x  3 root root 4096 mai  2 08:53 raddb
[root@ordi etc]# cd raddb
[root@ordi raddb]# ll
total 336
-rw-r--r--  1 root root  422 mai  2 08:53 acct_users
-rw-r--r--  1 root root 3454 mai  2 08:53 attrs
drwxr-xr-x  3 root root 4096 mai  2 08:53 certs
-rw-r-----  1 root root  189 mai  2 08:53 clients
-rw-r-----  1 root root 2937 mai  2 08:53 clients.conf
-rw-r--r--  1 root root  952 mai  2 08:53 dictionary
-rw-r--r--  1 root root 9080 mai  2 08:53 eap.conf
-rw-r--r--  1 root root 8266 mai  2 08:53 experimental.conf
-rw-r--r--  1 root root 2396 mai  2 08:53 hints
-rw-r--r--  1 root root 1604 mai  2 08:53 huntgroups
-rw-r--r--  1 root root 2333 mai  2 08:53 ldap.attrmap
-rw-r--r--  1 root root 9330 mai  2 08:53 mssql.conf
-rw-r--r--  1 root root 1020 mai  2 08:53 naslist
```

```

-rw-r----- 1 root root 856 mai 2 08:53 naspasswd
-rw-r--r-- 1 root root 12267 mai 2 08:53 oraclesql.conf
-rw-r--r-- 1 root root 14156 mai 2 08:53 postgresql.conf
-rw-r--r-- 1 root root 531 mai 2 08:53 preproxy_users
-rw-r--r-- 1 root root 8862 mai 2 08:53 proxy.conf
-rw-r--r-- 1 root root 57887 mai 2 08:53 radiusd.conf
-rw-r--r-- 1 root root 187 mai 2 08:53 realms
-rw-r--r-- 1 root root 1405 mai 2 08:53 snmp.conf
-rw-r--r-- 1 root root 13892 mai 2 08:53 sql.conf
-rw-r--r-- 1 root root 6940 mai 2 08:53 users
-rw-r--r-- 1 root root 7267 mai 2 08:53 x99.conf
-rw-r--r-- 1 root root 4165 mai 2 08:53 x99passwd.sample

```

j) Ajout d'un utilisateur pour tester l'installation en local.

```

[root@ordi raddb]# vi users
[...]
"j.landre"      Auth-Type := Local, User-Password == "testpw"
                  Reply-Message = "Hello, %u"
[...]

```

k) Définition du mot de passe partagé entre le point d'accès et le serveur RADIUS.

```

[root@ordi raddb]# vi clients.conf
[...]
client 127.0.0.1 {
    #
    # The shared secret use to "encrypt" and "sign" packets between
    # the NAS and FreeRADIUS. You MUST change this secret from the
    # default, otherwise it's not a secret any more!
    #
    # The secret can be any string, up to 32 characters in length.
    #
    secret          = secretpartage

    #
    # The short name is used as an alias for the fully qualified
    # domain name, or the IP address.
    #
    shortname       = localhost

    #
    # the following three fields are optional, but may be used by
    # checkrad.pl for simultaneous use checks
    #

    #
    # The nastype tells 'checkrad.pl' which NAS-specific method to
    # use to query the NAS for simultaneous use.
    #
    # Permitted NAS types are:
    #
    #     cisco
    #     computone
    #     livingston
    #     max40xx
    #     multitech

```

```

#       netserver
#       pathras
#       patton
#       portslave
#       tc
#       usrhiper
#       other          # for all other types

#
nastype      = other      # localhost isn't usually a NAS...

#
# The following two configurations are for future use.
# The 'naspaswd' file is currently used to store the NAS
# login name and password, which is used by checkrad.pl
# when querying the NAS for simultaneous use.
#
# login        = !root
# password     = someadminpas
}
[...]
```

1) Lancement de « radiusd » pour tester le bon fonctionnement.

```

[root@ordi raddb]# /usr/local/freeradius-1.0.2/sbin/radiusd -X
Starting - reading configuration files ...
reread_config: reading radiusd.conf
Config: including file: /usr/local/freeradius-1.0.2/etc/raddb/proxy.conf
Config: including file: /usr/local/freeradius-1.0.2/etc/raddb/clients.conf
Config: including file: /usr/local/freeradius-1.0.2/etc/raddb/snmp.conf
Config: including file: /usr/local/freeradius-1.0.2/etc/raddb/eap.conf
Config: including file: /usr/local/freeradius-1.0.2/etc/raddb/sql.conf
main: prefix = "/usr/local/freeradius-1.0.2"
main: localstatedir = "/usr/local/freeradius-1.0.2/var"
main: logdir = "/usr/local/freeradius-1.0.2/var/log/radius"
main: libdir = "/usr/local/freeradius-1.0.2/lib"
main: radacctdir = "/usr/local/freeradius-1.0.2/var/log/radius/radacct"
main: hostname_lookups = no
main: max_request_time = 30
main: cleanup_delay = 5
main: max_requests = 1024
main: delete_blocked_requests = 0
main: port = 0
main: allow_core_dumps = no
main: log_stripped_names = no
[...]
Module: Loaded radutmp
radutmp: filename = "/usr/local/freeradius-1.0.2/var/log/radius/radutmp"
radutmp: username = "%{User-Name}"
radutmp: case_sensitive = yes
radutmp: check_with_nas = yes
radutmp: perm = 384
radutmp: callerid = yes
Module: Instantiated radutmp (radutmp)
Listening on authentication *:1812
Listening on accounting *:1813
Listening on proxy *:1814
Ready to process requests.
```

m) Lancement du test « radtest » dans une autre fenêtre shell.

```
[root@ordi ~]# /usr/local/freeradius-1.0.2/bin/radtest
Usage: radtest user passwd radius-server[:port] nas-port-number secret [ppphint]
[nasname]
[root@ordi raddb]# /usr/local/freeradius-1.0.2/bin/radtest j.landre testpw
localhost:1812 1812 secretpartage
Sending Access-Request of id 105 to 127.0.0.1:1812
    User-Name = "j.landre"
    User-Password = "testpw"
    NAS-IP-Address = ordi.u-bourgogne.fr
    NAS-Port = 1812
rad_recv: Access-Accept packet from host 127.0.0.1:1812, id=105, length=37
    Reply-Message = "Hello, j.landre"
```

n) Vérifier le message côté serveur RADIUS.

```
rad_recv: Access-Request packet from host 127.0.0.1:32768, id=105, length=60
    User-Name = "j.landre"
    User-Password = "testpw"
    NAS-IP-Address = 255.255.255.255
    NAS-Port = 1812
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 0
    modcall[authorize]: module "preprocess" returns ok for request 0
    modcall[authorize]: module "chap" returns noop for request 0
    modcall[authorize]: module "mschap" returns noop for request 0
        rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
        rlm_realm: No such realm "NULL"
    modcall[authorize]: module "suffix" returns noop for request 0
    rlm_eap: No EAP-Message, not doing EAP
    modcall[authorize]: module "eap" returns noop for request 0
    users: Matched entry j.landre at line 97
    modcall[authorize]: module "files" returns ok for request 0
modcall: group authorize returns ok for request 0
    rad_check_password: Found Auth-Type Local
auth: type Local
auth: user supplied User-Password matches local User-Password
radius_xlat: 'Hello, j.landre'
Sending Access-Accept of id 105 to 127.0.0.1:32768
    Reply-Message = "Hello, j.landre"
Finished request 0
Going to the next request
--- Walking the entire request list ---
Waking up in 6 seconds...
--- Walking the entire request list ---
Cleaning up request 0 ID 105 with timestamp 427683ca
Nothing to do. Sleeping until we see a request.
```

o) Installation de freeradius en tant que service.

```
[root@ordi ~]# vi /etc/rc.local
[...]
# ajout par jerome landre
/usr/local/freeradius-1.0.2/sbin/rc.radiusd start
[root@ordi ~]#
```

Ce test permet de voir que le serveur freeRADIUS est correctement installé et fonctionne en répondant à une requête pour un utilisateur local. Il se chargera automatiquement à chaque démarrage du serveur grâce au fichier « /etc/rc.local ».

5.2.3. Plateforme de test de freeRADIUS

Afin de tester l'installation de freeradius, il a été nécessaire de mettre en place une infrastructure Wi-Fi de base comportant:

- Le serveur RADIUS,
- Un commutateur Cisco 3500 24 ports supportant 802.1x et les VLANs,
- Un point d'accès Cisco AP1100 relié au commutateur en mode TRUNK 802.1Q (tous VLANs),
- Un client Wi-Fi avec carte Cisco Aironet 350 supportant LEAP, TLS, TTLS et PEAP,
- Un ordinateur fixe relié en filaire sur un port configuré en VLAN 100.

La figure 20 donne le schéma de notre installation de test.

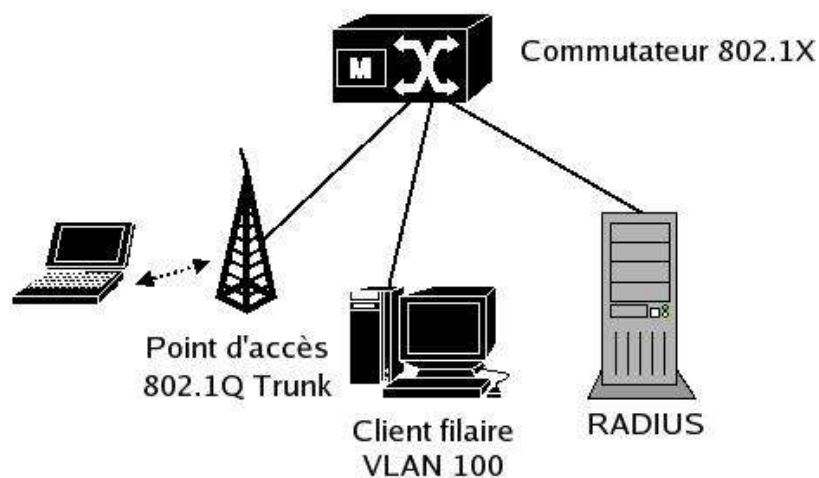


Figure 20: Plateforme de test de freeradius.

Le client filaire est placé dans le VLAN 100 afin de placer le client Wi-Fi dans ce WVLAN. Le point d'accès Wi-Fi est placé en lien filaire Trunk 802.1Q ce qui signifie qu'il est multi-VLAN. Le serveur RADIUS a l'adresse IP « 10.0.0.100 », le point d'accès l'adresse « 10.0.0.189 » et le client l'adresse publique « 193.52.240.254 ».

Afin de tester le basculement des utilisateurs dans un WVLAN (Wireless Virtual Local Area Network), c'est-à-dire un réseau local virtuel, nous avons créé un autre utilisateur dans le fichier des utilisateurs de freeradius. Cet utilisateur a été associé avec trois attributs permettant de le placer dans un WVLAN après authentification.

```
[root@ordi ~]# vi /usr/local/freeradius-1.0.2/etc/raddb/users
[...]
j.landre      Auth-Type := EAP, User-Password == "testpw"
               Tunnel-Type = 13,
               Tunnel-Medium-Type = 6,
               Tunnel-Private-Group-Id = 100
[...]
[root@ordi ~]#
```

L'attribut « Tunnel-Type » définit le type du tunnel, 13 signifie que le type du tunnel est « VLAN », « Tunnel-Medium-Type » donne le type du medium de transport, 6 signifie « IEEE-802 », « Tunnel-Private-Group-Id » définit le numéro du VLAN.

Ces valeurs d'attributs sont définies dans le fichier dictionnaire « /usr/local/freeradius-1.0.2/share/freeradius/dictionary.tunnel »:

```
[root@ordi ~]# cd /usr/local/freeradius-1.0.2/share/freeradius
[root@ordi freeradius]# less dictionary.tunnel
[...]
VALUE          Tunnel-Type      VLAN      13
[...]
VALUE          Tunnel-Medium-Type IEEE-802    6
[...]
```

Cette configuration va nous permettre de tester les EAP et le basculement dans un VLAN d'un utilisateur après succès de l'authentification.

5.2.4. Fonctionnement de freeRADIUS

Freeradius est un serveur qui s'exécute sous Linux sous la forme d'un moniteur (*daemon*) qui écoute sur le port 1812 pour l'authentification et sur le port 1813 pour la gestion des comptes (*accounting*).

Les fichiers de configuration se trouvent dans « /usr/local/freeradius-1.0.2/etc/raddb » (dans notre installation).

```
[root@ordi ~]# cd /usr/local/freeradius-1.0.2/
[root@ordi freeradius-1.0.2]# ll
total 64
drwxr-xr-x  2 root root  4096 mai  2 08:53 bin
drwxr-xr-x  3 root root  4096 mai  2 08:52 etc
drwxr-xr-x  2 root root 12288 mai  2 08:52 lib
drwxr-xr-x  5 root root  4096 mai  2 08:52 man
drwxr-xr-x  2 root root  4096 mai  2 08:53 sbin
drwx----- 4 root root  4096 mai  2 08:53 share
drwxr-xr-x  4 root root  4096 mai  2 08:52 var
[root@ordi freeradius-1.0.2]# cd etc/
[root@ordi etc]# ll
total 8
drwxr-xr-x  3 root root 4096 mai  2 21:47 raddb
[root@ordi etc]# cd raddb/
[root@ordi raddb]# ll
total 336
-rw-r--r--  1 root root  422 mai  2 08:53 acct_users
-rw-r--r--  1 root root 3454 mai  2 08:53 attrs
drwxr-xr-x  3 root root  4096 mai  2 08:53 certs
-rw-r----- 1 root root  189 mai  2 08:53 clients
-rw-r----- 1 root root 2940 mai  2 21:47 clients.conf
-rw-r--r--  1 root root  952 mai  2 08:53 dictionary
-rw-r--r--  1 root root 9080 mai  2 08:53 eap.conf
-rw-r--r--  1 root root 8266 mai  2 08:53 experimental.conf
-rw-r--r--  1 root root 2396 mai  2 08:53 hints
-rw-r--r--  1 root root 1604 mai  2 08:53 huntgroups
-rw-r--r--  1 root root 2333 mai  2 08:53 ldap.attrmap
-rw-r--r--  1 root root 9330 mai  2 08:53 mssql.conf
-rw-r--r--  1 root root 1020 mai  2 08:53 naslist
-rw-r----- 1 root root  856 mai  2 08:53 naspasswd
-rw-r--r--  1 root root 12267 mai  2 08:53 oraclesql.conf
-rw-r--r--  1 root root 14156 mai  2 08:53 postgresql.conf
-rw-r--r--  1 root root  531 mai  2 08:53 preproxy_users
-rw-r--r--  1 root root 8862 mai  2 08:53 proxy.conf
```

```
-rw-r--r-- 1 root root 57887 mai  2 08:53 radiusd.conf
-rw-r--r-- 1 root root   187 mai  2 08:53 realms
-rw-r--r-- 1 root root  1405 mai  2 08:53 snmp.conf
-rw-r--r-- 1 root root 13892 mai  2 08:53 sql.conf
-rw-r--r-- 1 root root   7028 mai  2 09:16 users
-rw-r--r-- 1 root root   7267 mai  2 08:53 x99.conf
-rw-r--r-- 1 root root   4165 mai  2 08:53 x99passwd.sample
[root@ordi raddb]#
```

La description des entrées principales de raddb permet de mieux comprendre l'architecture de freeradius et les services qu'il propose.

Nom	Description
certs	Répertoire des certificats utilisés par EAP-TLS, EAP-TTLS et PEAP.
clients.conf	Fichier des points d'accès reconnus par RADIUS (même mot de passe partagé).
eap.conf	Fichier définissant les EAP utilisés par le système d'authentification.
ldap.attrmap	Correspondance entre les attributs LDAP et attributs RADIUS.
radiusd.conf	Fichier de configuration principal de freeradius.
sql.conf	Fichier de configuration pour MySQL.
users	Fichier des utilisateurs locaux (utilisé pour notre exemple).

Il y a quatre fichiers principaux:

- radiusd.conf: fichier de configuration général de freeradius.
- clients.conf: fichier d'identification des clients (points d'accès) qui auront le droit d'interroger le serveur freeradius.
- eap.conf: fichier qui définit les EAP supportés par freeradius.
- users: base locale des utilisateurs (à ne pas utiliser sauf pour des démonstrations comme dans ce manuel).

Dans la suite de ce manuel, nous allons installer OpenSSL, nécessaire pour la gestion des certificats SSL et ensuite passer à la configuration des différents EAP.

5.2.5. Installation d'OpenSSL

OpenSSL est une implémentation libre de la norme SSL (Secure Socket Layer) qui permet de mettre en place une gestion de certificats numériques. OpenSSL sera utilisée pour la génération et l'exportation vers les clients de certificats numériques servant à identifier les utilisateurs ou à crypter les données pour EAP-TLS, EAP-TTLS et EAP-PEAP.

OpenSSL est librement téléchargeable sur le site officiel « <http://www.openssl.org> ». La version utilisée dans la suite porte le numéro 0.9.7g. Son installation a été réalisée à partir des sources pour une meilleure maîtrise de l'installation.

a) Décompression de l'archive compressée « .tar.gz »:

```
[root@ordi ~]# cd /tmp
[root@ordi tmp]# ll
total 3068
-rw-r--r-- 1 root root 3132217 mai  6 21:47 openssl-0.9.7g.tar.gz
[root@ordi tmp]# tar xvzf openssl-0.9.7g.tar.gz
openssl-0.9.7g/apps/
openssl-0.9.7g/apps/app_rand.c
```

```

openssl-0.9.7g/apps/apps.c
openssl-0.9.7g/apps/apps.h
openssl-0.9.7g/apps/asn1pars.c
openssl-0.9.7g/apps/ca.c
openssl-0.9.7g/apps/ca-cert.srl
openssl-0.9.7g/apps/CA.com
openssl-0.9.7g/apps/ca-key.pem
openssl-0.9.7g/apps/CA.pl
openssl-0.9.7g/apps/CA.pl.in
openssl-0.9.7g/apps/ca-req.pem
openssl-0.9.7g/apps/CA.sh
openssl-0.9.7g/apps/cert.pem
[...]
openssl-0.9.7g/util/x86asm.sh
openssl-0.9.7g/VMS/
openssl-0.9.7g/VMS/install.com
openssl-0.9.7g/VMS/mkshared.com
openssl-0.9.7g/VMS/multinet_shr.opt
openssl-0.9.7g/VMS/openssl_utils.com
openssl-0.9.7g/VMS/socketshr_shr.opt
openssl-0.9.7g/VMS/tcpip_shr_decc.opt
openssl-0.9.7g/VMS/test-includes.com
openssl-0.9.7g/VMS/TOD0
openssl-0.9.7g/VMS/ucx_shr_decc_log.opt
openssl-0.9.7g/VMS/ucx_shr_decc.opt
openssl-0.9.7g/VMS/ucx_shr_vaxc.opt
openssl-0.9.7g/VMS/WISHLIST.TXT
openssl-0.9.7g/VMS/VMSify-conf.pl
[root@ordi tmp]# ll
total 3076
drwxr-xr-x  21 root root    4096 mai  6 21:50 openssl-0.9.7g
-rw-r--r--   1 root root 3132217 mai  6 21:47 openssl-0.9.7g.tar.gz
[root@ordi tmp]#

```

b) Déplacement dans le répertoire openSSL.

```

[root@ordi tmp]# cd openssl-0.9.7g
[root@ordi openssl-0.9.7g]# ll
total 1048
drwxr-xr-x  4 root root    4096 avr 11 17:17 apps
drwxr-xr-x  2 root root    4096 avr 11 17:16 bugs
drwxr-xr-x  4 root root    4096 avr 11 17:16 certs
-rw-r--r--  1 root root 290580 avr 11 17:10 CHANGES
-rw-r--r--  1 root root  42751 déc 23 1998 CHANGES.SSLeay
-rwxr-xr-x  1 root root   25402 avr  7 22:26 config
-rwxr-xr-x  1 root root   85428 avr  7 18:06 Configure
drwxr-xr-x 46 root root    4096 avr 11 17:17 crypto
drwxr-xr-x 15 root root    4096 avr 11 17:16 demos
drwxr-xr-x  6 root root    4096 avr 11 17:17 doc
-rw-r--r--  1 root root   9539 oct 20 2004 e_os2.h
-rw-r--r--  1 root root  17295 jan 14 17:22 e_os.h
-rw-r--r--  1 root root  35747 avr 11 17:10 FAQ
drwxr-xr-x  9 root root    4096 avr 11 17:17 fips
drwxr-xr-x  3 root root    4096 avr 11 17:17 include
-rw-r--r--  1 root root  13301 mai 11 2004 INSTALL
-rw-r--r--  1 root root   2757 mai 27 2004 install.com
-rw-r--r--  1 root root   2053 jan 14 17:24 INSTALL.DJGPP
-rw-r--r--  1 root root   3264 oct  1 2001 INSTALL.MacOS
-rw-r--r--  1 root root    744 jui 17 2002 INSTALL.OS2
-rw-r--r--  1 root root  11363 sep  7 2001 INSTALL.VMS

```



```

-rw-r--r-- 1 root root 10134 mai 11 2004 INSTALL.W32
-rw-r--r-- 1 root root 2409 déc 3 2002 INSTALL.WCE
-rw-r--r-- 1 root root 6279 mar 17 2004 LICENSE
drwxr-xr-x 3 root root 4096 avr 11 17:16 MacOS
-rw-r--r-- 1 root root 33783 avr 11 17:17 Makefile
-rw-r--r-- 1 root root 33783 avr 11 17:17 Makefile.bak
-rw-r--r-- 1 root root 33851 mar 15 10:46 Makefile.org
-rwxr-xr-x 1 root root 26776 août 9 2004 makevms.com
drwxr-xr-x 2 root root 4096 avr 11 17:17 ms
-rw-r--r-- 1 root root 14633 avr 11 17:05 NEWS
-rw-r--r-- 1 root root 137 fév 28 1999 openssl.doxy
-rw-r--r-- 1 root root 7858 avr 11 17:10 openssl.spec
drwxr-xr-x 2 root root 4096 avr 11 17:17 os2
drwxr-xr-x 2 root root 4096 avr 11 17:17 perl
-rw-r--r-- 1 root root 4958 déc 20 14:20 PROBLEMS
-rw-r--r-- 1 root root 7912 avr 11 17:10 README
-rw-r--r-- 1 root root 7699 déc 8 2000 README.ASN1
-rw-r--r-- 1 root root 16100 jui 8 2002 README.ENGINE
drwxr-xr-x 2 root root 4096 avr 11 17:17 shlib
drwxr-xr-x 2 root root 4096 avr 11 17:17 ssl
drwxr-xr-x 2 root root 4096 avr 11 17:17 test
drwxr-xr-x 5 root root 4096 avr 11 17:17 times
drwxr-xr-x 2 root root 4096 avr 11 17:17 tools
drwxr-xr-x 3 root root 4096 avr 11 17:17 util
drwxr-xr-x 2 root root 4096 avr 11 17:16 VMS
[root@ordi openssl-0.9.7g]#

```

c) Configuration de l'installation dans « /usr/local/openssl-0.9.7g ».

```

[root@ordi openssl-0.9.7g]# ./config -prefix=/usr/local/openssl-0.9.7g shared
Operating system: i686-whatever-linux2
Configuring for linux-pentium
Configuring for linux-pentium
IsWindows=0
CC                =gcc
CFLAG              =-fPIC -DOPENSSL_THREADS -D_REENTRANT -DDSO_DLFCN -DHAVE_DLFCN_H
-DOPENSSL_NO_KRB5 -DL_ENDIAN -DTERMIO -O3 -fomit-frame-pointer -mcpu=pentium
-Wall -DSHA1_ASM -DMD5_ASM -DRMD160_ASM
EX_LIBS            =-ldl
BN_ASM             =asm/bn86-elf.o asm/co86-elf.o
DES_ENC            =asm/dx86-elf.o asm/yx86-elf.o
BF_ENC            =asm/bx86-elf.o
CAST_ENC          =c_enc.o
RC4_ENC           =asm/rx86-elf.o
RC5_ENC           =asm/r586-elf.o
MD5_OBJ_ASM       =asm/mx86-elf.o
SHA1_OBJ_ASM      =asm/sx86-elf.o
RMD160_OBJ_ASM    =asm/rm86-elf.o
[...]
make[1]: Entering directory `/tmp/tmp/openssl-0.9.7g/tools'
make[1]: Rien à faire pour « links ».
make[1]: Leaving directory `/tmp/tmp/openssl-0.9.7g/tools'
generating dummy tests (if needed)...
make[1]: Entering directory `/tmp/tmp/openssl-0.9.7g/test'
make[1]: Rien à faire pour « generate ».
make[1]: Leaving directory `/tmp/tmp/openssl-0.9.7g/test'

Configured for linux-pentium.
[root@ordi openssl-0.9.7g]#

```

d) Lancement de la compilation.

```
[root@ordi openssl-0.9.7g]# make
making all in crypto...
make[1]: Entering directory `/tmp/tmp/openssl-0.9.7g/crypto'
( echo "#ifndef MK1MF_BUILD"; \
echo ' /* auto-generated by crypto/Makefile for crypto/cversion.c */'; \
echo ' #define CFLAGS "gcc -fPIC -DOPENSSL_THREADS -D_REENTRANT -DDSO_DLFCN
-DHAVE_DLFCN_H -DOPENSSL_NO_KRB5 -DL_ENDIAN -DTERMIO -O3 -fomit-frame-pointer
-mcpu=pentium -Wall -DSHA1_ASM -DMD5_ASM -DRMD160_ASM"; \
echo ' #define PLATFORM "linux-pentium"; \
echo " #define DATE \"`LC_ALL=C LC_TIME=C date`\""; \
echo '#endif' ) >buildinf.h
gcc -I. -I.. -I../include -fPIC -DOPENSSL_THREADS -D_REENTRANT -DDSO_DLFCN
-DHAVE_DLFCN_H -DOPENSSL_NO_KRB5 -DL_ENDIAN -DTERMIO -O3 -fomit-frame-pointer
-mcpu=pentium -Wall -DSHA1_ASM -DMD5_ASM -DRMD160_ASM -c -o cryptlib.o
cryptlib.c
[...]
+ gcc -o dummytest -I.. -I../include -fPIC -DOPENSSL_THREADS -D_REENTRANT
-DDSO_DLFCN -DHAVE_DLFCN_H -DOPENSSL_NO_KRB5 -DL_ENDIAN -DTERMIO -O3 -fomit-
frame-pointer -mcpu=pentium -Wall -DSHA1_ASM -DMD5_ASM -DRMD160_ASM dummytest.o
-L.. -lssl -L.. -lcrypto -ldl
make[1]: Leaving directory `/tmp/tmp/openssl-0.9.7g/test'
making all in tools...
make[1]: Entering directory `/tmp/tmp/openssl-0.9.7g/tools'
make[1]: Rien à faire pour « all ».
make[1]: Leaving directory `/tmp/tmp/openssl-0.9.7g/tools'
[root@ordi openssl-0.9.7g]#
```

e) Installation des exécutables.

```
[root@ordi openssl-0.9.7g]# make install
making all in crypto...
make[1]: Entering directory `/tmp/tmp/openssl-0.9.7g/crypto'
making all in crypto/objects...
make[2]: Entering directory `/tmp/tmp/openssl-0.9.7g/crypto/objects'
make[2]: Rien à faire pour « all ».
make[2]: Leaving directory `/tmp/tmp/openssl-0.9.7g/crypto/objects'
making all in crypto/md2...
make[2]: Entering directory `/tmp/tmp/openssl-0.9.7g/crypto/md2'
make[2]: Rien à faire pour « all ».
make[2]: Leaving directory `/tmp/tmp/openssl-0.9.7g/crypto/md2'
making all in crypto/md4...
make[2]: Entering directory `/tmp/tmp/openssl-0.9.7g/crypto/md4'
make[2]: Rien à faire pour « all ».
make[2]: Leaving directory `/tmp/tmp/openssl-0.9.7g/crypto/md4'
making all in crypto/md5...
[...]
installing libssl.so.0.9.7
make[1]: Entering directory `/usr/local/openssl-0.9.7g/lib'
+ rm -f libcrypto.so.0
+ ln -s libcrypto.so.0.9.7 libcrypto.so.0
+ rm -f libcrypto.so
+ ln -s libcrypto.so.0 libcrypto.so
+ rm -f libssl.so.0
+ ln -s libssl.so.0.9.7 libssl.so.0
+ rm -f libssl.so
+ ln -s libssl.so.0 libssl.so
make[1]: Leaving directory `/usr/local/openssl-0.9.7g/lib'
```

OpenSSL shared libraries have been installed in:
/usr/local/openssl-0.9.7g

If this directory is not in a standard system path for dynamic/shared libraries, then you will have problems linking and executing applications that use OpenSSL libraries UNLESS:

- * you link with static (archive) libraries. If you are truly paranoid about security, you should use static libraries.
- * you use the GNU libtool code during linking (<http://www.gnu.org/software/libtool/libtool.html>)
- * you use pkg-config during linking (this requires that PKG_CONFIG_PATH includes the path to the OpenSSL shared library directory), and make use of -R or -rpath. (<http://www.freedesktop.org/software/pkgconfig/>)
- * you specify the system-wide link path via a command such as crle(1) on Solaris systems.
- * you add the OpenSSL shared library directory to /etc/ld.so.conf and run ldconfig(8) on Linux systems.
- * you define the LD_LIBRARY_PATH, LIBPATH, SHLIB_PATH (HP), DYLD_LIBRARY_PATH (MacOS X) or PATH (Cygwin and DJGPP) environment variable and add the OpenSSL shared library directory to it.

One common tool to check the dynamic dependencies of an executable or dynamic library is ldd(1) on most UNIX systems.

See any operating system documentation and manpages about shared libraries for your version of UNIX. The following manpages may be helpful: ld(1), ld.so(1), ld.so.1(1) [Solaris], dld.sl(1) [HP], ldd(1), crle(1) [Solaris], pldd(1) [Solaris], ldconfig(8) [Linux], chatr(1) [HP].

```
cp openssl.pc /usr/local/openssl-0.9.7g/lib/pkgconfig
chmod 644 /usr/local/openssl-0.9.7g/lib/pkgconfig/openssl.pc
[root@ordi openssl-0.9.7g]#
```

f) Validation de la recherche des librairies dans ce nouveau répertoire.

```
[root@ordi openssl-0.9.7g]# cd /usr/local/
[root@ordi local]# ll
total 120
drwxr-xr-x  2 root root 4096 avr 14 11:30 bin
drwxr-xr-x  3 root root 4096 jan  5 08:33 doc
drwxr-xr-x  2 root root 4096 août 12 2004 etc
drwxr-xr-x  9 root root 4096 mai  2 08:52 freeradius-1.0.2
drwxr-xr-x  2 root root 4096 août 12 2004 games
drwxr-xr-x  3 root root 4096 avr 14 11:43 gtkscan
drwxr-xr-x  3 root root 4096 mar 30 08:59 include
drwxr-xr-x  4 root root 4096 avr 12 21:23 lib
drwxr-xr-x  2 root root 4096 août 12 2004 libexec
drwxr-xr-x  3 root root 4096 jan  5 08:33 man
drwxr-xr-x  6 root root 4096 mai  6 22:03 openssl-0.9.7g
drwxr-xr-x  6 root root 4096 avr  8 09:43 openssl-certgen
drwxr-xr-x  2 root root 4096 avr 12 21:38 sbin
drwxr-xr-x  7 root root 4096 mar 30 08:59 share
drwxr-xr-x  2 root root 4096 août 12 2004 src
[root@ordi local]# cd openssl-certgen/
[root@ordi openssl-certgen]# ll
total 32
drwxr-xr-x  2 root root 4096 avr  8 09:43 bin
```

```

drwxr-xr-x  3 root root 4096 avr  8 09:43 include
drwxr-xr-x  3 root root 4096 avr  8 09:43 lib
drwxr-xr-x  6 root root 4096 avr  8 09:54 ssl
[root@ordi openssl-certgen]# ll lib
total 3456
-rw-r--r--  1 root root 1822660 avr  8 09:43 libcrypto.a
lrwxrwxrwx  1 root root      14 avr  8 09:43 libcrypto.so -> libcrypto.so.0
lrwxrwxrwx  1 root root      18 avr  8 09:43 libcrypto.so.0 ->
libcrypto.so.0.9.7
-r-xr-xr-x  1 root root 1163130 avr  8 09:43 libcrypto.so.0.9.7
-rw-r--r--  1 root root 280456 avr  8 09:43 libssl.a
lrwxrwxrwx  1 root root      11 avr  8 09:43 libssl.so -> libssl.so.0
lrwxrwxrwx  1 root root      15 avr  8 09:43 libssl.so.0 -> libssl.so.0.9.7
-r-xr-xr-x  1 root root 211589 avr  8 09:43 libssl.so.0.9.7
drwxr-xr-x  2 root root  4096 avr  8 09:43 pkgconfig
[root@ordi openssl-certgen]#

```

g) Reconfiguration du chargeur de librairies dynamiques.

```

[root@ordi openssl-certgen]# vi /etc/ld.so.conf
[...]
/usr/local/freeradius-1.0.2/lib
/usr/local/openssl-certgen/lib
[root@ordi openssl-certgen]# ldconfig -v | grep ssl
/usr/local/openssl-certgen/lib:
    libssl.so.0.9.7 -> libssl.so.0.9.7
    libssl.so.4 -> libssl.so.0.9.7a
    libssl3.so -> libssl3.so
[root@ordi openssl-certgen]#

```

h) Vérification du bon déroulement de l'installation.

```

[root@ordi openssl-certgen]# /usr/local/openssl-0.9.7g/bin/openssl version
OpenSSL 0.9.7g 11 Apr 2005
[root@ordi openssl-certgen]#

```

i) Déplacement dans le répertoire de configuration.

```

[root@ordi openssl-certgen]# cd /usr/local/openssl-certgen/ssl
[root@ordi ssl]# ll
total 44
drwxr-xr-x  2 root root 4096 avr  8 09:43 certs
drwxr-xr-x  6 root root 4096 avr  8 09:42 man
drwxr-xr-x  2 root root 4096 avr  8 09:43 misc
-rw-r--r--  1 root root 7907 avr  8 09:54 openssl.cnf
drwxr-xr-x  2 root root 4096 avr  8 09:43 private
[root@ordi ssl]#

```

Nous allons maintenant aborder les différents EAP qui ont été testés dans notre établissement. Nous verrons les avantages et les inconvénients de tous ces EAP afin de justifier le choix que nous avons retenu pour l'I.U.T. du Creusot.

5.2.6. EAP-LEAP

LEAP a été mis au point par Cisco. Il fonctionne donc sans problème sur tous les matériels Cisco

mais on le trouve aussi dans les drivers de nombreux constructeurs. Par contre, il n'est pas supporté par Microsoft car il n'est pas inclus dans le client de Windows 2000 ou XP.

La configuration de LEAP est réalisée sur l'ordinateur client sur lequel il faut installer les derniers pilotes de la carte réseau. Ensuite, il faut choisir 802.1x et LEAP en méthode d'authentification. Cet EAP est le plus simple à mettre en oeuvre car il ne nécessite pas de certificats.

a) Edition de « radiusd.conf ».

```
[root@ordi raddb]# vi radiusd.conf
[...]
authorize {
[...]
    eap
[...]
    files
}
[...]
authenticate {
[...]
    eap
}
```

b) Modification du type EAP par défaut.

```
[root@ordi raddb]# vi eap.conf

[...]
eap {
    default_eap_type = leap
    leap {
[...]
    }
}
```

c) Exemple de trace (log) freeradius pour LEAP.

```
[root@ordi raddb]# radiusd -X
[...]
Module: Instantiated radutmp (radutmp)
Listening on authentication *:1812
Listening on accounting *:1813
Listening on proxy *:1814
Ready to process requests.
rad_recv: Access-Request packet from host 10.0.0.189:1645, id=171, length=141
    User-Name = "j.landre"
    Framed-MTU = 1400
    Called-Station-Id = "0011.bbbaa.bb96"
    Calling-Station-Id = "0012.f0cc.ddf9"
    Service-Type = Login-User
    Message-Authenticator = 0xc66f7f65f27b0454087da6745806e43f
    EAP-Message = 0x0202000d016a2e6c616e647265
    NAS-Port-Type = Wireless-802.11
    NAS-Port = 1447
    NAS-IP-Address = 10.0.0.189
    NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
```

```

modcall: entering group authorize for request 0
  modcall[authorize]: module "preprocess" returns ok for request 0
  modcall[authorize]: module "chap" returns noop for request 0
  modcall[authorize]: module "mschap" returns noop for request 0
    rlm_realm: No '@' in User-Name = "j.landred", looking up realm NULL
    rlm_realm: No such realm "NULL"
  modcall[authorize]: module "suffix" returns noop for request 0
  rlm_eap: EAP packet type response id 2 length 13
  rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
  modcall[authorize]: module "eap" returns updated for request 0
    users: Matched entry j.landred at line 115
  modcall[authorize]: module "files" returns ok for request 0
[...]
  rad_check_password: Found Auth-Type EAP
auth: type "EAP"
  Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 0
  rlm_eap: EAP Identity
  rlm_eap: processing type leap
  rlm_eap_leap: Stage 2
  rlm_eap_leap: Issuing AP Challenge
  rlm_eap_leap: Successfully initiated
  modcall[authenticate]: module "eap" returns handled for request 0
modcall: group authenticate returns handled for request 0
Sending Access-Challenge of id 171 to 10.0.0.189:1645
  Tunnel-Type:0 = VLAN
  Tunnel-Medium-Type:0 = IEEE-802
  Tunnel-Private-Group-Id:0 = "100"
  EAP-Message = 0x0103001811010008dd004956efe940976a2e6c616e647265
  Message-Authenticator = 0x00000000000000000000000000000000
  State = 0x5d042ee146c90cebe6264e09e1c257a1
Finished request 0
Going to the next request
--- Walking the entire request list ---
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:1645, id=172, length=178
  User-Name = "j.landred"
  Framed-MTU = 1400
  Called-Station-Id = "0011.bbbaa.bb96"
  Calling-Station-Id = "0012.f0cc.ddf9"
  Service-Type = Login-User
  Message-Authenticator = 0x7e3833dc078aa64cb03475e2818c4fc9
                                     EAP-Message =
0x02030020110100180c0d0a1fe4592fe8ba6ed809e79bfb99333bfc0021cf560f
  NAS-Port-Type = Wireless-802.11
  NAS-Port = 1447
  State = 0x5d042ee146c90cebe6264e09e1c257a1
  NAS-IP-Address = 10.0.0.189
  NAS-Identifier = "AP1100_TEST"
  Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 1
  modcall[authorize]: module "preprocess" returns ok for request 1
  modcall[authorize]: module "chap" returns noop for request 1
  modcall[authorize]: module "mschap" returns noop for request 1
    rlm_realm: No '@' in User-Name = "j.landred", looking up realm NULL
    rlm_realm: No such realm "NULL"
  modcall[authorize]: module "suffix" returns noop for request 1
  rlm_eap: EAP packet type response id 3 length 32
  rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
  modcall[authorize]: module "eap" returns updated for request 1
    users: Matched entry j.landred at line 115

```

```

modcall[authorize]: module "files" returns ok for request 1
rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 1
rlm_eap: Request found, released from the list
rlm_eap: EAP/leap
rlm_eap: processing type leap
rlm_eap_leap: Stage 4
rlm_eap_leap: NtChallengeResponse from AP is valid
rlm_eap: Underlying EAP-Type set EAP ID to 4
modcall[authenticate]: module "eap" returns ok for request 1
modcall: group authenticate returns ok for request 1
Sending Access-Challenge of id 172 to 10.0.0.189:1645
Tunnel-Type:0 = VLAN
Tunnel-Medium-Type:0 = IEEE-802
Tunnel-Private-Group-Id:0 = "100"
EAP-Message = 0x03040004
Message-Authenticator = 0x00000000000000000000000000000000
State = 0x7d518a9ac9b26261623359f86cb72f38
Finished request 1
Going to the next request
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:1645, id=173, length=162
User-Name = "j.landree"
Framed-MTU = 1400
Called-Station-Id = "0011.bbbaa.bb96"
Calling-Station-Id = "0012.f0cc.ddf9"
Service-Type = Login-User
Message-Authenticator = 0xda81d7b5eb31e38665dddfc7b8cbe95ab
EAP-Message = 0x010400101101000801000000a8090e10
NAS-Port-Type = Wireless-802.11
NAS-Port = 1447
State = 0x7d518a9ac9b26261623359f86cb72f38
NAS-IP-Address = 10.0.0.189
NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 2
modcall[authorize]: module "preprocess" returns ok for request 2
modcall[authorize]: module "chap" returns noop for request 2
modcall[authorize]: module "mschap" returns noop for request 2
rlm_realm: No '@' in User-Name = "j.landree", looking up realm NULL
rlm_realm: No such realm "NULL"
modcall[authorize]: module "suffix" returns noop for request 2
rlm_eap: EAP packet type request id 4 length 16
rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
modcall[authorize]: module "eap" returns updated for request 2
users: Matched entry j.landree at line 115
modcall[authorize]: module "files" returns ok for request 2
rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 2
rlm_eap: Request found, released from the list
rlm_eap: EAP/leap
rlm_eap: processing type leap
rlm_eap_leap: Stage 6
rlm_eap: Freeing handler
modcall[authenticate]: module "eap" returns handled for request 2
modcall: group authenticate returns handled for request 2
Sending Access-Accept of id 173 to 10.0.0.189:1645

```

```

Tunnel-Type:0 = VLAN
Tunnel-Medium-Type:0 = IEEE-802
Tunnel-Private-Group-Id:0 = "100"
Cisco-AVPair += "leap:session-
key=\205i\200r\354\200M\035\272\230\341\306\245\210\004o\315TZ\342"\270\236\\\30
3L\205\303W\356\250\233\305\355"
EAP-Message =
0x02050028110100189b847f655451abef106021ab919ea5762035af8359c5fa216a2e6c616e6472
65
Message-Authenticator = 0x00000000000000000000000000000000
User-Name = "j.landre"
Finished request 2
Going to the next request
Waking up in 6 seconds...
--- Walking the entire request list ---
Cleaning up request 0 ID 171 with timestamp 428a0eaf
Cleaning up request 1 ID 172 with timestamp 428a0eaf
Cleaning up request 2 ID 173 with timestamp 428a0eaf
Nothing to do. Sleeping until we see a request.

```

En a), on précise dans la partie autorisation (*authorize*) qu'on souhaite utiliser un EAP et un fichier texte (*files*) et dans la partie authentification (*authenticate*) qu'on authentifie les utilisateurs par EAP.

En b), on définit le type d'EAP par défaut en LEAP (*leap*) et on valide le module LEAP inclus dans *freeradius (leap{ })*.

En c), un utilisateur (j.landre) s'est connecté sur notre point d'accès et est basculé automatiquement sur le WVLAN 100.

EAP-LEAP est assez simple à mettre en oeuvre: pas de certificats à gérer, il permet de basculer l'utilisateur dans un WVLAN automatiquement (grâce aux attributs vus au paragraphe 5.2.3), il est assez robuste, il est supporté dans quelques appareils Wi-Fi du commerce (imprimantes, vidéo-projecteurs) mais dispose de quelques failles au niveau de la sécurité. Il est donc à employer avec précaution.

5.2.7. EAP-TLS

L'EAP TLS est basé sur des certificats utilisés pour crypter les échanges entre le client et le serveur *freeradius*. Ces certificats sont des certificats signés par une autorité de certification prouvant leur validité. On peut utiliser des certificats générés par des autorités de confiance, mais on peut aussi choisir de générer soit même ses certificats.

La génération des certificats est abordée dans la suite de cette partie. Elle n'est pas simple mais si on suit l'exemple, cela devrait bien se passer.

a) création du répertoire scripts.

```

[root@ordi ~]# cd /usr/local/freeradius-1.0.2/etc/raddb
[root@ordi raddb]# mkdir scripts
[root@ordi raddb]# ll
total 344
-rw-r--r-- 1 root root 422 mai 2 08:53 acct_users
-rw-r--r-- 1 root root 3454 mai 2 08:53 attrs
drwxr-xr-x 3 root root 4096 mai 2 08:53 certs
-rw-r----- 1 root root 189 mai 2 08:53 clients

```



```

-rw-r----- 1 root root 2940 mai 2 21:47 clients.conf
-rw-r--r-- 1 root root 952 mai 2 08:53 dictionary
-rw-r--r-- 1 root root 9080 mai 2 08:53 eap.conf
-rw-r--r-- 1 root root 8266 mai 2 08:53 experimental.conf
-rw-r--r-- 1 root root 2396 mai 2 08:53 hints
-rw-r--r-- 1 root root 1604 mai 2 08:53 huntgroups
-rw-r--r-- 1 root root 2333 mai 2 08:53 ldap.attrmap
-rw-r--r-- 1 root root 9330 mai 2 08:53 mssql.conf
-rw-r--r-- 1 root root 1020 mai 2 08:53 naslist
-rw-r----- 1 root root 856 mai 2 08:53 naspaswd
-rw-r--r-- 1 root root 12267 mai 2 08:53 oraclesql.conf
-rw-r--r-- 1 root root 14156 mai 2 08:53 postgresql.conf
-rw-r--r-- 1 root root 531 mai 2 08:53 preproxy_users
-rw-r--r-- 1 root root 8862 mai 2 08:53 proxy.conf
-rw-r--r-- 1 root root 57887 mai 2 08:53 radiusd.conf
-rw-r--r-- 1 root root 187 mai 2 08:53 realms
drwxr-xr-x 2 root root 4096 mai 10 17:20 scripts
-rw-r--r-- 1 root root 1405 mai 2 08:53 snmp.conf
-rw-r--r-- 1 root root 13892 mai 2 08:53 sql.conf
-rw-r--r-- 1 root root 7028 mai 2 09:16 users
-rw-r--r-- 1 root root 7267 mai 2 08:53 x99.conf
-rw-r--r-- 1 root root 4165 mai 2 08:53 x99passwd.sample
[root@ordi raddb]#

```

b) Remplissage du répertoire scripts.

```

[root@ordi raddb]# cd scripts
[root@ordi scripts]# cp /tmp/freeradius-1.0.2/scripts/certs.sh .
[root@ordi scripts]# cp /tmp/freeradius-1.0.2/scripts/CA.certs .
[root@ordi scripts]# cp /tmp/freeradius-1.0.2/scripts/CA.certs .
[root@ordi scripts]# cp /tmp/freeradius-1.0.2/scripts/xpextensions .
[root@ordi scripts]# ll
total 28
-rwxr-xr-x 1 root root 5312 mai 10 17:24 CA.certs
-rwxr-xr-x 1 root root 1085 mai 10 17:24 certs.sh
-rw-r--r-- 1 root root 148 mai 10 17:24 xpextensions
[root@ordi scripts]#

```

c) Modification de « certs.sh ».

```

[root@ordi scripts]# vi certs.sh
[...]
[ "$SSL" = "" ] && SSL=/usr/local/openssl-0.9.7g
export SSL
[...]
#
# Generate DH stuff...
#
${SSL}/bin/openssl gendh > dh
[...]
root@ordi scripts]#

```

d) Modification de « CA.certs ».

```

[root@ordi scripts]# vi CA.certs

```

```
[...]
[ "$SSL" = "" ] && SSL=/usr/local/openssl-0.9.7g
export SSL

#
# Edit the following variables for your organization.
#
COUNTRY="FR"
PROVINCE="bourgogne"
CITY="le_creusot"
ORGANIZATION="service_informatique"
ORG_UNIT=`hostname`
PASSWORD="mdpiutcreusot"

COMMON_NAME_CLIENT="certificat_client_iut_le_creusot"
EMAIL_CLIENT="servinfo@iutlecreusot.u-bourgogne.fr"
PASSWORD_CLIENT=$PASSWORD

COMMON_NAME_SERVER="certificat_server_iut_le_creusot"
EMAIL_SERVER="servinfo@iutlecreusot.u-bourgogne.fr"
PASSWORD_SERVER=$PASSWORD

COMMON_NAME_ROOT="certificat_racine_iut_le_creusot"
EMAIL_ROOT="servinfo@iutlecreusot.u-bourgogne.fr"
PASSWORD_ROOT=$PASSWORD

#
# lifetime, in days, of the certs
#
LIFETIME=1825 # 5 ans

[...]
rm -rf demoCA roo* cert* *.pem *.der
mkdir demoCA
echo "01" > demoCA/serial
[...]
echo "newreq.pem" | /usr/local/openssl-0.9.7g/ssl/misc/CA.pl -newca || exit 2
[...]
root@ordi scripts]#
```

e) Modification de « CA.pl ».

```
[root@ordi scripts]# vi /usr/local/openssl-0.9.7g/ssl/misc/CA.pl
[...]
        . "-out ${CATOP}/serial");
        #. "-next_serial -out ${CATOP}/serial");
[...]
[root@ordi scripts]#
```

f) Génération des certificats.

```
[root@ordi scripts]# ./certs.sh
Generating DH parameters, 512 bit long safe prime, generator 2
This is going to take a long time
.....+.....+.....+.....
.....+...+.....+.....+.....
```



```
-rw-r--r-- 1 root root 2157 mai 10 17:44 root.p12
-rw-r--r-- 1 root root 3132 mai 10 17:44 root.pem
[root@ordi certs]#
```

i) Modification de « /usr/local/freeradius-1.0.2/etc/raddb/eap.conf ».

```
[root@ordi certs]# cd ..
[root@ordi raddb]# vi eap.conf
[...]
default_eap_type = tls
[...]
tls {

    private_key_password = mdpiutcreusot
    private_key_file = ${raddbdir}/certs/cert-srv.pem

    # If Private key & Certificate are located in
    # the same file, then private_key_file &
    # certificate_file must contain the same file
    # name.
    certificate_file = ${raddbdir}/certs/cert-srv.pem

    # Trusted Root CA list
    CA_file = ${raddbdir}/certs/demoCA/cacert.pem

    dh_file = ${raddbdir}/certs/dh
    random_file = ${raddbdir}/certs/random

    #
    # This can never exceed the size of a RADIUS
    # packet (4096 bytes), and is preferably half
    # that, to accomodate other attributes in
    # RADIUS packet. On most APs the MAX packet
    # length is configured between 1500 - 1600
    # In these cases, fragment size should be
    # 1024 or less.
    #
    fragment_size = 1024

    # include_length is a flag which is
    # by default set to yes If set to
    # yes, Total Length of the message is
    # included in EVERY packet we send.
    # If set to no, Total Length of the
    # message is included ONLY in the
    # First packet of a fragment series.
    #
    include_length = yes

    # Check the Certificate Revocation List
    #
    # 1) Copy CA certificates and CRLs to same directory.
    # 2) Execute 'c_rehash <CA certs&CRLs Directory>'.
    # 'c_rehash' is OpenSSL's command.
    # 3) Add 'CA_path=<CA certs&CRLs directory>'
    # to radiusd.conf's tls section.
    # 4) uncomment the line below.
    # 5) Restart radiusd
    check_crl = no
```

```

# If check_cert_cn is set, the value will
# be xlat'ed and checked against the CN
# in the client certificate. If the values
# do not match, the certificate verification
# will fail rejecting the user.
#
#check_cert_cn = %{User-Name}
}
[...]
```

[root@ordi raddb]#

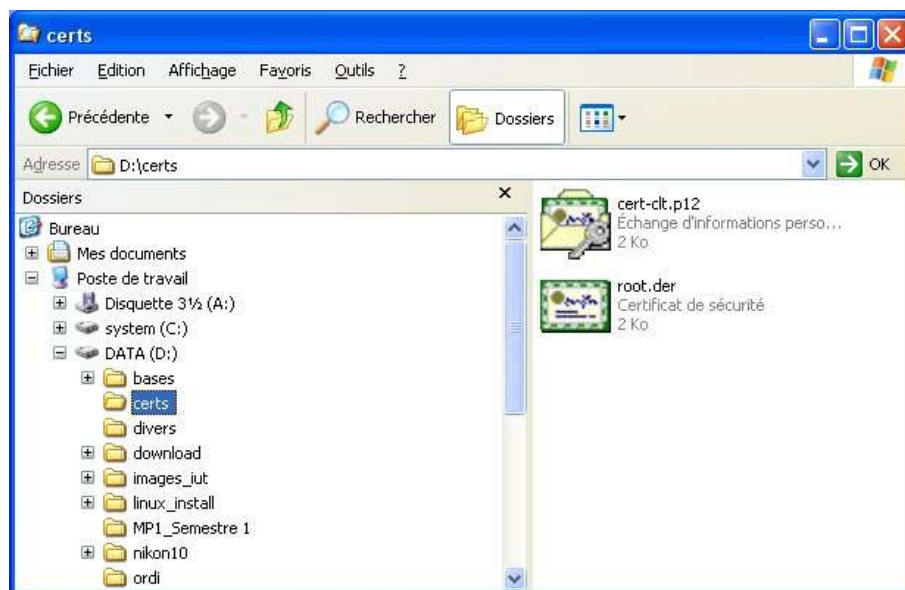
j) Redémarrage du service « radiusd ».

```

[root@ordi raddb]# /usr/local/freeradius-1.0.2/sbin/rc.radiusd restart
Arrêt du serveur RADIUS :           [ OK ]
Démarrage du serveur RADIUS :       [ OK ]
[root@ordi raddb]#
```

k) Copie de « root.der » sur le client windows XP dans les certificats racine.

Il est nécessaire de copier les fichiers « root.der » et « cert-ctl.p12 » sur le client par ftp, sur une disquette ou tout autre moyen.

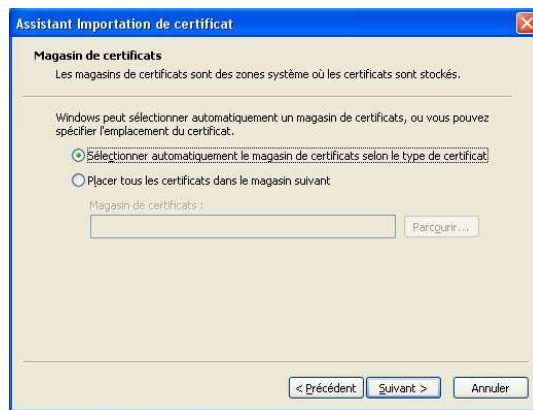






1) Copie de « *clt-client.p12* » sur le client.





m) Exemple de trace (log) freeradius pour EAP-TLS.

```
[root@ordi raddb]# /usr/local/freeradius-1.0.2/sbin/radiusd -X
Starting - reading configuration files ...
reread_config: reading radiusd.conf
Config: including file: /etc/raddb/proxy.conf
Config: including file: /etc/raddb/clients.conf
Config: including file: /etc/raddb/snmp.conf
Config: including file: /etc/raddb/eap.conf
Config: including file: /etc/raddb/sql.conf
main: prefix = "/usr"
main: localstatedir = "/usr/var"
main: logdir = "/usr/var/log/radius"
main: libdir = "/usr/lib"
main: radacctdir = "/usr/var/log/radius/radacct"
main: hostname_lookups = no
main: max_request_time = 30
```

```

main: cleanup_delay = 5
main: max_requests = 1024
main: delete_blocked_requests = 0
main: port = 0
main: allow_core_dumps = no
main: log_stripped_names = no
main: log_file = "/usr/var/log/radius/radius.log"
main: log_auth = no
main: log_auth_badpass = no
main: log_auth_goodpass = no
main: pidfile = "/usr/var/run/radiusd/radiusd.pid"
main: user = "(null)"
main: group = "(null)"
main: usercollide = no
main: lower_user = "no"
main: lower_pass = "no"
main: nospace_user = "no"
main: nospace_pass = "no"
main: checkrad = "/usr/sbin/checkrad"
main: proxy_requests = yes
proxy: retry_delay = 5
proxy: retry_count = 3
proxy: synchronous = no
proxy: default_fallback = yes
proxy: dead_time = 120
proxy: post_proxy_authorize = yes
proxy: wake_all_if_all_dead = no
security: max_attributes = 200
security: reject_delay = 1
security: status_server = no
main: debug_level = 0
read_config_files: reading dictionary
read_config_files: reading naslist
Using deprecated naslist file. Support for this will go away soon.
read_config_files: reading clients
read_config_files: reading realms
radiusd: entering modules setup
Module: Library search path is /usr/lib
Module: Loaded exec
exec: wait = yes
exec: program = "(null)"
exec: input_pairs = "request"
exec: output_pairs = "(null)"
exec: packet_type = "(null)"
rlm_exec: Wait=yes but no output defined. Did you mean output=none?
Module: Instantiated exec (exec)
Module: Loaded expr
Module: Instantiated expr (expr)
Module: Loaded PAP
pap: encryption_scheme = "crypt"
Module: Instantiated pap (pap)
Module: Loaded CHAP
Module: Instantiated chap (chap)
Module: Loaded MS-CHAP
mschap: use_mppe = yes
mschap: require_encryption = no
mschap: require_strong = no
mschap: with_ntdomain_hack = no
mschap: passwd = "(null)"
mschap: authtype = "MS-CHAP"
mschap: ntlm_auth = "(null)"

```

```

Module: Instantiated mschap (mschap)
Module: Loaded System
  unix: cache = no
  unix: passwd = "(null)"
  unix: shadow = "(null)"
  unix: group = "(null)"
  unix: radwtmp = "/usr/var/log/radius/radwtmp"
  unix: usegroup = no
  unix: cache_reload = 600
Module: Instantiated unix (unix)
Module: Loaded eap
  eap: default_eap_type = "tls"
  eap: timer_expire = 60
  eap: ignore_unknown_eap_types = no
  eap: cisco_accounting_username_bug = no
rlm_eap: Loaded and initialized type md5
rlm_eap: Loaded and initialized type leap
  gtc: challenge = "Password: "
  gtc: auth_type = "PAP"
rlm_eap: Loaded and initialized type gtc
  tls: rsa_key_exchange = no
  tls: dh_key_exchange = yes
  tls: rsa_key_length = 512
  tls: dh_key_length = 512
  tls: verify_depth = 0
  tls: CA_path = "(null)"
  tls: pem_file_type = yes
  tls: private_key_file = "/etc/raddb/certs/cert-srv.pem"
  tls: certificate_file = "/etc/raddb/certs/cert-srv.pem"
  tls: CA_file = "/etc/raddb/certs/demoCA/cacert.pem"
  tls: private_key_password = "mdpiutcreusot"
  tls: dh_file = "/etc/raddb/certs/dh"
  tls: random_file = "/etc/raddb/certs/random"
  tls: fragment_size = 1024
  tls: include_length = yes
  tls: check_crl = no
  tls: check_cert_cn = "(null)"
rlm_eap: Loaded and initialized type tls
  ttls: default_eap_type = "md5"
  ttls: copy_request_to_tunnel = yes
  ttls: use_tunneled_reply = yes
rlm_eap: Loaded and initialized type ttls
  peap: default_eap_type = "mschapv2"
  peap: copy_request_to_tunnel = no
  peap: use_tunneled_reply = no
  peap: proxy_tunneled_request_as_eap = yes
rlm_eap: Loaded and initialized type peap
  mschapv2: with_ntdomain_hack = no
rlm_eap: Loaded and initialized type mschapv2
Module: Instantiated eap (eap)
Module: Loaded preprocess
  preprocess: huntgroups = "/etc/raddb/huntgroups"
  preprocess: hints = "/etc/raddb/hints"
  preprocess: with_ascend_hack = no
  preprocess: ascend_channels_per_line = 23
  preprocess: with_ntdomain_hack = no
  preprocess: with_specialix_jetstream_hack = no
  preprocess: with_cisco_vsa_hack = no
Module: Instantiated preprocess (preprocess)
Module: Loaded realm
  realm: format = "suffix"

```

```

realm: delimiter = "@"
realm: ignore_default = no
realm: ignore_null = no
Module: Instantiated realm (suffix)
Module: Loaded files
  files: usersfile = "/etc/raddb/users"
  files: acctusersfile = "/etc/raddb/acct_users"
  files: preproxy_usersfile = "/etc/raddb/preproxy_users"
  files: compat = "no"
Module: Instantiated files (files)
Module: Loaded Acct-Unique-Session-Id
  acct_unique: key = "User-Name, Acct-Session-Id, NAS-IP-Address, Client-IP-Address, NAS-Port"
Module: Instantiated acct_unique (acct_unique)
Module: Loaded detail
  detail: detailfile = "/usr/var/log/radius/radacct/%{Client-IP-Address}/detail-%Y%m%d"
  detail: detailperm = 384
  detail: dirperm = 493
  detail: locking = no
Module: Instantiated detail (detail)
Module: Loaded radutmp
  radutmp: filename = "/usr/var/log/radius/radutmp"
  radutmp: username = "%{User-Name}"
  radutmp: case_sensitive = yes
  radutmp: check_with_nas = yes
  radutmp: perm = 384
  radutmp: callerid = yes
Module: Instantiated radutmp (radutmp)
Listening on authentication *:1812
Listening on accounting *:1813
Listening on proxy *:1814
Ready to process requests.
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=33, length=189
  User-Name = "certificat_server_iut_le_creusot"
  Framed-MTU = 1400
  Called-Station-Id = "0011.bbba.bb96"
  Calling-Station-Id = "0040.965d.e6ba"
  Service-Type = Login-User
  Message-Authenticator = 0x56339a6e2829ac1a0bbf100ec5313fa1
  EAP-Message =
0x0201002501636572746966696361745f7365727665725f6975745f6c655f63726575736f74
  NAS-Port-Type = Wireless-802.11
  NAS-Port = 1355
  NAS-IP-Address = 10.0.0.189
  NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 0
  modcall[authorize]: module "preprocess" returns ok for request 0
  modcall[authorize]: module "chap" returns noop for request 0
  modcall[authorize]: module "mschap" returns noop for request 0
  rlm_realm: No '@' in User-Name = "certificat_server_iut_le_creusot", looking
up realm NULL
  rlm_realm: No such realm "NULL"
  modcall[authorize]: module "suffix" returns noop for request 0
  rlm_eap: EAP packet type response id 1 length 37
  rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
  modcall[authorize]: module "eap" returns updated for request 0
  users: Matched entry DEFAULT at line 409
  modcall[authorize]: module "files" returns ok for request 0
modcall: group authorize returns updated for request 0

```

```

rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 0
rlm_eap: EAP Identity
rlm_eap: processing type tls
rlm_eap_tls: Requiring client certificate
rlm_eap_tls: Initiate
rlm_eap_tls: Start returned 1
modcall[authenticate]: module "eap" returns handled for request 0
modcall: group authenticate returns handled for request 0
Sending Access-Challenge of id 33 to 10.0.0.189:21657
EAP-Message = 0x0102000060d20
Message-Authenticator = 0x00000000000000000000000000000000
State = 0xcf2d29076a2d9096019b13402a2d7cf6
Finished request 0
Going to the next request
--- Walking the entire request list ---
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=34, length=276
User-Name = "certificat_server_iut_le_creusot"
Framed-MTU = 1400
Called-Station-Id = "0011.bbba.bb96"
Calling-Station-Id = "0040.965d.e6ba"
Service-Type = Login-User
Message-Authenticator = 0x60d797b3df91da1f7032e97f284210f8
EAP-Message =
0x0202006a0d8000000060160301005b010000570301425f94f32938da1e8bda7bf76d061c36f6bc
b233d82853d77f62d0dd39d0c51600003000390038003500160013000a00330032002f0066000500
040065006400630062006000150012000900140011000800030100
NAS-Port-Type = Wireless-802.11
NAS-Port = 1355
State = 0xcf2d29076a2d9096019b13402a2d7cf6
NAS-IP-Address = 10.0.0.189
NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 1
modcall[authorize]: module "preprocess" returns ok for request 1
modcall[authorize]: module "chap" returns noop for request 1
modcall[authorize]: module "mschap" returns noop for request 1
rlm_realm: No '@' in User-Name = "certificat_server_iut_le_creusot", looking
up realm NULL
rlm_realm: No such realm "NULL"
modcall[authorize]: module "suffix" returns noop for request 1
rlm_eap: EAP packet type response id 2 length 106
rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
modcall[authorize]: module "eap" returns updated for request 1
users: Matched entry DEFAULT at line 409
modcall[authorize]: module "files" returns ok for request 1
modcall: group authorize returns updated for request 1
rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 1
rlm_eap: Request found, released from the list
rlm_eap: EAP/tls
rlm_eap: processing type tls
rlm_eap_tls: Authenticate
rlm_eap_tls: processing TLS
rlm_eap_tls: Length Included
eaptls_verify returned 11

```

```

    (other): before/accept initialization
    TLS_accept: before/accept initialization
    rlm_eap_tls: <<< TLS 1.0 Handshake [length 005b], ClientHello
    TLS_accept: SSLv3 read client hello A
    rlm_eap_tls: >>> TLS 1.0 Handshake [length 004a], ServerHello
    TLS_accept: SSLv3 write server hello A
    rlm_eap_tls: >>> TLS 1.0 Handshake [length 07de], Certificate
    TLS_accept: SSLv3 write certificate A
    rlm_eap_tls: >>> TLS 1.0 Handshake [length 010d], ServerKeyExchange
    TLS_accept: SSLv3 write key exchange A
    rlm_eap_tls: >>> TLS 1.0 Handshake [length 00f0], CertificateRequest
    TLS_accept: SSLv3 write certificate request A
    TLS_accept: SSLv3 flush data
    TLS_accept:error in SSLv3 read client certificate A
In SSL Handshake Phase
In SSL Accept mode
    eaptls_process returned 13
    modcall[authenticate]: module "eap" returns handled for request 1
modcall: group authenticate returns handled for request 1
Sending Access-Challenge of id 34 to 10.0.0.189:21657
    EAP-Message
0x0103040a0dc000000a39160301004a0200000460301425f93707c475059666924bb65a011547d40
c59f42b0dec0971f174f1eb2e3fe204e90da86f1fc7f5e3ed7cf6187ecdc1c02f7da60f13e1551cb
9e6dc35e1a83ff00390016030107de0b0007da0007d700034b30820347308202b0a0030201020201
02300d06092a864886f70d01010405003081dc310c45678643dc3453466446523112301006035504
081309626f7572676f676e65311330110603550407140a6c655f63726575736f74311d301b060355
040a1414736572766963655f696e666f726d61746971756531253023060355040b131c7261646975
7363726575736f742e752d626f75
    EAP-Message
0x72676f676e652e66723129302706035504031420636572746966696361745f636c69656e745f69
75745f6c655f63726575736f743133303106092a864886f70d010901162473657276696e666f4069
75746c6563726575736f742e752d626f7572676f676e652e6672301e170d30353034313530393436
30385a170d3036303431353039343630385a3081dc310c45678643dc345346644652311230100603
5504081309626f7572676f676e65311330110603550407140a6c655f63726575736f74311d301b06
0355040a1414736572766963655f696e666f726d61746971756531253023060355040b131c726164
69757363726575736f742e752d62
    EAP-Message
0x6f7572676f676e652e66723129302706035504031420636572746966696361745f726163696e65
5f6975745f6c655f63726575736f743133303106092a864886f70d010901162473657276696e666f
406975746c6563726575736f742e752d626f7572676f676e652e667230819f300d06092a864886f7
0d010101050003818d0030818902818100c867154256f402ab95dc557cefc74fe0e82923b9b106c1
0af632d2bdc94da40fba69eb556204bbc16107a30b3881e99210d8b65dcef6eb9489b82e22cc2d89
82a6b259b2b81681504e22021da05c9367b73c8b24c2a77f2060b175f7cfa9a1b859b6e920344d69
dfe4dfcf6088b1da9a73b84208a2
    EAP-Message
0x8c547fe1bbb45109639d510203010001a317301530130603551d25040c300a06082b0601050507
0301300d06092a864886f70d01010405000381810092e230835ef0fca505bfc2e81c22e85aee6827
9018555826f53903e28c9b3dd8513f8b17d652d3679d551f58c94a7e0978936ecd0abf9a68e8df70
abfe8f1fffd6ae96f46534af6738668fd30df2729d5c4f9aff9a444f45bec4b4a3dbb727299cb5c
532662341281c4144821dad195820617035225ee06bebed5fadba35ce00048630820482308203eb
a00302010202090099840e9f0189a17d300d06092a864886f70d01010405003081dc310c45678643
dc34534664465231123010060355
    EAP-Message = 0x04081309626f7572676f676e65311330110603550407
    Message-Authenticator = 0x00000000000000000000000000000000
    State = 0x81a4d4240d6b5fa1850180d7464567e3
Finished request 1
Going to the next request
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=35, length=176
    User-Name = "certificat_server_iut_le_creusot"
    Framed-MTU = 1400

```

```

Called-Station-Id = "0011.bbbaa.bb96"
Calling-Station-Id = "0040.965d.e6ba"
Service-Type = Login-User
Message-Authenticator = 0x667f46927ac4bb2e2755a947227c4d35
EAP-Message = 0x020300060d00
NAS-Port-Type = Wireless-802.11
NAS-Port = 1355
State = 0x81a4d4240d6b5fa1850180d7464567e3
NAS-IP-Address = 10.0.0.189
NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 2
modcall[authorize]: module "preprocess" returns ok for request 2
modcall[authorize]: module "chap" returns noop for request 2
modcall[authorize]: module "mschap" returns noop for request 2
    rlm_realm: No '@' in User-Name = "certificat_server_iut_le_creusot", looking
up realm NULL
    rlm_realm: No such realm "NULL"
modcall[authorize]: module "suffix" returns noop for request 2
rlm_eap: EAP packet type response id 3 length 6
rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
modcall[authorize]: module "eap" returns updated for request 2
    users: Matched entry DEFAULT at line 409
modcall[authorize]: module "files" returns ok for request 2
modcall: group authorize returns updated for request 2
    rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 2
    rlm_eap: Request found, released from the list
    rlm_eap: EAP/tls
    rlm_eap: processing type tls
    rlm_eap_tls: Authenticate
    rlm_eap_tls: processing TLS
rlm_eap_tls: Received EAP-TLS ACK message
    rlm_eap_tls: ack handshake fragment handler
    eaptls_verify returned 1
    eaptls_process returned 13
modcall[authenticate]: module "eap" returns handled for request 2
modcall: group authenticate returns handled for request 2
Sending Access-Challenge of id 35 to 10.0.0.189:21657
    EAP-Message
0x0104040a0dc000000a39140a6c655f63726575736f74311d301b060355040a1414736572766963
655f696e666f726d61746971756531253023060355040b131c72616469757363726575736f742e75
2d626f7572676f676e652e66723129302706035504031420636572746966696361745f636c69656e
745f6975745f6c655f63726575736f743133303106092a864886f70d010901162473657276696e66
6f406975746c6563726575736f742e752d626f7572676f676e652e6672301e170d30353034313530
39343630375a170d3130303431343039343630375a3081dc310c45678643dc345346644652311230
1006035504081309626f7572676f
    EAP-Message
0x676e65311330110603550407140a6c655f63726575736f74311d301b060355040a141473657276
6963655f696e666f726d61746971756531253023060355040b131c72616469757363726575736f74
2e752d626f7572676f676e652e66723129302706035504031420636572746966696361745f636c69
656e745f6975745f6c655f63726575736f743133303106092a864886f70d01090116247365727669
6e666f406975746c6563726575736f742e752d626f7572676f676e652e667230819f300d06092a86
4886f70d0101050003818d0030818902818100a8eb16e223d41d8d33afeadde8e38edb9780fa0f
9ff12174fab90b981e9a1760a641
    EAP-Message
0x8f9de7e869156ee5f0fedfe1bb33512d21cacdd613a29302b6fe2bd2a889b7279bcf325b9c6f92
a39e8bb57fd922643ac08e7db560a90786721219537ec765d01bb70235b302ccc09dbd3db76ddf5a
9322491c94ea6e63851a4208c180090203010001a382014830820144301d0603551d0e041604145d

```

db0a9044d02c70c097f6bf9cbe83c49d4a6ead308201130603551d230482010a3082010680145ddb
0a9044d02c70c097f6bf9cbe83c49d4a6eada181e2a481df3081dc310c45678643dc345346644652
3112301006035504081309626f7572676f676e65311330110603550407140a6c655f63726575736f
74311d301b060355040a14147365

EAP-Message

=

0x72766963655f696e666f726d61746971756531253023060355040b131c72616469757363726575
736f742e752d626f7572676f676e652e66723129302706035504031420636572746966696361745f
636c69656e745f6975745f6c655f63726575736f743133303106092a864886f70d01090116247365
7276696e666f406975746c6563726575736f742e752d626f7572676f676e652e667282090099840e
9f0189a17d300c0603551d13040530030101ff300d06092a864886f70d0101040500038181000691
46dd03cf8dd1355041cd843e3acf951b6919d79f52f0cf53926b97f36dd864882a16f0bdee5a66bd
6bb22025992d707c4a2b0bb4d26e

EAP-Message = 0x3eb53e78a4f0e1f588573bcdba484fa2b7894f632146

Message-Authenticator = 0x00000000000000000000000000000000

State = 0x5d78d37f4556721b58c0a6a09a35c93f

Finished request 2

Going to the next request

Waking up in 6 seconds...

rad_recv: Access-Request packet from host 10.0.0.189:21657, id=36, length=176

User-Name = "certificat_server_iut_le_creusot"

Framed-MTU = 1400

Called-Station-Id = "0011.bbba.bb96"

Calling-Station-Id = "0040.965d.e6ba"

Service-Type = Login-User

Message-Authenticator = 0xc41d369b6033e4903346c036ac417096

EAP-Message = 0x020400060d00

NAS-Port-Type = Wireless-802.11

NAS-Port = 1355

State = 0x5d78d37f4556721b58c0a6a09a35c93f

NAS-IP-Address = 10.0.0.189

NAS-Identifier = "AP1100_TEST"

Processing the authorize section of radiusd.conf

modcall: entering group authorize for request 3

modcall[authorize]: module "preprocess" returns ok for request 3

modcall[authorize]: module "chap" returns noop for request 3

modcall[authorize]: module "mschap" returns noop for request 3

rlm_realm: No '@' in User-Name = "certificat_server_iut_le_creusot", looking
up realm NULL

rlm_realm: No such realm "NULL"

modcall[authorize]: module "suffix" returns noop for request 3

rlm_eap: EAP packet type response id 4 length 6

rlm_eap: No EAP Start, assuming it's an on-going EAP conversation

modcall[authorize]: module "eap" returns updated for request 3

users: Matched entry DEFAULT at line 409

modcall[authorize]: module "files" returns ok for request 3

modcall: group authorize returns updated for request 3

rad_check_password: Found Auth-Type EAP

auth: type "EAP"

Processing the authenticate section of radiusd.conf

modcall: entering group authenticate for request 3

rlm_eap: Request found, released from the list

rlm_eap: EAP/tls

rlm_eap: processing type tls

rlm_eap_tls: Authenticate

rlm_eap_tls: processing TLS

rlm_eap_tls: Received EAP-TLS ACK message

rlm_eap_tls: ack handshake fragment handler

eaptls_verify returned 1

eaptls_process returned 13

modcall[authenticate]: module "eap" returns handled for request 3

modcall: group authenticate returns handled for request 3

Sending Access-Challenge of id 36 to 10.0.0.189:21657

EAP-Message

0x010502430d8000000a399079f10914df6b1ff53e09acf78ab059b2d30c50e8cb6a51fad047cb1855a88e8cadbd48f8d34ef6082425338d9f08ec2371160301010d0c0001090040e4e1abf8258723f61d7fbb1dc1f5da079e7f6977a7b25436daea70a7fda0f8de4f86e7fd6372cb0882a7ff9d1f703920c4152ef1e03db020083a41c66ffff243000102004033cd1283fb8940790b8c13996ebe4489a3872c8ff91244bf7667168a5686d77fa49abb0f1fddf7c80a8f8d4ffd3b4187d0ca7612e8194110d6cc2414727f3d8e0080164dac8e983631905567260c6c757548eae5267817a4ca4489bdcbaa32c77cd74cb173bcfddefa4f48951e64b0c7d6

EAP-Message

0x532b1a374d2567643b67a51bae508de4289816484998046dc22b06edd3bfb137f00f8810a808ee572a2f7efeb709092137ceda3524314974b1e0f598dfc342f2c611a2b12e0723781ef97e22998666623a16030100f00d0000e8040304010200e100df3081dc310c45678643dc3453466446523112301006035504081309626f7572676f676e65311330110603550407140a6c655f63726575736f74311d301b060355040a1414736572766963655f696e666f726d61746971756531253023060355040b131c72616469757363726575736f742e752d626f7572676f676e652e66723129302706035504031420636572746966696361745f636c69656e

EAP-Message

0x745f6975745f6c655f63726575736f743133303106092a864886f70d010901162473657276696e666f406975746c6563726575736f742e752d626f7572676f676e652e66720e000000

Message-Authenticator = 0x00000000000000000000000000000000

State = 0x7cedcc3b85b6689b8832d4dc3a64cf93

Finished request 3

Going to the next request

Waking up in 6 seconds...

rad_recv: Access-Request packet from host 10.0.0.189:21657, id=37, length=1676

User-Name = "certificat_server_iut_le_creusot"

Framed-MTU = 1400

Called-Station-Id = "0011.bbba.bb96"

Calling-Station-Id = "0040.965d.e6ba"

Service-Type = Login-User

Message-Authenticator = 0x0864e140b2c307160f9720a53e7ad6ee

EAP-Message

0x020505d80dc0000008f416030107de0b0007da0007d700034b30820347308202b0a003020102020101300d06092a864886f70d01010405003081dc310c45678643dc3453466446523112301006035504081309626f7572676f676e65311330110603550407140a6c655f63726575736f74311d301b060355040a1414736572766963655f696e666f726d61746971756531253023060355040b131c72616469757363726575736f742e752d626f7572676f676e652e66723129302706035504031420636572746966696361745f636c69656e745f6975745f6c655f63726575736f743133303106092a864886f70d010901162473657276696e666f4069

EAP-Message

0x75746c6563726575736f742e752d626f7572676f676e652e6672301e170d3035303431353039343630375a170d3036303431353039343630375a3081dc310c45678643dc3453466446523112301006035504081309626f7572676f676e65311330110603550407140a6c655f63726575736f74311d301b060355040a1414736572766963655f696e666f726d61746971756531253023060355040b131c72616469757363726575736f742e752d626f7572676f676e652e66723129302706035504031420636572746966696361745f7365727665725f6975745f6c655f63726575736f743133303106092a864886f70d010901162473657276696e666f

EAP-Message

0x406975746c6563726575736f742e752d626f7572676f676e652e667230819f300d06092a864886f70d0101050003818d0030818902818100d0853227647f576120022e80a454dfb42988ca45b5a3a8325f4187130320b139b3579d5b65681a0fc1b2285966b1e1607408c39a4efcafa80b0d2e24224cd905ccba3d89efdb5bd5b406db17ec30b146ee9b089aed2e3983fe17881ebce610b4f944af86e95d2dd1ce9cf79603e25c7068bf3b3fcf7954fe382532e64a3cd310203010001a317301530130603551d25040c300a06082b06010505070302300d06092a864886f70d01010405000381810080c6eb9f3b2054e79ad2e3ba983f3d7995b2b1

EAP-Message

0xb2cd51e14285b550b18a6902f875ea0a2bc304768eb88fa712d91672b1d7e1053a118f86b93f00a560aeb47908e7635d109b797bc20908bf5f77c56f16e7e4c07c017b17d4cc3d414baf8a5134d0d605b95fbb7e112b1c6b5578c45d5c45b3da24fa5145d4672c1f3af03d53d500048630820482308203eba00302010202090099840e9f0189a17d300d06092a864886f70d01010405003081dc310c456786

43dc3453466446523112301006035504081309626f7572676f676e65311330110603550407140a6c
655f63726575736f74311d301b060355040a1414736572766963655f696e666f726d617469717565
31253023060355040b131c726164

EAP-Message

=

0x69757363726575736f742e752d626f7572676f676e652e66723129302706035504031420636572
746966696361745f636c69656e745f6975745f6c655f63726575736f743133303106092a864886f7
0d010901162473657276696e666f406975746c6563726575736f742e752d626f7572676f676e652e
6672301e170d3035303431353039343630375a170d3130303431343039343630375a3081dc310c45
678643dc3453466446523112301006035504081309626f7572676f676e6531133011060355040714
0a6c655f63726575736f74311d301b060355040a1414736572766963655f696e666f726d61746971
756531253023060355040b131c72

EAP-Message

=

0x616469757363726575736f742e752d626f7572676f676e652e6672312930270603550403142063
6572746966696361745f636c69656e745f6975745f6c655f63726575736f743133303106092a8648
86f70d010901162473657276696e666f406975746c6563726575736f742e752d626f7572676f676e
652e667230819f300d06092a864886f70d010101050003818d0030818902818100a8eb16e223d41d
8d33afeadde8e38edb9780fa0f9ff12174fab90b981e9a1760a6418f9de7e869156ee5f0fedfe1bb
33512d21cacdd613a29302b6fe2bd2a889b7279bcf325b9c6f92a39e8bb57fd9

NAS-Port-Type = Wireless-802.11

NAS-Port = 1355

State = 0x7cedcc3b85b6689b8832d4dc3a64cf93

NAS-IP-Address = 10.0.0.189

NAS-Identifier = "AP1100_TEST"

Processing the authorize section of radiusd.conf

modcall: entering group authorize for request 4

modcall[authorize]: module "preprocess" returns ok for request 4

modcall[authorize]: module "chap" returns noop for request 4

modcall[authorize]: module "mschap" returns noop for request 4

rlm_realm: No '@' in User-Name = "certificat_server_iut_le_creusot", looking
up realm NULL

rlm_realm: No such realm "NULL"

modcall[authorize]: module "suffix" returns noop for request 4

rlm_eap: EAP packet type response id 5 length 253

rlm_eap: No EAP Start, assuming it's an on-going EAP conversation

modcall[authorize]: module "eap" returns updated for request 4

users: Matched entry DEFAULT at line 409

modcall[authorize]: module "files" returns ok for request 4

modcall: group authorize returns updated for request 4

rad_check_password: Found Auth-Type EAP

auth: type "EAP"

Processing the authenticate section of radiusd.conf

modcall: entering group authenticate for request 4

rlm_eap: Request found, released from the list

rlm_eap: EAP/tls

rlm_eap: processing type tls

rlm_eap_tls: Authenticate

rlm_eap_tls: processing TLS

rlm_eap_tls: Received EAP-TLS First Fragment of the message

eaptls_verify returned 9

eaptls_process returned 13

modcall[authenticate]: module "eap" returns handled for request 4

modcall: group authenticate returns handled for request 4

Sending Access-Challenge of id 37 to 10.0.0.189:21657

EAP-Message = 0x010600060d00

Message-Authenticator = 0x00000000000000000000000000000000

State = 0x7629f43fff2fc70836a7f4591fe26f71

Finished request 4

Going to the next request

Waking up in 6 seconds...

rad_recv: Access-Request packet from host 10.0.0.189:21657, id=38, length=988

User-Name = "certificat_server_iut_le_creusot"

```

Framed-MTU = 1400
Called-Station-Id = "0011.bbaa.bb96"
Calling-Station-Id = "0040.965d.e6ba"
Service-Type = Login-User
Message-Authenticator = 0xa53af79fbafd178439f5a62fedd5fdc3
EAP-Message =
0x0206032c0d0022643ac08e7db560a90786721219537ec765d01bb70235b302ccc09dbd3db76ddf
5a9322491c94ea6e63851a4208c180090203010001a382014830820144301d0603551d0e04160414
5ddb0a9044d02c70c097f6bf9cbe83c49d4a6ead308201130603551d230482010a3082010680145d
db0a9044d02c70c097f6bf9cbe83c49d4a6ead181e2a481df3081dc310c45678643dc3453466446
523112301006035504081309626f7572676f676e65311330110603550407140a6c655f6372657573
6f74311d301b060355040a1414736572766963655f696e666f726d61746971756531253023060355
040b131c72616469757363726575
EAP-Message =
0x736f742e752d626f7572676f676e652e6672312930270603550403142063657274696669636174
5f636c69656e745f6975745f6c655f63726575736f743133303106092a864886f70d010901162473
657276696e666f406975746c6563726575736f742e752d626f7572676f676e652e66728209009984
0e9f0189a17d300c0603551d13040530030101ff300d06092a864886f70d01010405000381810006
9146dd03cf8dd1355041cd843e3acf951b6919d79f52f0cf53926b97f36dd864882a16f0bdee5a66
bd6bb22025992d707c4a2b0bb4d26e3eb53e78a4f0e1f588573bcdba484fa2b7894f6321469079f1
0914df6b1ff53e09acf78ab059b2
EAP-Message =
0xd30c50e8cb6a51fad047cb1855a88e8cadbd48f8d34ef6082425338d9f08ec2371160301004610
0000420040b43cfa491536da2b2d5502f60a94868656d8b638a44f21efe9280005bb3b364153db17
a38d7afb2eeb2f3c909962ab55416428ddc2e46d9cd71f1a84c0cd5aa816030100860f0000820080
31605c57c6d432bfb760898297b255177ca143b0423afc094d4e8bbfb7086c8bcacb59e05207ff7f
06ff934f9f458a56065233b70b867e30a021e3cb3031cd62601e9cdad99af7d6ac433cfd433d74bf
86bdb221637f1de6680a528de53ac5ca7a2e2e2a59da37f2ab30f7ade4527bab31dcfa5148e76ff2
c48f16e0230bf8b0140301000101
EAP-Message =
0x1603010030ae316e319069c4af78f3b7ad4169cc91b8379e007394be6ac64f130a6ec3827a2355
7021713c998978d536f04e050733
NAS-Port-Type = Wireless-802.11
NAS-Port = 1355
State = 0x7629f43fff2fc70836a7f4591fe26f71
NAS-IP-Address = 10.0.0.189
NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 5
modcall[authorize]: module "preprocess" returns ok for request 5
modcall[authorize]: module "chap" returns noop for request 5
modcall[authorize]: module "mschap" returns noop for request 5
rlm_realm: No '@' in User-Name = "certificat_server_iut_le_creusot", looking
up realm NULL
rlm_realm: No such realm "NULL"
modcall[authorize]: module "suffix" returns noop for request 5
rlm_eap: EAP packet type response id 6 length 253
rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
modcall[authorize]: module "eap" returns updated for request 5
users: Matched entry DEFAULT at line 409
modcall[authorize]: module "files" returns ok for request 5
modcall: group authorize returns updated for request 5
rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 5
rlm_eap: Request found, released from the list
rlm_eap: EAP/tls
rlm_eap: processing type tls
rlm_eap_tls: Authenticate
rlm_eap_tls: processing TLS

```

```

eaptls_verify returned 7
rlm_eap_tls: Done initial handshake
rlm_eap_tls: <<< TLS 1.0 Handshake [length 07de], Certificate
chain-depth=1,
error=0
--> User-Name = certificat_server_iut_le_creusot
--> BUF-Name = certificat_client_iut_le_creusot
-->          subject = /
C=FR/ST=bourgogne/L=le_creusot/O=service_informatique/OU=radiuscreusot.u-
bourgogne.fr/CN=certificat_client_iut_le_creusot/emailAddress=servinfo@iutlecreu
sot.u-bourgogne.fr
-->          issuer = /
C=FR/ST=bourgogne/L=le_creusot/O=service_informatique/OU=radiuscreusot.u-
bourgogne.fr/CN=certificat_client_iut_le_creusot/emailAddress=servinfo@iutlecreu
sot.u-bourgogne.fr
--> verify return:1
chain-depth=0,
error=0
--> User-Name = certificat_server_iut_le_creusot
--> BUF-Name = certificat_server_iut_le_creusot
-->          subject = /
C=FR/ST=bourgogne/L=le_creusot/O=service_informatique/OU=radiuscreusot.u-
bourgogne.fr/CN=certificat_server_iut_le_creusot/emailAddress=servinfo@iutlecreu
sot.u-bourgogne.fr
-->          issuer = /
C=FR/ST=bourgogne/L=le_creusot/O=service_informatique/OU=radiuscreusot.u-
bourgogne.fr/CN=certificat_client_iut_le_creusot/emailAddress=servinfo@iutlecreu
sot.u-bourgogne.fr
--> verify return:1
  TLS_accept: SSLv3 read client certificate A
  rlm_eap_tls: <<< TLS 1.0 Handshake [length 0046], ClientKeyExchange
  TLS_accept: SSLv3 read client key exchange A
  rlm_eap_tls: <<< TLS 1.0 Handshake [length 0086], CertificateVerify
  TLS_accept: SSLv3 read certificate verify A
  rlm_eap_tls: <<< TLS 1.0 ChangeCipherSpec [length 0001]
  rlm_eap_tls: <<< TLS 1.0 Handshake [length 0010], Finished
  TLS_accept: SSLv3 read finished A
  rlm_eap_tls: >>> TLS 1.0 ChangeCipherSpec [length 0001]
  TLS_accept: SSLv3 write change cipher spec A
  rlm_eap_tls: >>> TLS 1.0 Handshake [length 0010], Finished
  TLS_accept: SSLv3 write finished A
  TLS_accept: SSLv3 flush data
  (other): SSL negotiation finished successfully
SSL Connection Established
eaptls_process returned 13
modcall[authenticate]: module "eap" returns handled for request 5
modcall: group authenticate returns handled for request 5
Sending Access-Challenge of id 38 to 10.0.0.189:21657
  EAP-Message =
0x010700450d8000000003b1403010001011603010030f8e0dcbd99c31c84f8073abee0dc69355270
6b4d51390c144b4ac84e579d03ab093a5ba311c6b39f5e32a3d9856c8499
  Message-Authenticator = 0x00000000000000000000000000000000
  State = 0xa95614684b4029d23b1d2a6dc964c7ae
Finished request 5
Going to the next request
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=39, length=176
  User-Name = "certificat_server_iut_le_creusot"
  Framed-MTU = 1400
  Called-Station-Id = "0011.bbba.bb96"
  Calling-Station-Id = "0040.965d.e6ba"

```

```

Service-Type = Login-User
Message-Authenticator = 0xb0cf04f8680729ef180ea87c587532e5
EAP-Message = 0x020700060d00
NAS-Port-Type = Wireless-802.11
NAS-Port = 1355
State = 0xa95614684b4029d23b1d2a6dc964c7ae
NAS-IP-Address = 10.0.0.189
NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 6
  modcall[authorize]: module "preprocess" returns ok for request 6
  modcall[authorize]: module "chap" returns noop for request 6
  modcall[authorize]: module "mschap" returns noop for request 6
    rlm_realm: No '@' in User-Name = "certificat_server_iut_le_creusot", looking
up realm NULL
    rlm_realm: No such realm "NULL"
  modcall[authorize]: module "suffix" returns noop for request 6
  rlm_eap: EAP packet type response id 7 length 6
  rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
  modcall[authorize]: module "eap" returns updated for request 6
    users: Matched entry DEFAULT at line 409
  modcall[authorize]: module "files" returns ok for request 6
modcall: group authorize returns updated for request 6
  rad_check_password: Found Auth-Type EAP
auth: type "EAP"
  Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 6
  rlm_eap: Request found, released from the list
  rlm_eap: EAP/tls
  rlm_eap: processing type tls
  rlm_eap_tls: Authenticate
  rlm_eap_tls: processing TLS
rlm_eap_tls: Received EAP-TLS ACK message
  rlm_eap_tls: ack handshake is finished
  eaptls_verify returned 3
  eaptls_process returned 3
  rlm_eap: Freeing handler
  modcall[authenticate]: module "eap" returns ok for request 6
modcall: group authenticate returns ok for request 6
Sending Access-Accept of id 39 to 10.0.0.189:21657
  MS-MPPE-Recv-Key
0x5853eabc39c683ec04605688c9aa1cc73cbb4eff50a460c0d684cfd7ed31a687
  MS-MPPE-Send-Key
0xdfc55e1d3eeea7d135b32eee6e7695522541cf22546b642988acd10f28958b53
  EAP-Message = 0x03070004
  Message-Authenticator = 0x00000000000000000000000000000000
  User-Name = "certificat_server_iut_le_creusot"
Finished request 6
Going to the next request
Waking up in 6 seconds...
--- Walking the entire request list ---
Cleaning up request 0 ID 33 with timestamp 425f9370
Cleaning up request 1 ID 34 with timestamp 425f9370
Cleaning up request 2 ID 35 with timestamp 425f9370
Cleaning up request 3 ID 36 with timestamp 425f9370
Cleaning up request 4 ID 37 with timestamp 425f9370
Cleaning up request 5 ID 38 with timestamp 425f9370
Cleaning up request 6 ID 39 with timestamp 425f9370
Nothing to do. Sleeping until we see a request.

```

EAP-TLS fonctionne avec les certificats que nous avons générés (à changer tous les 5 ans sur les clients et sur le serveur). Il est assez difficile à mettre en oeuvre en raison de la gestion des certificats. Il ne permet pas de basculer l'utilisateur dans un WVLAN comme EAP-LEAP, ce qui peut être un grave inconvénient pour l'architecture du réseau Wi-Fi.

5.2.8. EAP-PEAP

L'EAP-PEAP permet d'utiliser un certificat côté client pour effectuer l'authentification. Il est plus simple à mettre en oeuvre que EAP-TLS (bien sûr, il faut toujours générer les certificats).

En PEAP, on utilise les mêmes certificats que pour l'EAP-TLS, la procédure est beaucoup plus simple. Il suffit d'installer les certificats sur les clients comme indiqué en section k) du paragraphe précédent.

a) Modification de « /usr/local/freeradius-1.0.2/etc/raddb/eap.conf ».

```
[root@ordi raddb]# vi eap.conf
[...]
    default_eap_type = peap
[...]
    peap {
        # The tunneled EAP session needs a default
        # EAP type which is separate from the one for
        # the non-tunneled EAP module. Inside of the
        # PEAP tunnel, we recommend using MS-CHAPv2,
        # as that is the default type supported by
        # Windows clients.
        default_eap_type = mschapv2
    }
[...]
[root@ordi raddb]#
```

b) Exemple de trace (log) freeradius pour EAP-PEAP.

```
[root@ordi raddb]# /usr/local/freeradius-1.0.2/sbin/radiusd -X
Starting - reading configuration files ...
reread_config: reading radiusd.conf
Config: including file: /etc/raddb/proxy.conf
Config: including file: /etc/raddb/clients.conf
Config: including file: /etc/raddb/snmp.conf
Config: including file: /etc/raddb/eap.conf
Config: including file: /etc/raddb/sql.conf
main: prefix = "/usr"
main: localstatedir = "/usr/var"
main: logdir = "/usr/var/log/radius"
main: libdir = "/usr/lib"
main: radacctdir = "/usr/var/log/radius/radacct"
main: hostname_lookups = no
main: max_request_time = 30
main: cleanup_delay = 5
main: max_requests = 1024
main: delete_blocked_requests = 0
main: port = 0
main: allow_core_dumps = no
main: log_stripped_names = no
```

```

main: log_file = "/usr/var/log/radius/radius.log"
main: log_auth = no
main: log_auth_badpass = no
main: log_auth_goodpass = no
main: pidfile = "/usr/var/run/radiusd/radiusd.pid"
main: user = "(null)"
main: group = "(null)"
main: usercollide = no
main: lower_user = "no"
main: lower_pass = "no"
main: nospace_user = "no"
main: nospace_pass = "no"
main: checkrad = "/usr/sbin/checkrad"
main: proxy_requests = yes
proxy: retry_delay = 5
proxy: retry_count = 3
proxy: synchronous = no
proxy: default_fallback = yes
proxy: dead_time = 120
proxy: post_proxy_authorize = yes
proxy: wake_all_if_all_dead = no
security: max_attributes = 200
security: reject_delay = 1
security: status_server = no
main: debug_level = 0
read_config_files: reading dictionary
read_config_files: reading naslist
Using deprecated naslist file. Support for this will go away soon.
read_config_files: reading clients
read_config_files: reading realms
radiusd: entering modules setup
Module: Library search path is /usr/lib
Module: Loaded exec
exec: wait = yes
exec: program = "(null)"
exec: input_pairs = "request"
exec: output_pairs = "(null)"
exec: packet_type = "(null)"
rlm_exec: Wait=yes but no output defined. Did you mean output=none?
Module: Instantiated exec (exec)
Module: Loaded expr
Module: Instantiated expr (expr)
Module: Loaded PAP
pap: encryption_scheme = "crypt"
Module: Instantiated pap (pap)
Module: Loaded CHAP
Module: Instantiated chap (chap)
Module: Loaded MS-CHAP
mschap: use_mppe = yes
mschap: require_encryption = no
mschap: require_strong = no
mschap: with_ntdomain_hack = no
mschap: passwd = "(null)"
mschap: authtype = "MS-CHAP"
mschap: ntlm_auth = "(null)"
Module: Instantiated mschap (mschap)
Module: Loaded System
unix: cache = no
unix: passwd = "(null)"
unix: shadow = "(null)"
unix: group = "(null)"

```

```

unix: radwtmp = "/usr/var/log/radius/radwtmp"
unix: usegroup = no
unix: cache_reload = 600
Module: Instantiated unix (unix)
Module: Loaded eap
  eap: default_eap_type = "tls"
  eap: timer_expire = 60
  eap: ignore_unknown_eap_types = no
  eap: cisco_accounting_username_bug = no
rlm_eap: Loaded and initialized type md5
rlm_eap: Loaded and initialized type leap
  gtc: challenge = "Password: "
  gtc: auth_type = "PAP"
rlm_eap: Loaded and initialized type gtc
  tls: rsa_key_exchange = no
  tls: dh_key_exchange = yes
  tls: rsa_key_length = 512
  tls: dh_key_length = 512
  tls: verify_depth = 0
  tls: CA_path = "(null)"
  tls: pem_file_type = yes
  tls: private_key_file = "/etc/raddb/certs/cert-srv.pem"
  tls: certificate_file = "/etc/raddb/certs/cert-srv.pem"
  tls: CA_file = "/etc/raddb/certs/demoCA/cacert.pem"
  tls: private_key_password = "mdpiutcreusot"
  tls: dh_file = "/etc/raddb/certs/dh"
  tls: random_file = "/etc/raddb/certs/random"
  tls: fragment_size = 1024
  tls: include_length = yes
  tls: check_crl = no
  tls: check_cert_cn = "(null)"
rlm_eap: Loaded and initialized type tls
  ttls: default_eap_type = "md5"
  ttls: copy_request_to_tunnel = yes
  ttls: use_tunneled_reply = yes
rlm_eap: Loaded and initialized type ttls
  peap: default_eap_type = "mschapv2"
  peap: copy_request_to_tunnel = no
  peap: use_tunneled_reply = no
  peap: proxy_tunneled_request_as_eap = yes
rlm_eap: Loaded and initialized type peap
  mschapv2: with_ntdomain_hack = no
rlm_eap: Loaded and initialized type mschapv2
Module: Instantiated eap (eap)
Module: Loaded preprocess
  preprocess: huntgroups = "/etc/raddb/huntgroups"
  preprocess: hints = "/etc/raddb/hints"
  preprocess: with_ascend_hack = no
  preprocess: ascend_channels_per_line = 23
  preprocess: with_ntdomain_hack = no
  preprocess: with_specialix_jetstream_hack = no
  preprocess: with_cisco_vsa_hack = no
Module: Instantiated preprocess (preprocess)
Module: Loaded realm
  realm: format = "suffix"
  realm: delimiter = "@"
  realm: ignore_default = no
  realm: ignore_null = no
Module: Instantiated realm (suffix)
Module: Loaded files
  files: usersfile = "/etc/raddb/users"

```



```

files: acctusersfile = "/etc/raddb/acct_users"
files: preproxy_usersfile = "/etc/raddb/preproxy_users"
files: compat = "no"
Module: Instantiated files (files)
Module: Loaded Acct-Unique-Session-Id
    acct_unique: key = "User-Name, Acct-Session-Id, NAS-IP-Address, Client-IP-Address, NAS-Port"
Module: Instantiated acct_unique (acct_unique)
Module: Loaded detail
    detail: detailfile = "/usr/var/log/radius/radacct/%{Client-IP-Address}/detail-%Y%m%d"
    detail: detailperm = 384
    detail: dirperm = 493
    detail: locking = no
Module: Instantiated detail (detail)
Module: Loaded radutmp
    radutmp: filename = "/usr/var/log/radius/radutmp"
    radutmp: username = "%{User-Name}"
    radutmp: case_sensitive = yes
    radutmp: check_with_nas = yes
    radutmp: perm = 384
    radutmp: callerid = yes
Module: Instantiated radutmp (radutmp)
Listening on authentication *:1812
Listening on accounting *:1813
Listening on proxy *:1814
Ready to process requests.
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=40, length=151
    User-Name = "j.landre"
    Framed-MTU = 1400
    Called-Station-Id = "0011.bbba.bb96"
    Calling-Station-Id = "0040.965d.e6ba"
    Service-Type = Login-User
    Message-Authenticator = 0x7e8b3c4f23debea5b47db780f3e0aa9b
    EAP-Message = 0x0201001201662e6c61746875696c69657265
    NAS-Port-Type = Wireless-802.11
    NAS-Port = 1356
    NAS-IP-Address = 10.0.0.189
    NAS-Identifier = "AP1100_TEST"
    Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 0
    modcall[authorize]: module "preprocess" returns ok for request 0
    modcall[authorize]: module "chap" returns noop for request 0
    modcall[authorize]: module "mschap" returns noop for request 0
        rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
        rlm_realm: No such realm "NULL"
    modcall[authorize]: module "suffix" returns noop for request 0
    rlm_eap: EAP packet type response id 1 length 18
    rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
    modcall[authorize]: module "eap" returns updated for request 0
        users: Matched entry j.landre at line 315
    modcall[authorize]: module "files" returns ok for request 0
modcall: group authorize returns updated for request 0
    rad_check_password: Found Auth-Type EAP
auth: type "EAP"
    Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 0
    rlm_eap: EAP Identity
    rlm_eap: processing type tls
    rlm_eap_tls: Requiring client certificate
    rlm_eap_tls: Initiate

```

```

    rlm_eap_tls: Start returned 1
    modcall[authenticate]: module "eap" returns handled for request 0
modcall: group authenticate returns handled for request 0
Sending Access-Challenge of id 40 to 10.0.0.189:21657
    Tunnel-Type:0 = VLAN
    Tunnel-Medium-Type:0 = IEEE-802
    Tunnel-Private-Group-Id:0 = "100"
    EAP-Message = 0x010200060d20
    Message-Authenticator = 0x00000000000000000000000000000000
    State = 0x32f0373f4ccb91d37373274b0b7f1bf3
Finished request 0
Going to the next request
--- Walking the entire request list ---
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=41, length=157
    User-Name = "j.landre"
    Framed-MTU = 1400
    Called-Station-Id = "0011.bbbaa.bb96"
    Calling-Station-Id = "0040.965d.e6ba"
    Service-Type = Login-User
    Message-Authenticator = 0xb49adb35bbc3513cbab8cd3ff1ccb183
    EAP-Message = 0x020200060319
    NAS-Port-Type = Wireless-802.11
    NAS-Port = 1356
    State = 0x32f0373f4ccb91d37373274b0b7f1bf3
    NAS-IP-Address = 10.0.0.189
    NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 1
    modcall[authorize]: module "preprocess" returns ok for request 1
    modcall[authorize]: module "chap" returns noop for request 1
    modcall[authorize]: module "mschap" returns noop for request 1
        rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
        rlm_realm: No such realm "NULL"
    modcall[authorize]: module "suffix" returns noop for request 1
    rlm_eap: EAP packet type response id 2 length 6
    rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
    modcall[authorize]: module "eap" returns updated for request 1
        users: Matched entry j.landre at line 315
    modcall[authorize]: module "files" returns ok for request 1
modcall: group authorize returns updated for request 1
    rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 1
    rlm_eap: Request found, released from the list
    rlm_eap: EAP NAK
    rlm_eap: EAP-NAK asked for EAP-Type/peap
    rlm_eap: processing type tls
    rlm_eap_tls: Initiate
    rlm_eap_tls: Start returned 1
    modcall[authenticate]: module "eap" returns handled for request 1
modcall: group authenticate returns handled for request 1
Sending Access-Challenge of id 41 to 10.0.0.189:21657
    Tunnel-Type:0 = VLAN
    Tunnel-Medium-Type:0 = IEEE-802
    Tunnel-Private-Group-Id:0 = "100"
    EAP-Message = 0x010300061920
    Message-Authenticator = 0x00000000000000000000000000000000
    State = 0x48140b5f4a0e06f457ba4da2058edaed
Finished request 1

```

```

Going to the next request
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=42, length=257
  User-Name = "j.landre"
  Framed-MTU = 1400
  Called-Station-Id = "0011.bbbaa.bb96"
  Calling-Station-Id = "0040.965d.e6ba"
  Service-Type = Login-User
  Message-Authenticator = 0xe7b03274816987ff55c248b2f31c32ff
  EAP-Message =
0x0203006a198000000060160301005b010000570301425f95149c3e4370dadcd2686ca07726d5442
86d0e0df529ce3448239a1ee196800003000390038003500160013000a00330032002f0066000500
040065006400630062006000150012000900140011000800030100
  NAS-Port-Type = Wireless-802.11
  NAS-Port = 1356
  State = 0x48140b5f4a0e06f457ba4da2058edaed
  NAS-IP-Address = 10.0.0.189
  NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 2
modcall[authorize]: module "preprocess" returns ok for request 2
modcall[authorize]: module "chap" returns noop for request 2
modcall[authorize]: module "mschap" returns noop for request 2
  rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
  rlm_realm: No such realm "NULL"
modcall[authorize]: module "suffix" returns noop for request 2
  rlm_eap: EAP packet type response id 3 length 106
  rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
modcall[authorize]: module "eap" returns updated for request 2
  users: Matched entry j.landre at line 315
modcall[authorize]: module "files" returns ok for request 2
modcall: group authorize returns updated for request 2
  rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 2
  rlm_eap: Request found, released from the list
  rlm_eap: EAP/peap
  rlm_eap: processing type peap
  rlm_eap_peap: Authenticate
  rlm_eap_tls: processing TLS
rlm_eap_tls: Length Included
  eaptls_verify returned 11
  (other): before/accept initialization
  TLS_accept: before/accept initialization
  rlm_eap_tls: <<< TLS 1.0 Handshake [length 005b], ClientHello
  TLS_accept: SSLv3 read client hello A
  rlm_eap_tls: >>> TLS 1.0 Handshake [length 004a], ServerHello
  TLS_accept: SSLv3 write server hello A
  rlm_eap_tls: >>> TLS 1.0 Handshake [length 07de], Certificate
  TLS_accept: SSLv3 write certificate A
  rlm_eap_tls: >>> TLS 1.0 Handshake [length 010d], ServerKeyExchange
  TLS_accept: SSLv3 write key exchange A
  rlm_eap_tls: >>> TLS 1.0 Handshake [length 0004], ServerHelloDone
  TLS_accept: SSLv3 write server done A
  TLS_accept: SSLv3 flush data
  TLS_accept:error in SSLv3 read client certificate A
In SSL Handshake Phase
In SSL Accept mode
  eaptls_process returned 13
  rlm_eap_peap: EAPTLS_HANDLED

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modcall[authenticate]: module "eap" returns handled for request 2
modcall: group authenticate returns handled for request 2
Sending Access-Challenge of id 42 to 10.0.0.189:21657
  Tunnel-Type:0 = VLAN
  Tunnel-Medium-Type:0 = IEEE-802
  Tunnel-Private-Group-Id:0 = "100"
  EAP-Message =
0x0104040a19c00000094d160301004a020000460301425f93910246b7a1f305b8e3e45d88fdc6d4
ade5dac0cbccb0b798db4d46fbd920411c4ee1f18259ede90a0b21230f4c51cf8dc3f95b4d12ea7f
367c9d229a51b400390016030107de0b0007da0007d700034b30820347308202b0a0030201020201
02300d06092a864886f70d01010405003081dc310c45678643dc3453466446523112301006035504
081309626f7572676f676e65311330110603550407140a6c655f63726575736f74311d301b060355
040a1414736572766963655f696e666f726d61746971756531253023060355040b131c7261646975
7363726575736f742e752d626f75
  EAP-Message =
0x72676f676e652e66723129302706035504031420636572746966696361745f636c69656e745f69
75745f6c655f63726575736f743133303106092a864886f70d010901162473657276696e666f4069
75746c6563726575736f742e752d626f7572676f676e652e6672301e170d30353034313530393436
30385a170d3036303431353039343630385a3081dc310c45678643dc345346644652311230100603
5504081309626f7572676f676e65311330110603550407140a6c655f63726575736f74311d301b06
0355040a1414736572766963655f696e666f726d61746971756531253023060355040b131c726164
69757363726575736f742e752d62
  EAP-Message =
0x6f7572676f676e652e66723129302706035504031420636572746966696361745f726163696e65
5f6975745f6c655f63726575736f743133303106092a864886f70d010901162473657276696e666f
406975746c6563726575736f742e752d626f7572676f676e652e667230819f300d06092a864886f7
0d010101050003818d0030818902818100c867154256f402ab95dc557cefc74fe0e82923b9b106c1
0af632d2bdc94da40fba69eb556204bbc16107a30b3881e99210d8b65dcef6eb9489b82e22cc2d89
82a6b259b2b81681504e22021da05c9367b73c8b24c2a77f2060b175f7cfa9a1b859b6e920344d69
dfe4dfcf6088b1da9a73b84208a2
  EAP-Message =
0x8c547fe1bbb45109639d510203010001a317301530130603551d25040c300a06082b0601050507
0301300d06092a864886f70d01010405000381810092e230835ef0fca505bfc2e81c22e85aee6827
9018555826f53903e28c9b3dd8513f8b17d652d3679d551f58c94a7e0978936ecd0abf9a68e8df70
abfe8f1fffd6ae96f46534af6738668fd30dfef2729d5c4f9aff9a444f45bec4b4a3dbb727299cb5c
532662341281c4144821dad195820617035225ee06bebed5fadba35ce00048630820482308203eb
a00302010202090099840e9f0189a17d300d06092a864886f70d01010405003081dc310c45678643
dc34534664465231123010060355
  EAP-Message = 0x04081309626f7572676f676e65311330110603550407
  Message-Authenticator = 0x00000000000000000000000000000000
  State = 0x049293bce49d3738d9785d06ae28d82d
Finished request 2
Going to the next request
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=43, length=157
  User-Name = "j.landree"
  Framed-MTU = 1400
  Called-Station-Id = "0011.bbba.bb96"
  Calling-Station-Id = "0040.965d.e6ba"
  Service-Type = Login-User
  Message-Authenticator = 0x6dc3aa5d8a7c958cebce98c559cdf44f
  EAP-Message = 0x020400061900
  NAS-Port-Type = Wireless-802.11
  NAS-Port = 1356
  State = 0x049293bce49d3738d9785d06ae28d82d
  NAS-IP-Address = 10.0.0.189
  NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 3
  modcall[authorize]: module "preprocess" returns ok for request 3
  modcall[authorize]: module "chap" returns noop for request 3

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modcall[authorize]: module "mschap" returns noop for request 3
  rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
  rlm_realm: No such realm "NULL"
modcall[authorize]: module "suffix" returns noop for request 3
  rlm_eap: EAP packet type response id 4 length 6
  rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
modcall[authorize]: module "eap" returns updated for request 3
  users: Matched entry j.landre at line 315
modcall[authorize]: module "files" returns ok for request 3
modcall: group authorize returns updated for request 3
  rad_check_password: Found Auth-Type EAP
auth: type "EAP"
  Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 3
  rlm_eap: Request found, released from the list
  rlm_eap: EAP/peap
  rlm_eap: processing type peap
  rlm_eap_peap: Authenticate
  rlm_eap_tls: processing TLS
rlm_eap_tls: Received EAP-TLS ACK message
  rlm_eap_tls: ack handshake fragment handler
  eaptls_verify returned 1
  eaptls_process returned 13
  rlm_eap_peap: EAPTLS_HANDLED
modcall[authenticate]: module "eap" returns handled for request 3
modcall: group authenticate returns handled for request 3
Sending Access-Challenge of id 43 to 10.0.0.189:21657
  Tunnel-Type:0 = VLAN
  Tunnel-Medium-Type:0 = IEEE-802
  Tunnel-Private-Group-Id:0 = "100"
  EAP-Message
=
0x010504061940140a6c655f63726575736f74311d301b060355040a1414736572766963655f696e
666f726d61746971756531253023060355040b131c72616469757363726575736f742e752d626f75
72676f676e652e66723129302706035504031420636572746966696361745f636c69656e745f6975
745f6c655f63726575736f743133303106092a864886f70d010901162473657276696e666f406975
746c6563726575736f742e752d626f7572676f676e652e6672301e170d3035303431353039343630
375a170d3130303431343039343630375a3081dc310c45678643dc34534664465231123010060355
04081309626f7572676f676e6531
  EAP-Message
=
0x1330110603550407140a6c655f63726575736f74311d301b060355040a1414736572766963655f
696e666f726d61746971756531253023060355040b131c72616469757363726575736f742e752d62
6f7572676f676e652e66723129302706035504031420636572746966696361745f636c69656e745f
6975745f6c655f63726575736f743133303106092a864886f70d010901162473657276696e666f40
6975746c6563726575736f742e752d626f7572676f676e652e667230819f300d06092a864886f70d
010101050003818d0030818902818100a8eb16e223d41d8d33afeadde8e38edb9780fa0f9ff12174
fab90b981e9a1760a6418f9de7e8
  EAP-Message
=
0x69156ee5f0fedfe1bb33512d21cacdd613a29302b6fe2bd2a889b7279bcf325b9c6f92a39e8bb5
7fd922643ac08e7db560a90786721219537ec765d01bb70235b302ccc09dbd3db76ddf5a9322491c
94ea6e63851a4208c180090203010001a382014830820144301d0603551d0e041604145ddb0a9044
d02c70c097f6bf9cbe83c49d4a6ead308201130603551d230482010a3082010680145ddb0a9044d0
2c70c097f6bf9cbe83c49d4a6ead181e2a481df3081dc310c45678643dc34534664465231123010
06035504081309626f7572676f676e65311330110603550407140a6c655f63726575736f74311d30
1b060355040a1414736572766963
  EAP-Message
=
0x655f696e666f726d61746971756531253023060355040b131c72616469757363726575736f742e
752d626f7572676f676e652e66723129302706035504031420636572746966696361745f636c6965
6e745f6975745f6c655f63726575736f743133303106092a864886f70d010901162473657276696e
666f406975746c6563726575736f742e752d626f7572676f676e652e667282090099840e9f0189a1
7d300c0603551d13040530030101ff300d06092a864886f70d010104050003818100069146dd03cf
8dd1355041cd843e3acf951b6919d79f52f0cf53926b97f36dd864882a16f0bdee5a66bd6bb22025

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992d707c4a2b0bb4d26e3eb53e78
    EAP-Message = 0xa4f0e1f588573bcdba484fa2b7894f632146
    Message-Authenticator = 0x00000000000000000000000000000000
    State = 0xb8818d77f19d4598b38345cf230fc9a6
Finished request 3
Going to the next request
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=44, length=157
    User-Name = "j.landree"
    Framed-MTU = 1400
    Called-Station-Id = "0011.bbba.bb96"
    Calling-Station-Id = "0040.965d.e6ba"
    Service-Type = Login-User
    Message-Authenticator = 0x1d17542d9c1daf708179d800bcf52097
    EAP-Message = 0x020500061900
    NAS-Port-Type = Wireless-802.11
    NAS-Port = 1356
    State = 0xb8818d77f19d4598b38345cf230fc9a6
    NAS-IP-Address = 10.0.0.189
    NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 4
    modcall[authorize]: module "preprocess" returns ok for request 4
    modcall[authorize]: module "chap" returns noop for request 4
    modcall[authorize]: module "mschap" returns noop for request 4
        rlm_realm: No '@' in User-Name = "j.landree", looking up realm NULL
        rlm_realm: No such realm "NULL"
    modcall[authorize]: module "suffix" returns noop for request 4
    rlm_eap: EAP packet type response id 5 length 6
    rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
    modcall[authorize]: module "eap" returns updated for request 4
        users: Matched entry j.landree at line 315
    modcall[authorize]: module "files" returns ok for request 4
modcall: group authorize returns updated for request 4
    rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 4
    rlm_eap: Request found, released from the list
    rlm_eap: EAP/peap
    rlm_eap: processing type peap
    rlm_eap_peap: Authenticate
    rlm_eap_tls: processing TLS
rlm_eap_tls: Received EAP-TLS ACK message
    rlm_eap_tls: ack handshake fragment handler
    eaptls_verify returned 1
    eaptls_process returned 13
    rlm_eap_peap: EAPTLS_HANDLED
    modcall[authenticate]: module "eap" returns handled for request 4
modcall: group authenticate returns handled for request 4
Sending Access-Challenge of id 44 to 10.0.0.189:21657
    Tunnel-Type:0 = VLAN
    Tunnel-Medium-Type:0 = IEEE-802
    Tunnel-Private-Group-Id:0 = "100"
    EAP-Message =
0x0106015319009079f10914df6b1ff53e09acf78ab059b2d30c50e8cb6a51fad047cb1855a88e8c
adbd48f8d34ef6082425338d9f08ec2371160301010d0c0001090040e4e1abf8258723f61d7fbb1d
c1f5da079e7f6977a7b25436daea70a7fda0f8de4f86e7fd6372cb0882a7ff9d1f703920c4152ef1
e03db020083a41c66ffff2430001020040a824888768e7d138d64cb00942d12957169a84c268193a
10c60c360dd2dd94c7539a0dac6d1ba630f011c7526addd2626371c238b95cc75b3cf17c0102348b
b5008085554156a599ba2717871dd678ff19367bf597a214408f39f119d48b0117f40abb0932e045

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b58bc651d28f62375e9457c8cdce
  EAP-Message
0xe522e06941cac0e6f1ed440aff146b16f596176f41b2aa710c24a924401f670973bcbec31a6826
7d3c3620071dc17528661e26f30239ca310ee69f38794df1534b04755d882899dfa263afc9da1603
0100040e000000
  Message-Authenticator = 0x00000000000000000000000000000000
  State = 0x02dd7c3bd5fb9ac4d5ec0bff9cad481d
Finished request 4
Going to the next request
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=45, length=295
  User-Name = "j.landre"
  Framed-MTU = 1400
  Called-Station-Id = "0011.bbbaa.bb96"
  Calling-Station-Id = "0040.965d.e6ba"
  Service-Type = Login-User
  Message-Authenticator = 0x93e15b89a5e8e1e6493def9caef5ae70
  EAP-Message
0x02060090198000000086160301004610000042004013299fbc50c76d7740aaa1df6ac67bb3ef29
45ddea5f418275e826e29092521600b00c5c6186e39cee4649b04227de9819c6bd133e8ccd3ea90a
11eb568424a61403010001011603010030b0e6e7dfbc939db2f0fe67adeb6f5cec9fbd314c9fc585
adfa7065935381671b5daefeb9f10925cab024f112828d7226
  NAS-Port-Type = Wireless-802.11
  NAS-Port = 1356
  State = 0x02dd7c3bd5fb9ac4d5ec0bff9cad481d
  NAS-IP-Address = 10.0.0.189
  NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 5
  modcall[authorize]: module "preprocess" returns ok for request 5
  modcall[authorize]: module "chap" returns noop for request 5
  modcall[authorize]: module "mschap" returns noop for request 5
    rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
    rlm_realm: No such realm "NULL"
  modcall[authorize]: module "suffix" returns noop for request 5
  rlm_eap: EAP packet type response id 6 length 144
  rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
  modcall[authorize]: module "eap" returns updated for request 5
    users: Matched entry j.landre at line 315
  modcall[authorize]: module "files" returns ok for request 5
modcall: group authorize returns updated for request 5
  rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 5
  rlm_eap: Request found, released from the list
  rlm_eap: EAP/peap
  rlm_eap: processing type peap
  rlm_eap_peap: Authenticate
  rlm_eap_tls: processing TLS
rlm_eap_tls: Length Included
  eaptls_verify returned 11
  rlm_eap_tls: <<< TLS 1.0 Handshake [length 0046], ClientKeyExchange
    TLS_accept: SSLv3 read client key exchange A
  rlm_eap_tls: <<< TLS 1.0 ChangeCipherSpec [length 0001]
  rlm_eap_tls: <<< TLS 1.0 Handshake [length 0010], Finished
    TLS_accept: SSLv3 read finished A
  rlm_eap_tls: >>> TLS 1.0 ChangeCipherSpec [length 0001]
    TLS_accept: SSLv3 write change cipher spec A
  rlm_eap_tls: >>> TLS 1.0 Handshake [length 0010], Finished
    TLS_accept: SSLv3 write finished A

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    TLS_accept: SSLv3 flush data
    (other): SSL negotiation finished successfully
SSL Connection Established
    eaptls_process returned 13
    rlm_eap_peap: EAPTLS_HANDLED
    modcall[authenticate]: module "eap" returns handled for request 5
modcall: group authenticate returns handled for request 5
Sending Access-Challenge of id 45 to 10.0.0.189:21657
    Tunnel-Type:0 = VLAN
    Tunnel-Medium-Type:0 = IEEE-802
    Tunnel-Private-Group-Id:0 = "100"
    EAP-Message =
0x01070041190014030100010116030100309df984b4da998f099e0ecfb760d14ac842e8439efdf4
e8326ad12964ecfd104d7a0baa4ecd3713cc40af3b014415656a
    Message-Authenticator = 0x00000000000000000000000000000000
    State = 0x360a0426f0638ffc6c00dc19ab774cfb
Finished request 5
Going to the next request
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=46, length=157
    User-Name = "j.landre"
    Framed-MTU = 1400
    Called-Station-Id = "0011.bbba.bb96"
    Calling-Station-Id = "0040.965d.e6ba"
    Service-Type = Login-User
    Message-Authenticator = 0x1f1ef5402e2b4fdd8adb9009fb6bde9d
    EAP-Message = 0x020700061900
    NAS-Port-Type = Wireless-802.11
    NAS-Port = 1356
    State = 0x360a0426f0638ffc6c00dc19ab774cfb
    NAS-IP-Address = 10.0.0.189
    NAS-Identifier = "AP1100_TEST"
    Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 6
    modcall[authorize]: module "preprocess" returns ok for request 6
    modcall[authorize]: module "chap" returns noop for request 6
    modcall[authorize]: module "mschap" returns noop for request 6
    rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
    rlm_realm: No such realm "NULL"
    modcall[authorize]: module "suffix" returns noop for request 6
    rlm_eap: EAP packet type response id 7 length 6
    rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
    modcall[authorize]: module "eap" returns updated for request 6
    users: Matched entry j.landre at line 315
    modcall[authorize]: module "files" returns ok for request 6
modcall: group authorize returns updated for request 6
    rad_check_password: Found Auth-Type EAP
auth: type "EAP"
    Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 6
    rlm_eap: Request found, released from the list
    rlm_eap: EAP/peap
    rlm_eap: processing type peap
    rlm_eap_peap: Authenticate
    rlm_eap_tls: processing TLS
rlm_eap_tls: Received EAP-TLS ACK message
    rlm_eap_tls: ack handshake is finished
    eaptls_verify returned 3
    eaptls_process returned 3
    rlm_eap_peap: EAPTLS_SUCCESS
    modcall[authenticate]: module "eap" returns handled for request 6

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modcall: group authenticate returns handled for request 6
Sending Access-Challenge of id 46 to 10.0.0.189:21657
    Tunnel-Type:0 = VLAN
    Tunnel-Medium-Type:0 = IEEE-802
    Tunnel-Private-Group-Id:0 = "100"
    EAP-Message
0x01080050190017030100201381d032256f477b27d7d9275e4bc0b7505260d0f643cc40eaf963ef
3da0a5da170301002024b48e7910719281e664f0b7f662d5990f223660dcfc037d71be94a3fa3cb1
2a
    Message-Authenticator = 0x00000000000000000000000000000000
    State = 0x7d7befdfcda47625247ca05835e9e709
Finished request 6
Going to the next request
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=47, length=247
    User-Name = "j.landre"
    Framed-MTU = 1400
    Called-Station-Id = "0011.bbbaa.bb96"
    Calling-Station-Id = "0040.965d.e6ba"
    Service-Type = Login-User
    Message-Authenticator = 0x1374a5a2c9b4267ca36b80c7a1be2c4e
    EAP-Message
0x020800601900170301002088d67bda5e6f7c50581c64251e9b5db6be942fffd3988660c2837a06
f96abdd31703010030aee36502a9b0df967cd76950b42b298659cab8754ffad09f16c2969f87652d
23866e0ca2eef6ac7345c28b454b17ff4c
    NAS-Port-Type = Wireless-802.11
    NAS-Port = 1356
    State = 0x7d7befdfcda47625247ca05835e9e709
    NAS-IP-Address = 10.0.0.189
    NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 7
    modcall[authorize]: module "preprocess" returns ok for request 7
    modcall[authorize]: module "chap" returns noop for request 7
    modcall[authorize]: module "mschap" returns noop for request 7
    rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
    rlm_realm: No such realm "NULL"
    modcall[authorize]: module "suffix" returns noop for request 7
    rlm_eap: EAP packet type response id 8 length 96
    rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
    modcall[authorize]: module "eap" returns updated for request 7
    users: Matched entry j.landre at line 315
    modcall[authorize]: module "files" returns ok for request 7
modcall: group authorize returns updated for request 7
    rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 7
    rlm_eap: Request found, released from the list
    rlm_eap: EAP/peap
    rlm_eap: processing type peap
    rlm_eap_peap: Authenticate
    rlm_eap_tls: processing TLS
    eaptls_verify returned 7
    rlm_eap_tls: Done initial handshake
    eaptls_process returned 7
    rlm_eap_peap: EAPTLS_OK
    rlm_eap_peap: Session established. Decoding tunneled attributes.
    rlm_eap_peap: Identity - j.landre
    rlm_eap_peap: Tunneled data is valid.
PEAP: Got tunneled identity of j.landre

```

```

PEAP: Setting default EAP type for tunneled EAP session.
PEAP: Setting User-Name to j.landre
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 7
modcall[authorize]: module "preprocess" returns ok for request 7
modcall[authorize]: module "chap" returns noop for request 7
modcall[authorize]: module "mschap" returns noop for request 7
    rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
    rlm_realm: No such realm "NULL"
modcall[authorize]: module "suffix" returns noop for request 7
rlm_eap: EAP packet type response id 8 length 18
rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
modcall[authorize]: module "eap" returns updated for request 7
    users: Matched entry j.landre at line 315
modcall[authorize]: module "files" returns ok for request 7
modcall: group authorize returns updated for request 7
    rad_check_password: Found Auth-Type EAP
auth: type "EAP"
    Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 7
    rlm_eap: EAP Identity
    rlm_eap: processing type mschapv2
rlm_eap_mschapv2: Issuing Challenge
    modcall[authenticate]: module "eap" returns handled for request 7
modcall: group authenticate returns handled for request 7
    PEAP: Got tunneled Access-Challenge
    modcall[authenticate]: module "eap" returns handled for request 7
modcall: group authenticate returns handled for request 7
Sending Access-Challenge of id 47 to 10.0.0.189:21657
    Tunnel-Type:0 = VLAN
    Tunnel-Medium-Type:0 = IEEE-802
    Tunnel-Private-Group-Id:0 = "100"
    EAP-Message =
0x010900701900170301002091aeb9d9df08b231e046b07aab5f02bfa269b11ddc4f363c67928967
1f1a9fc61703010040245dd952530e52fadb163ef7dddf9a99918e0324364c22c1cd529ffa856792
88da78bbb65048ec80bed0d87a3764bd3266f5c1b324604f009564ec546c162a9a
    Message-Authenticator = 0x00000000000000000000000000000000
    State = 0x19764f28bfd32d6839fa243869f4275d
Finished request 7
Going to the next request
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=48, length=295
    User-Name = "j.landre"
    Framed-MTU = 1400
    Called-Station-Id = "0011.bbba.bb96"
    Calling-Station-Id = "0040.965d.e6ba"
    Service-Type = Login-User
    Message-Authenticator = 0x6dc048a14a4e0b0de847fddae4749389
    EAP-Message =
0x02090090190017030100209421b5535dfee3f687e84f294e43c9abc1e61520502b32fc079163cb
c6f44e57170301006012e7236d4c8641697b9395bd843517c0ef87e9743e484d4aa3f8efc563fff3
f08ea211e1d4c7489cb0efa1d393bff2ca98f76abd089b0978a61ff18c038355ddb00bd649cf3035
3965f9948dddf43310f5dc892d26f39b8955c876d3c0acb333
    NAS-Port-Type = Wireless-802.11
    NAS-Port = 1356
    State = 0x19764f28bfd32d6839fa243869f4275d
    NAS-IP-Address = 10.0.0.189
    NAS-Identifier = "AP1100_TEST"
    Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 8
    modcall[authorize]: module "preprocess" returns ok for request 8

```

```

modcall[authorize]: module "chap" returns noop for request 8
modcall[authorize]: module "mschap" returns noop for request 8
  rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
  rlm_realm: No such realm "NULL"
modcall[authorize]: module "suffix" returns noop for request 8
rlm_eap: EAP packet type response id 9 length 144
rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
modcall[authorize]: module "eap" returns updated for request 8
  users: Matched entry j.landre at line 315
modcall[authorize]: module "files" returns ok for request 8
modcall: group authorize returns updated for request 8
  rad_check_password: Found Auth-Type EAP
auth: type "EAP"
  Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 8
  rlm_eap: Request found, released from the list
  rlm_eap: EAP/peap
  rlm_eap: processing type peap
  rlm_eap_peap: Authenticate
  rlm_eap_tls: processing TLS
  eaptls_verify returned 7
  rlm_eap_tls: Done initial handshake
  eaptls_process returned 7
  rlm_eap_peap: EAPTLS_OK
  rlm_eap_peap: Session established. Decoding tunneled attributes.
  rlm_eap_peap: EAP type mschapv2
  rlm_eap_peap: Tunneled data is valid.
  PEAP: Setting User-Name to j.landre
  PEAP: Adding old state with f9 a0
  Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 8
  modcall[authorize]: module "preprocess" returns ok for request 8
  modcall[authorize]: module "chap" returns noop for request 8
  modcall[authorize]: module "mschap" returns noop for request 8
    rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
    rlm_realm: No such realm "NULL"
  modcall[authorize]: module "suffix" returns noop for request 8
  rlm_eap: EAP packet type response id 9 length 72
  rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
  modcall[authorize]: module "eap" returns updated for request 8
    users: Matched entry j.landre at line 315
  modcall[authorize]: module "files" returns ok for request 8
modcall: group authorize returns updated for request 8
  rad_check_password: Found Auth-Type EAP
auth: type "EAP"
  Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 8
  rlm_eap: Request found, released from the list
  rlm_eap: EAP/mschapv2
  rlm_eap: processing type mschapv2
  Processing the authenticate section of radiusd.conf
modcall: entering group Auth-Type for request 8
  rlm_mschap: Told to do MS-CHAPv2 for j.landre with NT-Password
  rlm_mschap: adding MS-CHAPv2 MPPE keys
  modcall[authenticate]: module "mschap" returns ok for request 8
modcall: group Auth-Type returns ok for request 8
MSCHAP Success
  modcall[authenticate]: module "eap" returns handled for request 8
modcall: group authenticate returns handled for request 8
  PEAP: Got tunneled Access-Challenge
  modcall[authenticate]: module "eap" returns handled for request 8

```

```

modcall: group authenticate returns handled for request 8
Sending Access-Challenge of id 48 to 10.0.0.189:21657
    Tunnel-Type:0 = VLAN
    Tunnel-Medium-Type:0 = IEEE-802
    Tunnel-Private-Group-Id:0 = "100"
    EAP-Message
0x010a008019001703010020cdd1ed0279f49aa2371e534ae567083cdf5a0f3ca81e1fbb73e7751
2c29202d17030100506f06e2f3ef4a7dda9de55836aaac3a957f2fd00c1e8c74293ac2b849d8755
42dfe91203b83b6b0ac53e438d88b42e072b30483bbc0821e9b98d8d77d72dd3efb5468da221892a
551352b1982cac7c69
    Message-Authenticator = 0x00000000000000000000000000000000
    State = 0x48d4e359b81fe9e76d518ee377660bd3
Finished request 8
Going to the next request
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=49, length=231
    User-Name = "j.landree"
    Framed-MTU = 1400
    Called-Station-Id = "0011.bbba.bb96"
    Calling-Station-Id = "0040.965d.e6ba"
    Service-Type = Login-User
    Message-Authenticator = 0x14c382132b39e0e9c6a1e2dc2ad3fee0
    EAP-Message
0x020a0050190017030100209a004a64f8db7b1216ed4716f6bd0acb3bf827f556c66f111bae5af5
f2a2678017030100206e53bfcaf647fa7242310707a6efded02f21aaf2ddc7a355303d396a76b068
e2
    NAS-Port-Type = Wireless-802.11
    NAS-Port = 1356
    State = 0x48d4e359b81fe9e76d518ee377660bd3
    NAS-IP-Address = 10.0.0.189
    NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 9
modcall[authorize]: module "preprocess" returns ok for request 9
modcall[authorize]: module "chap" returns noop for request 9
modcall[authorize]: module "mschap" returns noop for request 9
    rlm_realm: No '@' in User-Name = "j.landree", looking up realm NULL
    rlm_realm: No such realm "NULL"
modcall[authorize]: module "suffix" returns noop for request 9
    rlm_eap: EAP packet type response id 10 length 80
    rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
modcall[authorize]: module "eap" returns updated for request 9
    users: Matched entry j.landree at line 315
modcall[authorize]: module "files" returns ok for request 9
modcall: group authorize returns updated for request 9
    rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 9
    rlm_eap: Request found, released from the list
    rlm_eap: EAP/peap
    rlm_eap: processing type peap
    rlm_eap_peap: Authenticate
    rlm_eap_tls: processing TLS
    eaptls_verify returned 7
    rlm_eap_tls: Done initial handshake
    eaptls_process returned 7
    rlm_eap_peap: EAPTLS_OK
    rlm_eap_peap: Session established. Decoding tunneled attributes.
    rlm_eap_peap: EAP type mschapv2
    rlm_eap_peap: Tunneled data is valid.

```

```

PEAP: Setting User-Name to j.landre
PEAP: Adding old state with 4b 6d
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 9
modcall[authorize]: module "preprocess" returns ok for request 9
modcall[authorize]: module "chap" returns noop for request 9
modcall[authorize]: module "mschap" returns noop for request 9
    rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
    rlm_realm: No such realm "NULL"
modcall[authorize]: module "suffix" returns noop for request 9
    rlm_eap: EAP packet type response id 10 length 6
    rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
modcall[authorize]: module "eap" returns updated for request 9
    users: Matched entry j.landre at line 315
modcall[authorize]: module "files" returns ok for request 9
modcall: group authorize returns updated for request 9
    rad_check_password: Found Auth-Type EAP
auth: type "EAP"
    Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 9
    rlm_eap: Request found, released from the list
    rlm_eap: EAP/mschapv2
    rlm_eap: processing type mschapv2
    rlm_eap: Freeing handler
modcall[authenticate]: module "eap" returns ok for request 9
modcall: group authenticate returns ok for request 9
    PEAP: Tunneled authentication was successful.
    rlm_eap_peap: SUCCESS
modcall[authenticate]: module "eap" returns handled for request 9
modcall: group authenticate returns handled for request 9
Sending Access-Challenge of id 49 to 10.0.0.189:21657
    Tunnel-Type:0 = VLAN
    Tunnel-Medium-Type:0 = IEEE-802
    Tunnel-Private-Group-Id:0 = "100"
    EAP-Message =
0x010b0050190017030100205f5a0254eb0541d93d7a0d48f61e34fb9af4645ebc51ae92fc3ee8de
106095b3170301002039c66adf07de128ddee6d51c434c712fc46a6a6c8304669db768b5e9f856cd
f2
    Message-Authenticator = 0x00000000000000000000000000000000
    State = 0x7462dde1692fd19f67151f2136259fa
Finished request 9
Going to the next request
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=50, length=231
    User-Name = "j.landre"
    Framed-MTU = 1400
    Called-Station-Id = "0011.bbba.bb96"
    Calling-Station-Id = "0040.965d.e6ba"
    Service-Type = Login-User
    Message-Authenticator = 0xb3b59a4a0d8e29a5a97cebf9e81d98c2
    EAP-Message =
0x020b00501900170301002047e93d005cbe605f982231d387bc721bf346d70907d23fcbd177b7c9
19e1ca9c17030100201174885a2507e86a34e367f9786ec527f4703616c7652d7b86d2734d5719c6
4f
    NAS-Port-Type = Wireless-802.11
    NAS-Port = 1356
    State = 0x7462dde1692fd19f67151f2136259fa
    NAS-IP-Address = 10.0.0.189
    NAS-Identifier = "AP1100_TEST"
    Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 10

```

```

modcall[authorize]: module "preprocess" returns ok for request 10
modcall[authorize]: module "chap" returns noop for request 10
modcall[authorize]: module "mschap" returns noop for request 10
    rlm_realm: No '@' in User-Name = "j.landred", looking up realm NULL
    rlm_realm: No such realm "NULL"
modcall[authorize]: module "suffix" returns noop for request 10
rlm_eap: EAP packet type response id 11 length 80
rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
modcall[authorize]: module "eap" returns updated for request 10
    users: Matched entry j.landred at line 315
modcall[authorize]: module "files" returns ok for request 10
modcall: group authorize returns updated for request 10
    rad_check_password: Found Auth-Type EAP
auth: type "EAP"
    Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 10
    rlm_eap: Request found, released from the list
    rlm_eap: EAP/peap
    rlm_eap: processing type peap
    rlm_eap_peap: Authenticate
    rlm_eap_tls: processing TLS
    eaptls_verify returned 7
    rlm_eap_tls: Done initial handshake
    eaptls_process returned 7
    rlm_eap_peap: EAPTLS_OK
    rlm_eap_peap: Session established. Decoding tunneled attributes.
    rlm_eap_peap: Received EAP-TLV response.
    rlm_eap_peap: Tunneled data is valid.
    rlm_eap_peap: Success
    rlm_eap: Freeing handler
modcall[authenticate]: module "eap" returns ok for request 10
modcall: group authenticate returns ok for request 10
Sending Access-Accept of id 50 to 10.0.0.189:21657
    Tunnel-Type:0 = VLAN
    Tunnel-Medium-Type:0 = IEEE-802
    Tunnel-Private-Group-Id:0 = "100"
    MS-MPPE-Recv-Key
0x13fc098b88553d9b903d7e423ce7d3e288c8fd7fa49591f55931b1148d4fe4b3
    MS-MPPE-Send-Key
0x93932ebfe8bd3f9cc2f6f6ebf6b9c1dab345ef741ab09d80ad4244723ffa427c
    EAP-Message = 0x030b0004
    Message-Authenticator = 0x00000000000000000000000000000000
    User-Name = "j.landred"
Finished request 10
Going to the next request
Waking up in 6 seconds...
--- Walking the entire request list ---
Cleaning up request 0 ID 40 with timestamp 425f9391
Cleaning up request 1 ID 41 with timestamp 425f9391
Cleaning up request 2 ID 42 with timestamp 425f9391
Cleaning up request 3 ID 43 with timestamp 425f9391
Cleaning up request 4 ID 44 with timestamp 425f9391
Cleaning up request 5 ID 45 with timestamp 425f9391
Cleaning up request 6 ID 46 with timestamp 425f9391
Cleaning up request 7 ID 47 with timestamp 425f9391
Cleaning up request 8 ID 48 with timestamp 425f9391
Cleaning up request 9 ID 49 with timestamp 425f9391
Cleaning up request 10 ID 50 with timestamp 425f9391
Nothing to do. Sleeping until we see a request.

```

EAP-PEAP fonctionne parfaitement bien. Il permet de basculer l'utilisateur dans un WVLAN et est supporté de base dans Windows XP et Windows CE (utile pour les PDA). Il utilise la méthode d'authentification MSCHAPv2, ce qui permet d'authentifier l'utilisateur sur un serveur Active Directory, comme nous le verrons plus tard dans ce document.

5.2.9. EAP-TTLS

EAP-TTLS fournit un tunnel sécurisé entre le client et le serveur d'authentification. Dans ce tunnel, on utilise un autre EAP (quelconque) pour faire transiter l'information. Nous avons choisi EAP-MD5 (peu sécurisé en direct mais qui suffit dans un tunnel de bout en bout). La configuration est très simple (bien sûr, il faut toujours générer les certificats).

a) Modification de « /usr/local/freeradius-1.0.2/etc/raddb/eap.conf ».

```
[root@ordi raddb]# vi eap.conf
[...]
    default_eap_type = tls
[...]
    md5 {
    }
[...]
    ttls {
        # The tunneled EAP session needs a default
        # EAP type which is separate from the one for
        # the non-tunneled EAP module. Inside of the
        # TTLS tunnel, we recommend using EAP-MD5.
        # If the request does not contain an EAP
        # conversation, then this configuration entry
        # is ignored.
        default_eap_type = md5

        # The tunneled authentication request does
        # not usually contain useful attributes
        # like 'Calling-Station-Id', etc. These
        # attributes are outside of the tunnel,
        # and normally unavailable to the tunneled
        # authentication request.
        #
        # By setting this configuration entry to
        # 'yes', any attribute which NOT in the
        # tunneled authentication request, but
        # which IS available outside of the tunnel,
        # is copied to the tunneled request.
        #
        # allowed values: {no, yes}
        copy_request_to_tunnel = yes

        # The reply attributes sent to the NAS are
        # usually based on the name of the user
        # 'outside' of the tunnel (usually
        # 'anonymous'). If you want to send the
        # reply attributes based on the user name
        # inside of the tunnel, then set this
        # configuration entry to 'yes', and the reply
        # to the NAS will be taken from the reply to
        # the tunneled request.
        #
```

```

        # allowed values: {no, yes}
        use_tunneled_reply = yes
    }
[...]
```

[root@ordi raddb]#

b) Exemple de trace (log) freeradius pour EAP-TTLS.

```

[root@ordi raddb]# /usr/local/freeradius-1.0.2/sbin/radiusd -X
Starting - reading configuration files ...
reread_config: reading radiusd.conf
Config: including file: /etc/raddb/proxy.conf
Config: including file: /etc/raddb/clients.conf
Config: including file: /etc/raddb/snmp.conf
Config: including file: /etc/raddb/eap.conf
Config: including file: /etc/raddb/sql.conf
main: prefix = "/usr"
main: localstatedir = "/usr/var"
main: logdir = "/usr/var/log/radius"
main: libdir = "/usr/lib"
main: radacctdir = "/usr/var/log/radius/radacct"
main: hostname_lookups = no
main: max_request_time = 30
main: cleanup_delay = 5
main: max_requests = 1024
main: delete_blocked_requests = 0
main: port = 0
main: allow_core_dumps = no
main: log_stripped_names = no
main: log_file = "/usr/var/log/radius/radius.log"
main: log_auth = no
main: log_auth_badpass = no
main: log_auth_goodpass = no
main: pidfile = "/usr/var/run/radiusd/radiusd.pid"
main: user = "(null)"
main: group = "(null)"
main: usercollide = no
main: lower_user = "no"
main: lower_pass = "no"
main: nospace_user = "no"
main: nospace_pass = "no"
main: checkrad = "/usr/sbin/checkrad"
main: proxy_requests = yes
proxy: retry_delay = 5
proxy: retry_count = 3
proxy: synchronous = no
proxy: default_fallback = yes
proxy: dead_time = 120
proxy: post_proxy_authorize = yes
proxy: wake_all_if_all_dead = no
security: max_attributes = 200
security: reject_delay = 1
security: status_server = no
main: debug_level = 0
read_config_files: reading dictionary
read_config_files: reading naslist
Using deprecated naslist file. Support for this will go away soon.
read_config_files: reading clients
read_config_files: reading realms
radiusd: entering modules setup
```



```

Module: Library search path is /usr/lib
Module: Loaded exec
  exec: wait = yes
  exec: program = "(null)"
  exec: input_pairs = "request"
  exec: output_pairs = "(null)"
  exec: packet_type = "(null)"
rlm_exec: Wait=yes but no output defined. Did you mean output=none?
Module: Instantiated exec (exec)
Module: Loaded expr
Module: Instantiated expr (expr)
Module: Loaded PAP
  pap: encryption_scheme = "crypt"
Module: Instantiated pap (pap)
Module: Loaded CHAP
Module: Instantiated chap (chap)
Module: Loaded MS-CHAP
  mschap: use_mppe = yes
  mschap: require_encryption = no
  mschap: require_strong = no
  mschap: with_ntdomain_hack = no
  mschap: passwd = "(null)"
  mschap: authtype = "MS-CHAP"
  mschap: ntlm_auth = "(null)"
Module: Instantiated mschap (mschap)
Module: Loaded System
  unix: cache = no
  unix: passwd = "(null)"
  unix: shadow = "(null)"
  unix: group = "(null)"
  unix: radwtmp = "/usr/var/log/radius/radwtmp"
  unix: usegroup = no
  unix: cache_reload = 600
Module: Instantiated unix (unix)
Module: Loaded eap
  eap: default_eap_type = "tls"
  eap: timer_expire = 60
  eap: ignore_unknown_eap_types = no
  eap: cisco_accounting_username_bug = no
rlm_eap: Loaded and initialized type md5
rlm_eap: Loaded and initialized type leap
  gtc: challenge = "Password: "
  gtc: auth_type = "PAP"
rlm_eap: Loaded and initialized type gtc
  tls: rsa_key_exchange = no
  tls: dh_key_exchange = yes
  tls: rsa_key_length = 512
  tls: dh_key_length = 512
  tls: verify_depth = 0
  tls: CA_path = "(null)"
  tls: pem_file_type = yes
  tls: private_key_file = "/etc/raddb/certs/cert-srv.pem"
  tls: certificate_file = "/etc/raddb/certs/cert-srv.pem"
  tls: CA_file = "/etc/raddb/certs/demoCA/cacert.pem"
  tls: private_key_password = "mdpiutcreusot"
  tls: dh_file = "/etc/raddb/certs/dh"
  tls: random_file = "/etc/raddb/certs/random"
  tls: fragment_size = 1024
  tls: include_length = yes
  tls: check_crl = no
  tls: check_cert_cn = "(null)"

```

```

rlm_eap: Loaded and initialized type tls
  ttls: default_eap_type = "md5"
  ttls: copy_request_to_tunnel = yes
  ttls: use_tunneled_reply = yes
rlm_eap: Loaded and initialized type ttls
  peap: default_eap_type = "mschapv2"
  peap: copy_request_to_tunnel = no
  peap: use_tunneled_reply = no
  peap: proxy_tunneled_request_as_eap = yes
rlm_eap: Loaded and initialized type peap
  mschapv2: with_ntdomain_hack = no
rlm_eap: Loaded and initialized type mschapv2
Module: Instantiated eap (eap)
Module: Loaded preprocess
  preprocess: huntgroups = "/etc/raddb/huntgroups"
  preprocess: hints = "/etc/raddb/hints"
  preprocess: with_ascend_hack = no
  preprocess: ascend_channels_per_line = 23
  preprocess: with_ntdomain_hack = no
  preprocess: with_specialix_jetstream_hack = no
  preprocess: with_cisco_vsa_hack = no
Module: Instantiated preprocess (preprocess)
Module: Loaded realm
  realm: format = "suffix"
  realm: delimiter = "@"
  realm: ignore_default = no
  realm: ignore_null = no
Module: Instantiated realm (suffix)
Module: Loaded files
  files: usersfile = "/etc/raddb/users"
  files: acctusersfile = "/etc/raddb/acct_users"
  files: preproxy_usersfile = "/etc/raddb/preproxy_users"
  files: compat = "no"
Module: Instantiated files (files)
Module: Loaded Acct-Unique-Session-Id
  acct_unique: key = "User-Name, Acct-Session-Id, NAS-IP-Address, Client-IP-Address, NAS-Port"
Module: Instantiated acct_unique (acct_unique)
Module: Loaded detail
  detail: detailfile = "/usr/var/log/radius/radacct/%{Client-IP-Address}/detail-%Y%m%d"
  detail: detailperm = 384
  detail: dirperm = 493
  detail: locking = no
Module: Instantiated detail (detail)
Module: Loaded radutmp
  radutmp: filename = "/usr/var/log/radius/radutmp"
  radutmp: username = "%{User-Name}"
  radutmp: case_sensitive = yes
  radutmp: check_with_nas = yes
  radutmp: perm = 384
  radutmp: callerid = yes
Module: Instantiated radutmp (radutmp)
Listening on authentication *:1812
Listening on accounting *:1813
Listening on proxy *:1814
Ready to process requests.
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=52, length=151
  User-Name = "j.landre"
  Framed-MTU = 1400
  Called-Station-Id = "0011.bbbaa.bb96"

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Calling-Station-Id = "0040.965d.e6ba"
Service-Type = Login-User
Message-Authenticator = 0x8dc6ba6fa895bdf60db9f20cc1333021
EAP-Message = 0x0202001201662e6c61746875696c69657265
NAS-Port-Type = Wireless-802.11
NAS-Port = 1358
NAS-IP-Address = 10.0.0.189
NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 0
modcall[authorize]: module "preprocess" returns ok for request 0
modcall[authorize]: module "chap" returns noop for request 0
modcall[authorize]: module "mschap" returns noop for request 0
    rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
    rlm_realm: No such realm "NULL"
modcall[authorize]: module "suffix" returns noop for request 0
rlm_eap: EAP packet type response id 2 length 18
rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
modcall[authorize]: module "eap" returns updated for request 0
    users: Matched entry j.landre at line 315
modcall[authorize]: module "files" returns ok for request 0
modcall: group authorize returns updated for request 0
rad_check_password: Found Auth-Type EAP
auth: type "EAP"
    Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 0
    rlm_eap: EAP Identity
    rlm_eap: processing type tls
    rlm_eap_tls: Requiring client certificate
    rlm_eap_tls: Initiate
    rlm_eap_tls: Start returned 1
modcall[authenticate]: module "eap" returns handled for request 0
modcall: group authenticate returns handled for request 0
Sending Access-Challenge of id 52 to 10.0.0.189:21657
    Tunnel-Type:0 = VLAN
    Tunnel-Medium-Type:0 = IEEE-802
    Tunnel-Private-Group-Id:0 = "100"
    EAP-Message = 0x010300060d20
    Message-Authenticator = 0x00000000000000000000000000000000
    State = 0x59255c6666553d70b53624c470489ed6
Finished request 0
Going to the next request
--- Walking the entire request list ---
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=53, length=157
    User-Name = "j.landre"
    Framed-MTU = 1400
    Called-Station-Id = "0011.bbba.bb96"
    Calling-Station-Id = "0040.965d.e6ba"
    Service-Type = Login-User
    Message-Authenticator = 0x1231187edd9390696c68a70f10ba0d4d
    EAP-Message = 0x020300060315
    NAS-Port-Type = Wireless-802.11
    NAS-Port = 1358
    State = 0x59255c6666553d70b53624c470489ed6
    NAS-IP-Address = 10.0.0.189
    NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 1
modcall[authorize]: module "preprocess" returns ok for request 1
modcall[authorize]: module "chap" returns noop for request 1

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modcall[authorize]: module "mschap" returns noop for request 1
  rlm_realm: No '@' in User-Name = "j.landred", looking up realm NULL
  rlm_realm: No such realm "NULL"
modcall[authorize]: module "suffix" returns noop for request 1
  rlm_eap: EAP packet type response id 3 length 6
  rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
modcall[authorize]: module "eap" returns updated for request 1
  users: Matched entry j.landred at line 315
modcall[authorize]: module "files" returns ok for request 1
modcall: group authorize returns updated for request 1
  rad_check_password: Found Auth-Type EAP
auth: type "EAP"
  Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 1
  rlm_eap: Request found, released from the list
  rlm_eap: EAP NAK
  rlm_eap: EAP-NAK asked for EAP-Type/ttls
  rlm_eap: processing type tls
  rlm_eap_tls: Initiate
  rlm_eap_tls: Start returned 1
  modcall[authenticate]: module "eap" returns handled for request 1
modcall: group authenticate returns handled for request 1
Sending Access-Challenge of id 53 to 10.0.0.189:21657
  Tunnel-Type:0 = VLAN
  Tunnel-Medium-Type:0 = IEEE-802
  Tunnel-Private-Group-Id:0 = "100"
  EAP-Message = 0x0104000061520
  Message-Authenticator = 0x00000000000000000000000000000000
  State = 0xdfdf300c601f01371c5c76c1a2f436c1
Finished request 1
Going to the next request
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=51, length=151
  User-Name = "j.landred"
  Framed-MTU = 1400
  Called-Station-Id = "0011.bbbaa.bb96"
  Calling-Station-Id = "0040.965d.e6ba"
  Service-Type = Login-User
  Message-Authenticator = 0x471d3ab98e4c7bc7ff102c161e28371d
  EAP-Message = 0x0202001201662e6c61746875696c69657265
  NAS-Port-Type = Wireless-802.11
  NAS-Port = 1357
  NAS-IP-Address = 10.0.0.189
  NAS-Identifier = "AP1100_TEST"
  Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 2
  modcall[authorize]: module "preprocess" returns ok for request 2
  modcall[authorize]: module "chap" returns noop for request 2
  modcall[authorize]: module "mschap" returns noop for request 2
    rlm_realm: No '@' in User-Name = "j.landred", looking up realm NULL
    rlm_realm: No such realm "NULL"
  modcall[authorize]: module "suffix" returns noop for request 2
  rlm_eap: EAP packet type response id 2 length 18
  rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
modcall[authorize]: module "eap" returns updated for request 2
  users: Matched entry j.landred at line 315
  modcall[authorize]: module "files" returns ok for request 2
modcall: group authorize returns updated for request 2
  rad_check_password: Found Auth-Type EAP
auth: type "EAP"
  Processing the authenticate section of radiusd.conf

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modcall: entering group authenticate for request 2
  rlm_eap: EAP Identity
  rlm_eap: processing type tls
  rlm_eap_tls: Requiring client certificate
  rlm_eap_tls: Initiate
  rlm_eap_tls: Start returned 1
  modcall[authenticate]: module "eap" returns handled for request 2
modcall: group authenticate returns handled for request 2
Sending Access-Challenge of id 51 to 10.0.0.189:21657
  Tunnel-Type:0 = VLAN
  Tunnel-Medium-Type:0 = IEEE-802
  Tunnel-Private-Group-Id:0 = "100"
  EAP-Message = 0x0103000060d20
  Message-Authenticator = 0x00000000000000000000000000000000
  State = 0x07908b5feae3234f31684784a1fbd310
Finished request 2
Going to the next request
--- Walking the entire request list ---
Waking up in 1 seconds...
--- Walking the entire request list ---
Cleaning up request 0 ID 52 with timestamp 425f9414
Cleaning up request 1 ID 53 with timestamp 425f9414
Waking up in 5 seconds...
--- Walking the entire request list ---
Cleaning up request 2 ID 51 with timestamp 425f9419
Nothing to do. Sleeping until we see a request.
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=54, length=257
  User-Name = "j.landre"
  Framed-MTU = 1400
  Called-Station-Id = "0011.bbba.bb96"
  Calling-Station-Id = "0040.965d.e6ba"
  Service-Type = Login-User
  Message-Authenticator = 0x6fe73440a0f9ad25e71ab257d023d4c3
  EAP-Message =
0x0204006a158000000060160301005b010000570301425f95a8ff968a2d72f25ac9bcf29fd2d3b1
f0d5d9b7811e5d0260c9c242eb5a00003000390038003500160013000a00330032002f0066000500
040065006400630062006000150012000900140011000800030100
  NAS-Port-Type = Wireless-802.11
  NAS-Port = 1358
  State = 0xdfdf300c601f01371c5c76c1a2f436c1
  NAS-IP-Address = 10.0.0.189
  NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 3
  modcall[authorize]: module "preprocess" returns ok for request 3
  modcall[authorize]: module "chap" returns noop for request 3
  modcall[authorize]: module "mschap" returns noop for request 3
  rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
  rlm_realm: No such realm "NULL"
  modcall[authorize]: module "suffix" returns noop for request 3
  rlm_eap: EAP packet type response id 4 length 106
  rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
  modcall[authorize]: module "eap" returns updated for request 3
  users: Matched entry j.landre at line 315
  modcall[authorize]: module "files" returns ok for request 3
modcall: group authorize returns updated for request 3
  rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 3
  rlm_eap: Request found, released from the list

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rlm_eap: EAP/ttls
rlm_eap: processing type ttls
rlm_eap_ttls: Authenticate
rlm_eap_ttls: processing TLS
rlm_eap_ttls: Length Included
eaptls_verify returned 11
  (other): before/accept initialization
  TLS_accept: before/accept initialization
rlm_eap_ttls: <<< TLS 1.0 Handshake [length 005b], ClientHello
  TLS_accept: SSLv3 read client hello A
rlm_eap_ttls: >>> TLS 1.0 Handshake [length 004a], ServerHello
  TLS_accept: SSLv3 write server hello A
rlm_eap_ttls: >>> TLS 1.0 Handshake [length 07de], Certificate
  TLS_accept: SSLv3 write certificate A
rlm_eap_ttls: >>> TLS 1.0 Handshake [length 010d], ServerKeyExchange
  TLS_accept: SSLv3 write key exchange A
rlm_eap_ttls: >>> TLS 1.0 Handshake [length 0004], ServerHelloDone
  TLS_accept: SSLv3 write server done A
  TLS_accept: SSLv3 flush data
  TLS_accept:error in SSLv3 read client certificate A
In SSL Handshake Phase
In SSL Accept mode
  eaptls_process returned 13
  modcall[authenticate]: module "eap" returns handled for request 3
modcall: group authenticate returns handled for request 3
Sending Access-Challenge of id 54 to 10.0.0.189:21657
  Tunnel-Type:0 = VLAN
  Tunnel-Medium-Type:0 = IEEE-802
  Tunnel-Private-Group-Id:0 = "100"
  EAP-Message
0x0105040a15c00000094d160301004a020000460301425f94269d60abf3129ae1acb677b0cd7b82
e040dd58731a7b266a25743c0f7c20bea9661090f7d452ecb0da2756bee2ffeb7ae8f91306f3f13f
5d91bee88f15ac00390016030107de0b0007da0007d700034b30820347308202b0a0030201020201
02300d06092a864886f70d01010405003081dc310c45678643dc3453466446523112301006035504
081309626f7572676f676e65311330110603550407140a6c655f63726575736f74311d301b060355
040a1414736572766963655f696e666f726d61746971756531253023060355040b131c7261646975
7363726575736f742e752d626f75
  EAP-Message
0x72676f676e652e66723129302706035504031420636572746966696361745f636c69656e745f69
75745f6c655f63726575736f743133303106092a864886f70d010901162473657276696e666f4069
75746c6563726575736f742e752d626f7572676f676e652e6672301e170d30353034313530393436
30385a170d3036303431353039343630385a3081dc310c45678643dc345346644652311230100603
5504081309626f7572676f676e65311330110603550407140a6c655f63726575736f74311d301b06
0355040a1414736572766963655f696e666f726d61746971756531253023060355040b131c726164
69757363726575736f742e752d62
  EAP-Message
0x6f7572676f676e652e66723129302706035504031420636572746966696361745f726163696e65
5f6975745f6c655f63726575736f743133303106092a864886f70d010901162473657276696e666f
406975746c6563726575736f742e752d626f7572676f676e652e667230819f300d06092a864886f7
0d010101050003818d0030818902818100c867154256f402ab95dc557cefc74fe0e82923b9b106c1
0af632d2bdc94da40fba69eb556204bbc16107a30b3881e99210d8b65dcef6eb9489b82e22cc2d89
82a6b259b2b81681504e22021da05c9367b73c8b24c2a77f2060b175f7cfa9a1b859b6e920344d69
dfe4dfcf6088b1da9a73b84208a2
  EAP-Message
0x8c547fe1bbb45109639d510203010001a317301530130603551d25040c300a06082b0601050507
0301300d06092a864886f70d01010405000381810092e230835ef0fca505bfc2e81c22e85aee6827
9018555826f53903e28c9b3dd8513f8b17d652d3679d551f58c94a7e0978936ecd0abf9a68e8df70
abfe8f1fffd6ae96f46534af6738668fd30dfef2729d5c4f9aff9a444f45bec4b4a3dbb727299cb5c
532662341281c1414821dad195820617035225ee06bebed5fadba35ce00048630820482308203eb
a00302010202090099840e9f0189a17d300d06092a864886f70d01010405003081dc310c45678643
dc34534664465231123010060355

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EAP-Message = 0x04081309626f7572676f676e65311330110603550407
Message-Authenticator = 0x00000000000000000000000000000000
State = 0xc7a72350343d92c8045d9dc25802cc00
Finished request 3
Going to the next request
--- Walking the entire request list ---
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=55, length=157
  User-Name = "j.landre"
  Framed-MTU = 1400
  Called-Station-Id = "0011.bbba.bb96"
  Calling-Station-Id = "0040.965d.e6ba"
  Service-Type = Login-User
  Message-Authenticator = 0xd4f494010e0d65a18f446cf76b7032c8
  EAP-Message = 0x020500061500
  NAS-Port-Type = Wireless-802.11
  NAS-Port = 1358
  State = 0xc7a72350343d92c8045d9dc25802cc00
  NAS-IP-Address = 10.0.0.189
  NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 4
  modcall[authorize]: module "preprocess" returns ok for request 4
  modcall[authorize]: module "chap" returns noop for request 4
  modcall[authorize]: module "mschap" returns noop for request 4
    rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
    rlm_realm: No such realm "NULL"
  modcall[authorize]: module "suffix" returns noop for request 4
  rlm_eap: EAP packet type response id 5 length 6
  rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
  modcall[authorize]: module "eap" returns updated for request 4
    users: Matched entry j.landre at line 315
  modcall[authorize]: module "files" returns ok for request 4
modcall: group authorize returns updated for request 4
  rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 4
  rlm_eap: Request found, released from the list
  rlm_eap: EAP/ttls
  rlm_eap: processing type ttls
  rlm_eap_ttls: Authenticate
  rlm_eap_ttls: processing TLS
rlm_eap_ttls: Received EAP-TLS ACK message
  rlm_eap_ttls: ack handshake fragment handler
  eaptls_verify returned 1
  eaptls_process returned 13
  modcall[authenticate]: module "eap" returns handled for request 4
modcall: group authenticate returns handled for request 4
Sending Access-Challenge of id 55 to 10.0.0.189:21657
  Tunnel-Type:0 = VLAN
  Tunnel-Medium-Type:0 = IEEE-802
  Tunnel-Private-Group-Id:0 = "100"
  EAP-Message
=
0x0106040a15c00000094d140a6c655f63726575736f74311d301b060355040a1414736572766963
655f696e666f726d61746971756531253023060355040b131c72616469757363726575736f742e75
2d626f7572676f676e652e66723129302706035504031420636572746966696361745f636c69656e
745f6975745f6c655f63726575736f743133303106092a864886f70d010901162473657276696e66
6f406975746c6563726575736f742e752d626f7572676f676e652e6672301e170d30353034313530
39343630375a170d3130303431343039343630375a3081dc310c45678643dc345346644652311230
1006035504081309626f7572676f

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EAP-Message =
0x676e65311330110603550407140a6c655f63726575736f74311d301b060355040a141473657276
6963655f696e666f726d61746971756531253023060355040b131c72616469757363726575736f74
2e752d626f7572676f676e652e66723129302706035504031420636572746966696361745f636c69
656e745f6975745f6c655f63726575736f743133303106092a864886f70d01090116247365727669
6e666f406975746c6563726575736f742e752d626f7572676f676e652e667230819f300d06092a86
4886f70d010101050003818d0030818902818100a8eb16e223d41d8d33afeadde8e38edb9780fa0f
9ff12174fab90b981e9a1760a641

EAP-Message =
0x8f9de7e869156ee5f0fedfe1bb33512d21cacdd613a29302b6fe2bd2a889b7279bcf325b9c6f92
a39e8bb57fd922643ac08e7db560a90786721219537ec765d01bb70235b302ccc09dbd3db76ddf5a
9322491c94ea6e63851a4208c180090203010001a382014830820144301d0603551d0e041604145d
db0a9044d02c70c097f6bf9cbe83c49d4a6ead308201130603551d230482010a3082010680145ddb
0a9044d02c70c097f6bf9cbe83c49d4a6ead181e2a481df3081dc310c45678643dc345346644652
3112301006035504081309626f7572676f676e65311330110603550407140a6c655f63726575736f
74311d301b060355040a14147365

EAP-Message =
0x72766963655f696e666f726d61746971756531253023060355040b131c72616469757363726575
736f742e752d626f7572676f676e652e66723129302706035504031420636572746966696361745f
636c69656e745f6975745f6c655f63726575736f743133303106092a864886f70d01090116247365
7276696e666f406975746c6563726575736f742e752d626f7572676f676e652e667282090099840e
9f0189a17d300c0603551d13040530030101ff300d06092a864886f70d0101040500038181000691
46dd03cf8dd1355041cd843e3acf951b6919d79f52f0cf53926b97f36dd864882a16f0bdee5a66bd
6bb22025992d707c4a2b0bb4d26e

EAP-Message = 0x3eb53e78a4f0e1f588573bcdba484fa2b7894f632146
Message-Authenticator = 0x00000000000000000000000000000000
State = 0xe3e952a4b7ecf2a817b18ad4a1a48c86
Finished request 4
Going to the next request
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=56, length=157
  User-Name = "j.landred"
  Framed-MTU = 1400
  Called-Station-Id = "0011.bbba.bb96"
  Calling-Station-Id = "0040.965d.e6ba"
  Service-Type = Login-User
  Message-Authenticator = 0xe488c51aa13bab353a8b5b5684c9136f
  EAP-Message = 0x020600061500
  NAS-Port-Type = Wireless-802.11
  NAS-Port = 1358
  State = 0xe3e952a4b7ecf2a817b18ad4a1a48c86
  NAS-IP-Address = 10.0.0.189
  NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 5
  modcall[authorize]: module "preprocess" returns ok for request 5
  modcall[authorize]: module "chap" returns noop for request 5
  modcall[authorize]: module "mschap" returns noop for request 5
    rlm_realm: No '@' in User-Name = "j.landred", looking up realm NULL
    rlm_realm: No such realm "NULL"
  modcall[authorize]: module "suffix" returns noop for request 5
  rlm_eap: EAP packet type response id 6 length 6
  rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
  modcall[authorize]: module "eap" returns updated for request 5
    users: Matched entry j.landred at line 315
  modcall[authorize]: module "files" returns ok for request 5
modcall: group authorize returns updated for request 5
  rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 5

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rlm_eap: Request found, released from the list
rlm_eap: EAP/ttls
rlm_eap: processing type ttls
rlm_eap_ttls: Authenticate
rlm_eap_ttls: processing TLS
rlm_eap_ttls: Received EAP-TLS ACK message
rlm_eap_ttls: ack handshake fragment handler
eaptls_verify returned 1
eaptls_process returned 13
modcall[authenticate]: module "eap" returns handled for request 5
modcall: group authenticate returns handled for request 5
Sending Access-Challenge of id 56 to 10.0.0.189:21657
Tunnel-Type:0 = VLAN
Tunnel-Medium-Type:0 = IEEE-802
Tunnel-Private-Group-Id:0 = "100"
EAP-Message
0x01070157158000000094d9079f10914df6b1ff53e09acf78ab059b2d30c50e8cb6a51fad047cb18
55a88e8cadbd48f8d34ef6082425338d9f08ec2371160301010d0c0001090040e4e1abf8258723f6
1d7fbb1dc1f5da079e7f6977a7b25436daea70a7fda0f8de4f86e7fd6372cb0882a7ff9d1f703920
c4152ef1e03db020083a41c66ffff243000102004088f6d6d535cb8735289b6ad752039212087c34
7fff7150073d716d41e7fecbdde4060009bbe05e0864ea470343fb13bee4aed74537708bde0bd93
edecf77547008031090f565e2f4cef4b5827d937bb6b79edcf16cc274123aa62610cf22c25001f43
346cd220120143d0b7b7549b51b2
EAP-Message
0x9e3b7e9cfc7fc46445a16472b2df29f128bcf92e291771525477fa8108f4752aac2b7e7f71ea3a
a216e0593d1f7d1f4c8102dbe37a028d077da74dfa6f3d129e631d5d15133d3f2c6b3c756bba2d38
479d16030100040e000000
Message-Authenticator = 0x00000000000000000000000000000000
State = 0xf323702f5b1182b48c64a7ea491f997a
Finished request 5
Going to the next request
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=57, length=295
User-Name = "j.landre"
Framed-MTU = 1400
Called-Station-Id = "0011.bbba.bb96"
Calling-Station-Id = "0040.965d.e6ba"
Service-Type = Login-User
Message-Authenticator = 0x28e58faa7ebd9ba78716ef9fc1cf7ffb
EAP-Message
0x020700901580000000086160301004610000042004040c3ede27d0a9bd630aa73adac766d055c0d
20c6952bb7771a61064f7826367392055e9172d2730edfededbc0331368f267d5d43ea4b8b8a8f85
1927437ce72714030100010116030100303a681ef0ae7b1ed6ba03ca2efdddb06907c92efdea111b
e0d213017a3de8c736d870652dd516f1b08348b96de8efe552
NAS-Port-Type = Wireless-802.11
NAS-Port = 1358
State = 0xf323702f5b1182b48c64a7ea491f997a
NAS-IP-Address = 10.0.0.189
NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 6
modcall[authorize]: module "preprocess" returns ok for request 6
modcall[authorize]: module "chap" returns noop for request 6
modcall[authorize]: module "mschap" returns noop for request 6
rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
rlm_realm: No such realm "NULL"
modcall[authorize]: module "suffix" returns noop for request 6
rlm_eap: EAP packet type response id 7 length 144
rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
modcall[authorize]: module "eap" returns updated for request 6
users: Matched entry j.landre at line 315

```

```

modcall[authorize]: module "files" returns ok for request 6
modcall: group authorize returns updated for request 6
rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 6
rlm_eap: Request found, released from the list
rlm_eap: EAP/ttls
rlm_eap: processing type ttls
rlm_eap_ttls: Authenticate
rlm_eap_ttls: processing TLS
rlm_eap_ttls: Length Included
eaptls_verify returned 11
rlm_eap_ttls: <<< TLS 1.0 Handshake [length 0046], ClientKeyExchange
TLS_accept: SSLv3 read client key exchange A
rlm_eap_ttls: <<< TLS 1.0 ChangeCipherSpec [length 0001]
rlm_eap_ttls: <<< TLS 1.0 Handshake [length 0010], Finished
TLS_accept: SSLv3 read finished A
rlm_eap_ttls: >>> TLS 1.0 ChangeCipherSpec [length 0001]
TLS_accept: SSLv3 write change cipher spec A
rlm_eap_ttls: >>> TLS 1.0 Handshake [length 0010], Finished
TLS_accept: SSLv3 write finished A
TLS_accept: SSLv3 flush data
(other): SSL negotiation finished successfully
SSL Connection Established
eaptls_process returned 13
modcall[authenticate]: module "eap" returns handled for request 6
modcall: group authenticate returns handled for request 6
Sending Access-Challenge of id 57 to 10.0.0.189:21657
Tunnel-Type:0 = VLAN
Tunnel-Medium-Type:0 = IEEE-802
Tunnel-Private-Group-Id:0 = "100"
EAP-Message =
0x0108004515800000003b14030100010116030100309f7c4271f2c28dcfc99c74700de3a3cbc97d
d68455f24aef7d655ce8b9f4a090a0248cc091f9f5939a16ea59881e7560
Message-Authenticator = 0x00000000000000000000000000000000
State = 0x931e465d9d1f3f625b150ff8a30c286f
Finished request 6
Going to the next request
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=58, length=267
User-Name = "j.landre"
Framed-MTU = 1400
Called-Station-Id = "0011.bbba.bb96"
Calling-Station-Id = "0040.965d.e6ba"
Service-Type = Login-User
Message-Authenticator = 0xde3e7f8868e66b017d40da19493d7245
EAP-Message =
0x0208007415800000006a17030100200676b40996ebd5895f3cc45864eb8df4c148b62fafe43c5e
8857b867d7e9a85f1703010040d9b0a5dcbb92e6b27625641d6e1652abbde722afaa27c0316472b5
108dca36f2a33c7c791b8a6351c2b474ceb96ce270b318acbf65f094f2a11918940f0d7b51
NAS-Port-Type = Wireless-802.11
NAS-Port = 1358
State = 0x931e465d9d1f3f625b150ff8a30c286f
NAS-IP-Address = 10.0.0.189
NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 7
modcall[authorize]: module "preprocess" returns ok for request 7
modcall[authorize]: module "chap" returns noop for request 7
modcall[authorize]: module "mschap" returns noop for request 7

```

```

    rlm_realm: No '@' in User-Name = "j.landred", looking up realm NULL
    rlm_realm: No such realm "NULL"
modcall[authorize]: module "suffix" returns noop for request 7
    rlm_eap: EAP packet type response id 8 length 116
    rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
modcall[authorize]: module "eap" returns updated for request 7
    users: Matched entry j.landred at line 315
modcall[authorize]: module "files" returns ok for request 7
modcall: group authorize returns updated for request 7
    rad_check_password: Found Auth-Type EAP
auth: type "EAP"
    Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 7
    rlm_eap: Request found, released from the list
    rlm_eap: EAP/ttls
    rlm_eap: processing type ttls
    rlm_eap_ttls: Authenticate
    rlm_eap_ttls: processing TLS
rlm_eap_ttls: Length Included
    eaptls_verify returned 11
    eaptls_process returned 7
    rlm_eap_ttls: Session established. Proceeding to decode tunneled attributes.
    TTLS: Got tunneled identity of j.landred
    TTLS: Setting default EAP type for tunneled EAP session.
    Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 7
    modcall[authorize]: module "preprocess" returns ok for request 7
    modcall[authorize]: module "chap" returns noop for request 7
    modcall[authorize]: module "mschap" returns noop for request 7
        rlm_realm: No '@' in User-Name = "j.landred", looking up realm NULL
        rlm_realm: No such realm "NULL"
    modcall[authorize]: module "suffix" returns noop for request 7
    rlm_eap: EAP packet type response id 0 length 18
    rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
modcall[authorize]: module "eap" returns updated for request 7
    users: Matched entry j.landred at line 315
modcall[authorize]: module "files" returns ok for request 7
modcall: group authorize returns updated for request 7
    rad_check_password: Found Auth-Type EAP
auth: type "EAP"
    Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 7
    rlm_eap: EAP Identity
    rlm_eap: processing type md5
rlm_eap_md5: Issuing Challenge
    modcall[authenticate]: module "eap" returns handled for request 7
modcall: group authenticate returns handled for request 7
    TTLS: Got tunneled Access-Challenge
    modcall[authenticate]: module "eap" returns handled for request 7
modcall: group authenticate returns handled for request 7
Sending Access-Challenge of id 58 to 10.0.0.189:21657
    Tunnel-Type:0 = VLAN
    Tunnel-Medium-Type:0 = IEEE-802
    Tunnel-Private-Group-Id:0 = "100"
    EAP-Message
0x01090074158000000006a1703010020227c1e95c3a570df1807cf8591e2185604af6f1827855130
24459c1ddebd1a7c1703010040100a7f9cf469f46a58afecc1ac25b9c18111948d1677b3e76eeaae
dd20804f4762f43704ddb78e0468f4e851e9ee80fb78fd89b18273d10be95a793f7b50a44b
    Message-Authenticator = 0x00000000000000000000000000000000
    State = 0x1b1fe6b254c1b8a41ecf0d932dac6bc0
Finished request 7

```

```

Going to the next request
Waking up in 6 seconds...
rad_recv: Access-Request packet from host 10.0.0.189:21657, id=59, length=283
  User-Name = "j.landre"
  Framed-MTU = 1400
  Called-Station-Id = "0011.bbbaa.bb96"
  Calling-Station-Id = "0040.965d.e6ba"
  Service-Type = Login-User
  Message-Authenticator = 0x400be9db3d861e2c23bcf856a4acef98
  EAP-Message =
0x0209008415800000007a17030100202066297c66751eee794cebc5fbddc1747b53209f75d71c5c
4b2d8e64aba85b0b1703010050615ae181637032cb3b84b19710cdcd8fcde480187dd85a4812b083
5f66534f56b92c2486d69fd4058603c8360a22c2e685f5f218acc74fb5a1d86f1f4f06b20667c7ef
0ca7e2be0d0f7d5bb20d4697d9
  NAS-Port-Type = Wireless-802.11
  NAS-Port = 1358
  State = 0x1b1fe6b254c1b8a41ecf0d932dac6bc0
  NAS-IP-Address = 10.0.0.189
  NAS-Identifier = "AP1100_TEST"
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 8
  modcall[authorize]: module "preprocess" returns ok for request 8
  modcall[authorize]: module "chap" returns noop for request 8
  modcall[authorize]: module "mschap" returns noop for request 8
    rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
    rlm_realm: No such realm "NULL"
  modcall[authorize]: module "suffix" returns noop for request 8
  rlm_eap: EAP packet type response id 9 length 132
  rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
  modcall[authorize]: module "eap" returns updated for request 8
    users: Matched entry j.landre at line 315
  modcall[authorize]: module "files" returns ok for request 8
modcall: group authorize returns updated for request 8
  rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 8
  rlm_eap: Request found, released from the list
  rlm_eap: EAP/ttls
  rlm_eap: processing type ttls
  rlm_eap_ttls: Authenticate
  rlm_eap_ttls: processing TLS
rlm_eap_ttls: Length Included
  eaptls_verify returned 11
  eaptls_process returned 7
  rlm_eap_ttls: Session established. Proceeding to decode tunneled attributes.
  TTLS: Adding old state with f3 88
Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 8
  modcall[authorize]: module "preprocess" returns ok for request 8
  modcall[authorize]: module "chap" returns noop for request 8
  modcall[authorize]: module "mschap" returns noop for request 8
    rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
    rlm_realm: No such realm "NULL"
  modcall[authorize]: module "suffix" returns noop for request 8
  rlm_eap: EAP packet type response id 1 length 35
  rlm_eap: No EAP Start, assuming it's an on-going EAP conversation
  modcall[authorize]: module "eap" returns updated for request 8
    users: Matched entry j.landre at line 315
  modcall[authorize]: module "files" returns ok for request 8
modcall: group authorize returns updated for request 8

```

```

rad_check_password: Found Auth-Type EAP
auth: type "EAP"
Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 8
rlm_eap: Request found, released from the list
rlm_eap: EAP/md5
rlm_eap: processing type md5
rlm_eap: Freeing handler
modcall[authenticate]: module "eap" returns ok for request 8
modcall: group authenticate returns ok for request 8
TTLS: Got tunneled Access-Accept
rlm_eap: Freeing handler
TTLS: Freeing handler for user j.landre
modcall[authenticate]: module "eap" returns ok for request 8
modcall: group authenticate returns ok for request 8
Sending Access-Accept of id 59 to 10.0.0.189:21657
Tunnel-Type:0 = VLAN
Tunnel-Medium-Type:0 = IEEE-802
Tunnel-Private-Group-Id:0 = "100"
Tunnel-Type:0 = VLAN
Tunnel-Medium-Type:0 = IEEE-802
Tunnel-Private-Group-Id:0 = "100"
Message-Authenticator = 0x00000000000000000000000000000000
User-Name = "j.landre"
MS-MPPE-Recv-Key                                     =
0x91fed0f2ce03d1c466f86d27031fd42d3a91eca7e7a02379730e607b025ae200
MS-MPPE-Send-Key                                     =
0x977fffd071504325e010458994cb66c365ee9b1700692ba82745f153b9635ca05
EAP-Message = 0x03090004
Finished request 8
Going to the next request
Waking up in 6 seconds...
--- Walking the entire request list ---
Cleaning up request 3 ID 54 with timestamp 425f9426
Cleaning up request 4 ID 55 with timestamp 425f9426
Cleaning up request 5 ID 56 with timestamp 425f9426
Cleaning up request 6 ID 57 with timestamp 425f9426
Cleaning up request 7 ID 58 with timestamp 425f9426
Cleaning up request 8 ID 59 with timestamp 425f9426

```

EAP-TTLS fonctionne et permet le basculement de l'utilisateur dans un WVLAN particulier. Il est très sûr par la garantie du cryptage de bout en bout (tunnel).

5.2.10. Résumé des différents EAP

Voici un bref résumé de l'expérience que nous avons des EAP après les différents essais menés à l'aide de notre base de tests. Plus un EAP est simple à mettre en oeuvre, plus il possède d'étoiles.

EAP	certificats	WVLAN	Active Directory	Simplicité
EAP-LEAP	Non	Oui	Non	* * * * *
EAP-TLS	Oui	Non	Non	*
EAP-PEAP	Oui	Oui	Oui	* * *
EAP-TTLS	Oui	Oui	Non (Oui si MSCHAPv2)	* * *

EAP-LEAP est de loin le plus simple à installer, car il ne nécessite pas de certificats. Il autorise le basculement des utilisateurs dans un WVLAN. Les autres EAP nécessitent la mise en place de certificats, lourds à gérer sur les différents clients. Par contre, ils autorisent l'authentification des utilisateurs sur un Active Directory grâce à MSCHAPv2.

5.2.11. EAP et WPA

Afin d'améliorer encore la sécurité du réseau, on peut ajouter aux points d'accès la prise en charge WPA ou WPA2. Cette protection assure une identification mutuelle du client et du serveur ainsi que la rotation des clés qui deviennent de vraies clés dynamiques. Cette modification s'effectue simplement sur les points d'accès en paramétrant WPA au lieu de WEP.

Au niveau du client, il suffit de télécharger le dernier pilote de la carte Wi-Fi (souvent nécessaire) si la carte ne supporte pas WPA. Ensuite, lors de la configuration du Wi-Fi sur ce client, il faut préciser au pilote d'utiliser WPA au lieu de WEP. Le changement est très simple pour l'utilisateur.

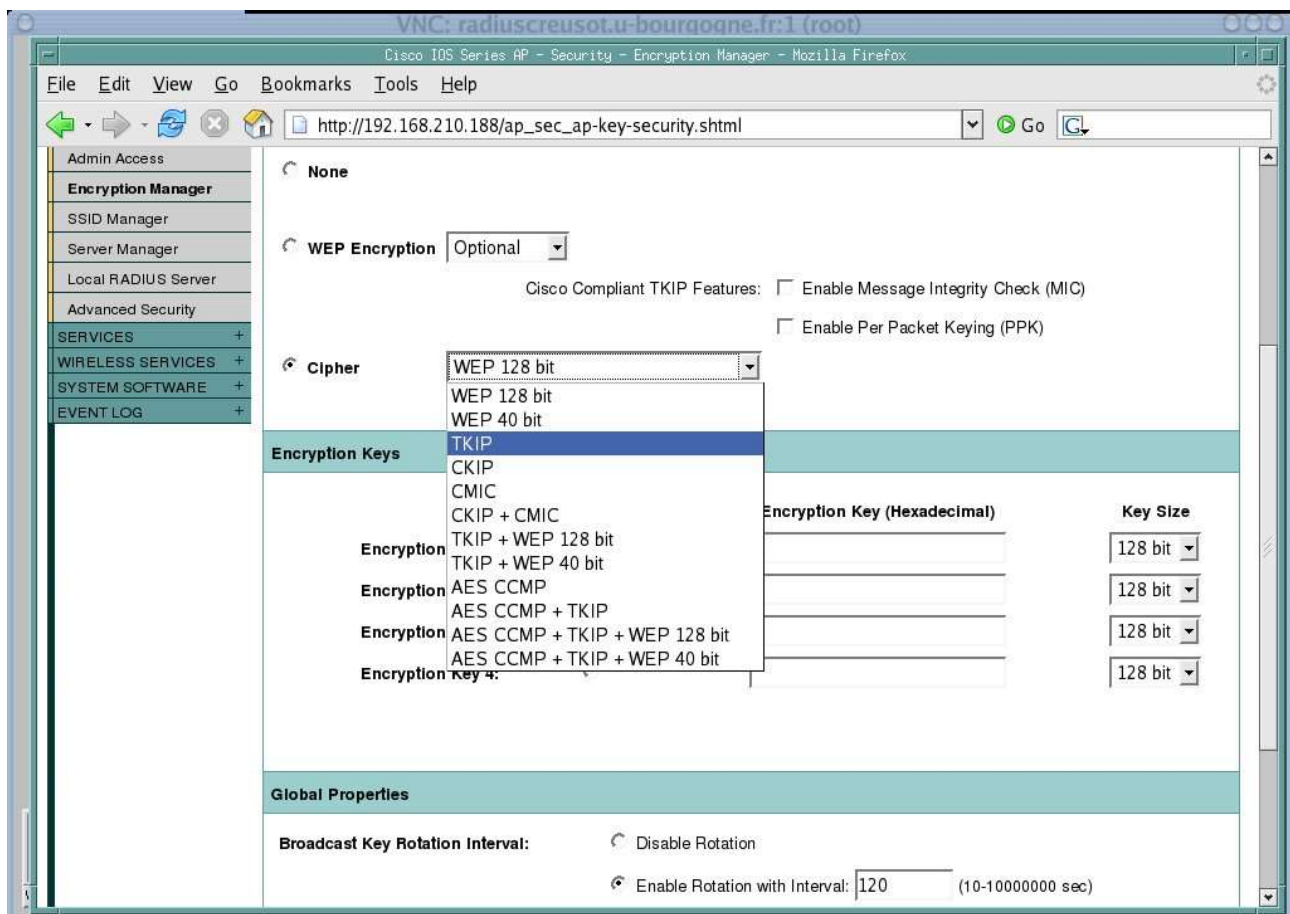


Figure 21: Mise en place de WPA et WPA2.

6. MySQL, LDAP et Active Directory

Il est possible et même recommandé d'utiliser une base d'utilisateurs dynamique dans laquelle les mots de passe sont cryptés. Dans la partie précédente, nous avons volontairement utilisé une base d'utilisateurs sous la forme d'un fichier local afin d'illustrer le comportement de freeradius. Cependant, cette base est statique et toute modification ou ajout d'utilisateur ne sera prise en compte qu'au redémarrage de freeradius (`# /usr/local/freeradius-1.0.2/sbin/rc.radiusd restart`).

Dans la partie suivante, nous allons étudier trois modes de gestion dynamique des utilisateurs: une base de données MySQL, un annuaire OpenLDAP et un annuaire Active Directory.

6.1. MySQL

MySQL est une base de données gratuite (mais pas libre) qui propose un moteur puissant et efficace de gestion de bases de données relationnelles. Avec freeradius, MySQL permet de gérer la partie authentification des utilisateurs, mais aussi les autorisations et la partie gestion de compte. Dans la suite, un utilisateur local testlocal sera utilisé pour vérifier l'installation.

MySQL gère les groupes d'utilisateurs, ainsi, on peut basculer un utilisateur dans un WLAN à l'aide des attributs en configurant un groupe d'utilisateur. C'est ce qui est fait dans cet exemple afin de montrer l'utilisation des groupes.

La procédure d'installation et de configuration est décrite ci-dessous.

a) Décompression de l'archive compressée « .tar.gz »:

```
[root@ordi ~]# cd /tmp
[root@ordi tmp]# ll
total 23624
-rw-r--r-- 1 root root 24154248 mai 11 17:06 mysql-5.0.4-beta.tar.gz
[root@ordi tmp]# tar xvzf mysql-5.0.4-beta.tar.gz
mysql-5.0.4-beta/
mysql-5.0.4-beta/bdb/
mysql-5.0.4-beta/bdb/Makefile.in
mysql-5.0.4-beta/bdb/btree/
mysql-5.0.4-beta/bdb/btree/bt_reclaim.c
mysql-5.0.4-beta/bdb/btree/bt_stat.c
mysql-5.0.4-beta/bdb/btree/bt_delete.c
mysql-5.0.4-beta/bdb/btree/bt_rec.c
mysql-5.0.4-beta/bdb/btree/bt_compare.c
mysql-5.0.4-beta/bdb/btree/bt_cursor.c
[...]
mysql-5.0.4-beta/server-tools/instance-manager/guardian.cc
mysql-5.0.4-beta/server-tools/instance-manager/guardian.h
mysql-5.0.4-beta/server-tools/instance-manager/parse_output.cc
mysql-5.0.4-beta/server-tools/instance-manager/parse_output.h
mysql-5.0.4-beta/server-tools/instance-manager/mysql_manager_error.h
[root@ordi tmp]# ll
total 23632
drwxrwxrwx 38 503 users 4096 avr 16 21:10 mysql-5.0.4-beta
-rw-r--r-- 1 root root 24154248 mai 11 17:06 mysql-5.0.4-beta.tar.gz
[root@ordi tmp]#
```

b) Déplacement dans le répertoire de MySQL.

```

[root@ordi tmp]# cd mysql-5.0.4-beta
[root@ordi mysql-5.0.4-beta]# ll
total 3016
-rw-r--r-- 1 503 users 244713 avr 16 21:02 aclocal.m4
drwxrwxrwx 48 503 users 4096 avr 16 21:10 bdb
drwxrwxrwx 2 503 users 4096 avr 16 21:10 BUILD
-rw-r--r-- 1 503 users 203998 avr 16 21:01 ChangeLog
drwxrwxrwx 2 503 users 4096 avr 16 21:10 client
drwxrwxrwx 4 503 users 4096 avr 16 21:10 cmd-line-utils
drwxrwxrwx 3 503 users 4096 avr 16 21:10 config
-rwxr-xr-x 1 503 users 43536 avr 16 21:02 config.guess
-rw-r--r-- 1 503 users 27121 avr 16 21:02 config.h.in
-rwxr-xr-x 1 503 users 31108 avr 16 21:02 config.sub
-rwxr-xr-x 1 503 users 1240927 avr 16 21:02 configure
-rw-r--r-- 1 503 users 84022 avr 16 21:01 configure.in
-rw-r--r-- 1 503 users 19099 avr 16 21:04 COPYING
drwxrwxrwx 2 503 users 4096 avr 16 21:10 debug
-rwxr-xr-x 1 503 users 14841 avr 16 21:02 depcomp
drwxrwxrwx 4 503 users 4096 avr 16 21:10 Docs
-rw-r--r-- 1 503 users 5164 avr 16 21:04 EXCEPTIONS-CLIENT
drwxrwxrwx 2 503 users 4096 avr 16 21:10 extra
drwxrwxrwx 2 503 users 4096 avr 16 21:10 heap
drwxrwxrwx 2 503 users 4096 avr 16 21:10 include
drwxrwxrwx 32 503 users 4096 avr 16 21:10 innobase
-rwxr-xr-x 1 503 users 9208 avr 16 21:02 install-sh
-rw-r--r-- 1 503 users 403578 avr 16 21:04 INSTALL-SOURCE
drwxrwxrwx 2 503 users 4096 avr 16 21:10 libmysql
drwxrwxrwx 3 503 users 4096 avr 16 21:10 libmysqld
drwxrwxrwx 2 503 users 4096 avr 16 21:10 libmysql_r
-rwxr-xr-x 1 503 users 99497 avr 16 21:01 ltconfig
-rw-r--r-- 1 503 users 183730 avr 16 21:02 ltmain.sh
-rw-r--r-- 1 503 users 3477 avr 16 21:01 Makefile.am
-rw-r--r-- 1 503 users 27798 avr 16 21:02 Makefile.in
drwxrwxrwx 2 503 users 4096 avr 16 21:10 man
-rwxr-xr-x 1 503 users 10678 avr 16 21:02 missing
-rwxr-xr-x 1 503 users 3421 avr 16 21:02 mkinstalldirs
drwxrwxrwx 2 503 users 4096 avr 16 21:10 myisam
drwxrwxrwx 2 503 users 4096 avr 16 21:10 myisammrg
drwxrwxrwx 8 503 users 4096 avr 16 21:10 mysql-test
drwxrwxrwx 2 503 users 4096 avr 16 21:10 mysys
drwxrwxrwx 9 503 users 4096 avr 16 21:10 ndb
drwxrwxrwx 3 503 users 4096 avr 16 21:10 netware
drwxrwxrwx 3 503 users 4096 avr 16 21:10 os2
drwxrwxrwx 3 503 users 4096 avr 16 21:10 pstack
-rw-r--r-- 1 503 users 1937 avr 16 21:01 README
drwxrwxrwx 2 503 users 4096 avr 16 21:10 regex
drwxrwxrwx 2 503 users 4096 avr 16 21:10 scripts
drwxrwxrwx 3 503 users 4096 avr 16 21:10 server-tools
drwxrwxrwx 4 503 users 4096 avr 16 21:10 sql
drwxrwxrwx 5 503 users 4096 avr 16 21:10 sql-bench
drwxrwxrwx 2 503 users 4096 avr 16 21:10 sql-common
drwxrwxrwx 2 503 users 4096 avr 16 21:10 SSL
drwxrwxrwx 2 503 users 4096 avr 16 21:10 strings
drwxrwxrwx 3 503 users 4096 avr 16 21:10 support-files
drwxrwxrwx 2 503 users 4096 avr 16 21:10 tests
drwxrwxrwx 2 503 users 4096 avr 16 21:10 tools
drwxrwxrwx 2 503 users 4096 avr 16 21:10 vio
drwxrwxrwx 2 503 users 4096 avr 16 21:10 zlib
[root@ordi mysql-5.0.4-beta]#

```


c) Configuration de l'installation dans « /usr/local/mysql-5.0.4 ».

```
[root@ordi mysql-5.0.4-beta]# ./configure --prefix=/usr/local/mysql-5.0.4
checking build system type... i686-pc-linux-gnu
checking host system type... i686-pc-linux-gnu
checking target system type... i686-pc-linux-gnu
checking for a BSD-compatible install... /usr/bin/install -c
checking whether build environment is sane... yes
checking for gawk... gawk
checking whether make sets $(MAKE)... yes
checking "character sets"... default: latin1, collation: latin1_swedish_ci;
compiled in: latin1 latin1 utf8
checking whether to compile national Unicode collations... yes
[...]
```

```
config.status: creating ib_config.h
config.status: ib_config.h is unchanged
config.status: executing depfiles commands
```

MySQL has a Web site at <http://www.mysql.com/> which carries details on the latest release, upcoming features, and other information to make your work or play with MySQL more productive. There you can also find information about mailing lists for MySQL discussion.

Remember to check the platform specific part of the reference manual for hints about installing MySQL on your platform. Also have a look at the files in the Docs directory.

Thank you for choosing MySQL!

```
[root@ordi mysql-5.0.4-beta]#
```

c) Lancement de la compilation.

```
[root@ordi mysql-5.0.4-beta]# make
make all-recursive
make[1]: Entering directory `/tmp/tmp/mysql-5.0.4-beta'
Making all in .
make[2]: Entering directory `/tmp/tmp/mysql-5.0.4-beta'
cd libmysql_r; make link_sources
make[3]: Entering directory `/tmp/tmp/mysql-5.0.4-beta/libmysql_r'
set -x; \
for f in `cd ../libmysql && echo *.ch`; do \
  rm -f ./$f; \
  /bin/ln -s ../libmysql/$f ./$f; \
done
++ cd ../libmysql
++ echo array.c bchange.c bcmp.c bmove.c bmove_upp.c charset.c charset-def.c
client.c client_settings.h conf_to_src.c ctype-big5.c ctype-bin.c ctype.c ctype-
cp932.c ctype-czech.c ctype-eucjpms.c ctype-euc_kr.c ctype-extra.c ctype-
gb2312.c ctype-gbk.c ctype-latin1.c ctype-mb.c ctype-simple.c ctype-sjis.c
ctype-tis620.c ctype-uca.c ctype-ucs2.c ctype-ujis.c ctype-utf8.c ctype-
win1250ch.c dbug.c default.c errmsg.c errors.c get_password.c hash.c int2str.c
is_prefix.c libmysql.c list.c llstr.c longlong2str.c manager.c md5.c mf_cache.c
mf_dirname.c mf_fn_ext.c mf_format.c mf_iocache2.c mf_iocache.c mf_loadpath.c
mf_pack.c mf_path.c mf_tempfile.c mf_unixpath.c mf_wcomp.c mulalloc.c my_alloc.c
my_compress.c my_create.c my_delete.c my_div.c my_error.c my_file.c my_fopen.c
my_fstream.c my_gethostbyname.c my_getopt.c my_getwd.c my_init.c my_lib.c
my_malloc.c my_messnc.c my_net.c my_once.c my_open.c my_port.c my_pread.c
```

```

my_pthread.c my_read.c my_realloc.c my_rename.c my_seek.c my_sleep.c my_static.c
my_static.h my_strtoll10.c my_symlink.c mysys_priv.h my_thr_init.c my_time.c
my_vsnprintf.c my_write.c net.c pack.c password.c safemalloc.c sha1.c str2int.c
strcend.c strcont.c strend.c strfill.c string.c strinstr.c strmake.c strmov.c
strnln.c strnmov.c strtoc.c strtod.c strtoll.c strtoull.c strxmouv.c strxnouv.c
thr_mutex.c typelib.c vio.c vio_priv.h viosocket.c vioss.c viossfactories.c
xml.c
[...]
g++ -O3 -DDEBUG_OFF -fno-implicit-templates -fno-exceptions -fno-rtti -rdynamic
-o mysqlmanager command.o mysqlmanager.o manager.o log.o thread_registry.o
listener.o protocol.o mysql_connection.o user_map.o messages.o commands.o
factory.o instance.o instance_map.o instance_options.o buffer.o parse.o
guardian.o parse_output.o liboptions.a libnet.a ../../vio/libvio.a ../../
mysys/libmysys.a ../../strings/libmystrings.a ../../debug/libdebug.a -lz -lpthread
-lcrypt -lnsl -lm -lpthread
make[3]: Leaving directory `/tmp/tmp/mysql-5.0.4-beta/server-tools/instance-
manager'
make[3]: Entering directory `/tmp/tmp/mysql-5.0.4-beta/server-tools'
make[3]: Rien à faire pour « all-am ».
make[3]: Leaving directory `/tmp/tmp/mysql-5.0.4-beta/server-tools'
make[2]: Leaving directory `/tmp/tmp/mysql-5.0.4-beta/server-tools'
make[1]: Leaving directory `/tmp/tmp/mysql-5.0.4-beta'
[root@ordi mysql-5.0.4-beta]#

```

c) Installation des exécutables.

```

[root@ordi mysql-5.0.4-beta]# make install
Making install in .
make[1]: Entering directory `/tmp/tmp/mysql-5.0.4-beta'
make[2]: Entering directory `/tmp/tmp/mysql-5.0.4-beta'
make[2]: Rien à faire pour « install-exec-am ».
make[2]: Rien à faire pour « install-data-am ».
make[2]: Leaving directory `/tmp/tmp/mysql-5.0.4-beta'
make[1]: Leaving directory `/tmp/tmp/mysql-5.0.4-beta'
Making install in include
make[1]: Entering directory `/tmp/tmp/mysql-5.0.4-beta/include'
make install-am
make[2]: Entering directory `/tmp/tmp/mysql-5.0.4-beta/include'
make[3]: Entering directory `/tmp/tmp/mysql-5.0.4-beta/include'
make[3]: Rien à faire pour « install-exec-am ».
test -z "/usr/local/mysql-5.0.4/include/mysql" || mkdir -p -- .
"/usr/local/mysql-5.0.4/include/mysql"
/usr/bin/install -c -m 644 'my_debug.h' '/usr/local/mysql-
5.0.4/include/mysql/my_debug.h'
[...]
make[3]: Rien à faire pour « install-data-am ».
make[3]: Leaving directory `/tmp/tmp/mysql-5.0.4-beta/server-tools'
make[2]: Leaving directory `/tmp/tmp/mysql-5.0.4-beta/server-tools'
make[1]: Leaving directory `/tmp/tmp/mysql-5.0.4-beta/server-tools'
[root@ordi mysql-5.0.4-beta]#

```

d) Fin de l'installation et configuration.

```

[root@ordi mysql-5.0.4-beta]# groupadd mysql
[root@ordi mysql-5.0.4-beta]# useradd -g mysql mysql
[root@ordi mysql-5.0.4-beta]# cp support-files/my-medium.cnf /etc/my.cnf
[root@ordi mysql-5.0.4-beta]# cd /usr/local/mysql-5.0.4/

```

```
[root@ordi mysql-5.0.4]# bin/mysql_install_db --user=mysql
Installing all prepared tables
Fill help tables
```

To start mysqld at boot time you have to copy support-files/mysql.server to the right place for your system

PLEASE REMEMBER TO SET A PASSWORD FOR THE MySQL root USER !

To do so, start the server, then issue the following commands:

```
/usr/local/mysql-5.0.4/bin/mysqladmin -u root password 'new-password'
```

```
/usr/local/mysql-5.0.4/bin/mysqladmin -u root -h ordi.u-bourgogne.fr password 'new-password'
```

See the manual for more instructions.

You can start the MySQL daemon with:

```
cd /usr/local/mysql-5.0.4 ; /usr/local/mysql-5.0.4/bin/mysqld_safe &
```

You can test the MySQL daemon with the benchmarks in the 'sql-bench' directory:

```
cd sql-bench ; perl run-all-tests
```

Please report any problems with the /usr/local/mysql-5.0.4/bin/mysqlbug script!

The latest information about MySQL is available on the web at

<http://www.mysql.com>

Support MySQL by buying support/licenses at <https://order.mysql.com>

```
[root@ordi mysql-5.0.4]# chown -R root .
```

```
[root@ordi mysql-5.0.4]# chown -R mysql var
```

```
[root@ordi mysql-5.0.4]# chgrp -R mysql .
```

```
[root@ordi mysql-5.0.4]# bin/mysqld_safe --user=mysql &
```

```
[1] 5096
```

```
[root@ordi mysql-5.0.4]# Starting mysqld daemon with databases from /usr/local/mysql-5.0.4/var
```

```
[root@ordi mysql-5.0.4]#
```

e) Lancement de MySQL (juste pour tester).

```
[root@ordi mysql-5.0.4]# mysql
```

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 1 to server version: 5.0.4-beta-log

Type 'help;' or '\h' for help. Type '\c' to clear the buffer.

```
mysql> use mysql
```

Database changed

```
mysql> show tables;
```

```
+-----+
| Tables_in_mysql |
+-----+
| columns_priv    |
| db              |
| func            |
| help_category   |
| help_keyword    |
| help_relation   |
| help_topic      |
| host            |
| proc            |
| procs_priv      |
| tables_priv     |
```

```
| time_zone |
| time_zone_leap_second |
| time_zone_name |
| time_zone_transition |
| time_zone_transition_type |
| user |
+-----+
17 rows in set (0,00 sec)
```

```
mysql> quit
[root@ordi mysql-5.0.4]#
```

f) Changement du mot de passe de l'utilisateur « root » de MySQL.

```
[root@ordi mysql-5.0.4]# bin/mysqladmin -u root flush-privileges password
"radiusmysqlrootpassword"
[root@ordi mysql-5.0.4]#
```

g) Création d'un fichier texte contenant les commandes MySQL nécessaires à la création et au remplissage des tables.

```
[root@ordi mysql-5.0.4]# vi /usr/local/freeradius-1.0.2/etc/raddb/radius.sql
```

```
drop database if exists radius;
```

```
create database radius;
```

```
use radius;
```

```
#####
# db_mysql.sql                                rlm_sql - FreeRADIUS SQL Module      #
#                                                                                        #
# Database schema for MySQL rlm_sql module                                          #
#                                                                                        #
# To load:                                                                            #
# mysql -uroot -prootpass radius < db_mysql.sql                                  #
#                                                                                        #
# Mike Machado <mike@innercite.com>                                                #
#####
#
# Table structure for table 'radacct'
```

```
CREATE TABLE radacct (
  RadAcctId bigint(21) NOT NULL auto_increment,
  AcctSessionId varchar(32) NOT NULL default '',
  AcctUniqueId varchar(32) NOT NULL default '',
  UserName varchar(64) NOT NULL default '',
  Realm varchar(64) default '',
  NASIPAddress varchar(15) NOT NULL default '',
  NASPortId int(12) default NULL,
  NASPortType varchar(32) default NULL,
  AcctStartTime datetime NOT NULL default '0000-00-00 00:00:00',
  AcctStopTime datetime NOT NULL default '0000-00-00 00:00:00',
  AcctSessionTime int(12) default NULL,
  AcctAuthentic varchar(32) default NULL,
  ConnectInfo_start varchar(32) default NULL,
```

```

ConnectInfo_stop varchar(32) default NULL,
AcctInputOctets bigint(12) default NULL,
AcctOutputOctets bigint(12) default NULL,
CalledStationId varchar(50) NOT NULL default '',
CallingStationId varchar(50) NOT NULL default '',
AcctTerminateCause varchar(32) NOT NULL default '',
ServiceType varchar(32) default NULL,
FramedProtocol varchar(32) default NULL,
FramedIPAddress varchar(15) NOT NULL default '',
AcctStartDelay int(12) default NULL,
AcctStopDelay int(12) default NULL,
PRIMARY KEY (RadAcctId),
KEY UserName (UserName),
KEY FramedIPAddress (FramedIPAddress),
KEY AcctSessionId (AcctSessionId),
KEY AcctUniqueId (AcctUniqueId),
KEY AcctStartTime (AcctStartTime),
KEY AcctStopTime (AcctStopTime),
KEY NASIPAddress (NASIPAddress)
) ;

```

```

#
# Table structure for table 'radcheck'
#

```

```

CREATE TABLE radcheck (
  id int(11) unsigned NOT NULL auto_increment,
  UserName varchar(64) NOT NULL default '',
  Attribute varchar(32) NOT NULL default '',
  op char(2) NOT NULL DEFAULT '==',
  Value varchar(253) NOT NULL default '',
  PRIMARY KEY (id),
  KEY UserName (UserName(32))
) ;

```

```

#
# Table structure for table 'radgroupcheck'
#

```

```

CREATE TABLE radgroupcheck (
  id int(11) unsigned NOT NULL auto_increment,
  GroupName varchar(64) NOT NULL default '',
  Attribute varchar(32) NOT NULL default '',
  op char(2) NOT NULL DEFAULT '==',
  Value varchar(253) NOT NULL default '',
  PRIMARY KEY (id),
  KEY GroupName (GroupName(32))
) ;

```

```

#
# Table structure for table 'radgroupreply'
#

```

```

CREATE TABLE radgroupreply (
  id int(11) unsigned NOT NULL auto_increment,
  GroupName varchar(64) NOT NULL default '',
  Attribute varchar(32) NOT NULL default '',
  op char(2) NOT NULL DEFAULT '=',
  Value varchar(253) NOT NULL default '',
  prio int unsigned NOT NULL default '0',
  PRIMARY KEY (id),

```

```

    KEY GroupName (GroupName(32))
) ;

#
# Table structure for table 'radreply'
#

CREATE TABLE radreply (
  id int(11) unsigned NOT NULL auto_increment,
  UserName varchar(64) NOT NULL default '',
  Attribute varchar(32) NOT NULL default '',
  op char(2) NOT NULL DEFAULT '=',
  Value varchar(253) NOT NULL default '',
  PRIMARY KEY (id),
  KEY UserName (UserName(32))
) ;

#
# Table structure for table 'usergroup'
#

CREATE TABLE usergroup (
  id int(11) unsigned NOT NULL auto_increment,
  UserName varchar(64) NOT NULL default '',
  GroupName varchar(64) NOT NULL default '',
  PRIMARY KEY (id),
  KEY UserName (UserName(32))
) ;

#####
#
# The next two tables are commented out because they are not
# currently used in the server.
#

#
# Table structure for table 'dictionary'
#
#CREATE TABLE dictionary (
# id int(10) DEFAULT '0' NOT NULL auto_increment,
# Type varchar(30),
# Attribute varchar(64),
# Value varchar(64),
# Format varchar(20),
# Vendor varchar(32),
# PRIMARY KEY (id)
#);

#
# Table structure for table 'nas'
#
#CREATE TABLE nas (
# id int(10) DEFAULT '0' NOT NULL auto_increment,
# nasname varchar(128),
# shortname varchar(32),
# ipaddr varchar(15),
# type varchar(30),
# ports int(5),
# secret varchar(60),
# community varchar(50),

```

```

# snmp varchar(10),
# PRIMARY KEY (id)
#);

# remplissage de la table --- par jerome landre ---

insert into usergroup values (1,'etudiant0','etudiant');
insert into usergroup values (2,'jerome','personnel');
insert into usergroup values (3,'adm','admin');

insert into radcheck values (1,'etudiant0','Password','=','etudiant00');
insert into radcheck values (2,'jerome','Password','=','jerome00');
insert into radcheck values (3,'adm','Password','=','admin00');
insert into radcheck values (4,'testlocal','Password','=','local');

insert into radgroupreply values (1, 'personnel', 'Tunnel-Type', ':=', '13', 0);
insert into radgroupreply values (2, 'personnel', 'Tunnel-Medium-Type', ':=', '6', 0);
insert into radgroupreply values (3, 'personnel', 'Tunnel-Private-Group-Id', ':=', '100', 0);
insert into radgroupreply values (4, 'etudiant', 'Tunnel-Type', ':=', '13', 0);
insert into radgroupreply values (5, 'etudiant', 'Tunnel-Medium-Type', ':=', '6', 0);
insert into radgroupreply values (6, 'etudiant', 'Tunnel-Private-Group-Id', ':=', '110', 0);
insert into radgroupreply values (7, 'admin', 'Tunnel-Type', ':=', '13', 0);
insert into radgroupreply values (8, 'admin', 'Tunnel-Medium-Type', ':=', '6', 0);
insert into radgroupreply values (9, 'admin', 'Tunnel-Private-Group-Id', ':=', '1', 0);

insert into radgroupcheck values (1,'etudiant','Auth-Type',':=','eap');
insert into radgroupcheck values (2,'personnel','Auth-Type',':=','eap');
insert into radgroupcheck values (3,'admin','Auth-Type',':=','eap');
insert into radgroupcheck values (4,'testlocal','Auth-Type',':=','local');

# Definition de l'utilisateur radiusmysql et de ses droits.

use mysql;

delete from user where user='radiusmysql';

insert into user values ('localhost','radiusmysql',password('radiusmysqlpw'),
'Y','Y','Y','Y','Y','Y','N','N','N','N','N','N','N','N','N','N','N','N','N','N',
'N','N','N','','',' ',0,0,0);

delete from db where db='radius';

insert into db values ('localhost','radius','radiusmysql','Y','Y','Y','Y',
'Y','Y','N','N','N','N','N','N','N','N');

[root@ordi mysql-5.0.4]#

```

h) Lancement du fichier script MySQL précédent.

```

[root@ordi mysql-5.0.4]# cd /usr/local/freeradius-1.0.2/etc/raddb
[root@ordi raddb]# mysql -u root -p < radius.sql
Enter password:
[root@ordi raddb]#

```

i) Configuration de « radiusd.conf ».

```
[root@ordi raddb]# vi radiusd.conf
[...]
authorize {
[...]
    eap

    #
    # Read the 'users' file
    files

    #
    # Look in an SQL database. The schema of the database
    # is meant to mirror the "users" file.
    #
    # See "Authorization Queries" in sql.conf
    sql
[...]
accounting {
[...]
    #
    # Log traffic to an SQL database.
    #
    # See "Accounting queries" in sql.conf
    sql
[...]
}
```

j) Configuration de « sql.conf ».

```
[root@ordi ~]# cd /usr/local/freeradius-1.0.2/etc/raddb/
[root@ordi raddb]# vi sql.conf
#
# Configuration for the SQL module, when using MySQL.
#
# The database schema is available at:
#
#   src/radiusd/src/modules/rlm_sql/drivers/rlm_sql_mysql/db_mysql.sql
#
# If you are using PostgreSQL, please use 'postgresql.conf', instead.
# If you are using Oracle, please use 'oracle.conf', instead.
# If you are using MS-SQL, please use 'mssql.conf', instead.
#
# $Id: sql.conf,v 1.41.2.1 2004/06/10 00:45:01 phampson Exp $
#
sql {

    # Database type
    # Current supported are: rlm_sql_mysql, rlm_sql_postgresql,
    # rlm_sql_iodbc, rlm_sql_oracle, rlm_sql_unixodbc, rlm_sql_freetds
    driver = "rlm_sql_mysql"

    # Connect info
    server = "localhost"
    login = "radiusmysql"
    password = "radiusmysqlpw"

    # Database table configuration
}
```



```
radius_db = "radius"
[...]
```

k) Test de fonctionnement.

```
[root@ordi raddb]# cd ../../
[root@ordi freeradius-1.0.2]# /usr/local/freeradius-1.0.2/sbin/radiusd -X
Starting - reading configuration files ...
reread_config: reading radiusd.conf
Config: including file: /usr/local/freeradius-1.0.2/etc/raddb/proxy.conf
Config: including file: /usr/local/freeradius-1.0.2/etc/raddb/clients.conf
Config: including file: /usr/local/freeradius-1.0.2/etc/raddb/snmp.conf
Config: including file: /usr/local/freeradius-1.0.2/etc/raddb/eap.conf
Config: including file: /usr/local/freeradius-1.0.2/etc/raddb/sql.conf
main: prefix = "/usr/local/freeradius-1.0.2"
main: localstatedir = "/usr/local/freeradius-1.0.2/var"
main: logdir = "/usr/local/freeradius-1.0.2/var/log/radius"
main: libdir = "/usr/local/freeradius-1.0.2/lib"
main: radacctdir = "/usr/local/freeradius-1.0.2/var/log/radius/radacct"
main: hostname_lookups = no
main: max_request_time = 30
main: cleanup_delay = 5
main: max_requests = 1024
main: delete_blocked_requests = 0
main: port = 0
main: allow_core_dumps = no
main: log_stripped_names = no
main: log_file = "/usr/local/freeradius-1.0.2/var/log/radius/radius.log"
main: log_auth = no
main: log_auth_badpass = no
main: log_auth_goodpass = no
main: pidfile = "/usr/local/freeradius-1.0.2/var/run/radiusd/radiusd.pid"
main: user = "(null)"
main: group = "(null)"
main: usercollide = no
main: lower_user = "no"
main: lower_pass = "no"
main: nospace_user = "no"
main: nospace_pass = "no"
main: checkrad = "/usr/local/freeradius-1.0.2/sbin/checkrad"
main: proxy_requests = yes
proxy: retry_delay = 5
proxy: retry_count = 3
proxy: synchronous = no
proxy: default_fallback = yes
proxy: dead_time = 120
proxy: post_proxy_authorize = yes
proxy: wake_all_if_all_dead = no
security: max_attributes = 200
security: reject_delay = 1
security: status_server = no
main: debug_level = 0
read_config_files: reading dictionary
read_config_files: reading naslist
Using deprecated naslist file. Support for this will go away soon.
read_config_files: reading clients
read_config_files: reading realms
radiusd: entering modules setup
Module: Library search path is /usr/local/freeradius-1.0.2/lib
```

```

Module: Loaded exec
exec: wait = yes
exec: program = "(null)"
exec: input_pairs = "request"
exec: output_pairs = "(null)"
exec: packet_type = "(null)"
rlm_exec: Wait=yes but no output defined. Did you mean output=none?
Module: Instantiated exec (exec)
Module: Loaded expr
Module: Instantiated expr (expr)
Module: Loaded PAP
pap: encryption_scheme = "crypt"
Module: Instantiated pap (pap)
Module: Loaded CHAP
Module: Instantiated chap (chap)
Module: Loaded MS-CHAP
mschap: use_mppe = yes
mschap: require_encryption = no
mschap: require_strong = no
mschap: with_ntdomain_hack = no
mschap: passwd = "(null)"
mschap: authtype = "MS-CHAP"
mschap: ntlm_auth = "(null)"
Module: Instantiated mschap (mschap)
Module: Loaded System
unix: cache = no
unix: passwd = "(null)"
unix: shadow = "(null)"
unix: group = "(null)"
unix: radwtmp = "/usr/local/freeradius-1.0.2/var/log/radius/radwtmp"
unix: usegroup = no
unix: cache_reload = 600
Module: Instantiated unix (unix)
Module: Loaded eap
eap: default_eap_type = "md5"
eap: timer_expire = 60
eap: ignore_unknown_eap_types = no
eap: cisco_accounting_username_bug = no
rlm_eap: Loaded and initialized type md5
rlm_eap: Loaded and initialized type leap
gtc: challenge = "Password: "
gtc: auth_type = "PAP"
rlm_eap: Loaded and initialized type gtc
mschapv2: with_ntdomain_hack = no
rlm_eap: Loaded and initialized type mschapv2
Module: Instantiated eap (eap)
Module: Loaded preprocess
preprocess: huntgroups = "/usr/local/freeradius-1.0.2/etc/raddb/huntgroups"
preprocess: hints = "/usr/local/freeradius-1.0.2/etc/raddb/hints"
preprocess: with_ascend_hack = no
preprocess: ascend_channels_per_line = 23
preprocess: with_ntdomain_hack = no
preprocess: with_specialix_jetstream_hack = no
preprocess: with_cisco_vsa_hack = no
Module: Instantiated preprocess (preprocess)
Module: Loaded realm
realm: format = "suffix"
realm: delimiter = "@"
realm: ignore_default = no
realm: ignore_null = no
Module: Instantiated realm (suffix)

```

```

Module: Loaded SQL
sql: driver = "rlm_sql_mysql"
sql: server = "localhost"
sql: port = ""
sql: login = "radiusmysql"
sql: password = "radiusmysqlpw"
sql: radius_db = "radius"
sql: acct_table = "radacct"
sql: acct_table2 = "radacct"
sql: authcheck_table = "radcheck"
sql: authreply_table = "radreply"
sql: groupcheck_table = "radgroupcheck"
sql: groupreply_table = "radgroupreply"
sql: usergroup_table = "usergroup"
sql: nas_table = "nas"
sql: dict_table = "dictionary"
sql: sqltrace = no
sql: sqltracefile = "/usr/local/freeradius-1.0.2/var/log/radius/sqltrace.sql"
sql: readclients = no
sql: deletestalesessions = yes
sql: num_sql_socks = 5
sql: sql_user_name = "%{User-Name}"
sql: default_user_profile = ""
sql: query_on_not_found = no
sql: authorize_check_query = "SELECT id,UserName,Attribute,Value,op FROM
radcheck WHERE Username = '%{SQL-User-Name}' ORDER BY id"
sql: authorize_reply_query = "SELECT id,UserName,Attribute,Value,op FROM
radreply WHERE Username = '%{SQL-User-Name}' ORDER BY id"
sql: authorize_group_check_query = "SELECT
radgroupcheck.id,radgroupcheck.GroupName,radgroupcheck.Attribute,radgroupcheck.V
alue,radgroupcheck.op FROM radgroupcheck,usergroup WHERE usergroup.Username =
'%{SQL-User-Name}' AND usergroup.GroupName = radgroupcheck.GroupName ORDER BY
radgroupcheck.id"
sql: authorize_group_reply_query = "SELECT
radgroupreply.id,radgroupreply.GroupName,radgroupreply.Attribute,radgroupreply.V
alue,radgroupreply.op FROM radgroupreply,usergroup WHERE usergroup.Username =
'%{SQL-User-Name}' AND usergroup.GroupName = radgroupreply.GroupName ORDER BY
radgroupreply.id"
sql: accounting_onoff_query = "UPDATE radacct SET AcctStopTime='%S',
AcctSessionTime=unix_timestamp('%S') - unix_timestamp(AcctStartTime),
AcctTerminateCause='%{Acct-Terminate-Cause}', AcctStopDelay = '%{Acct-Delay-
Time}' WHERE AcctSessionTime=0 AND AcctStopTime=0 AND NASIPAddress= '%{NAS-IP-
Address}' AND AcctStartTime <= '%S'"
sql: accounting_update_query = "UPDATE radacct ? SET FramedIPAddress = '%
{Framed-IP-Address}', ? AcctSessionTime = '%{Acct-Session-Time}', ?
AcctInputOctets = '%{Acct-Input-Octets}', ? AcctOutputOctets = '%{Acct-Output-
Octets}' ? WHERE AcctSessionId = '%{Acct-Session-Id}' ? AND UserName = '%{SQL-
User-Name}' ? AND NASIPAddress= '%{NAS-IP-Address}'"
sql: accounting_update_query_alt = "INSERT into radacct (AcctSessionId,
AcctUniqueId, UserName, Realm, NASIPAddress, NASPortId, NASPortType,
AcctStartTime, AcctSessionTime, AcctAuthentic, ConnectInfo_start,
AcctInputOctets, AcctOutputOctets, CalledStationId, CallingStationId,
ServiceType, FramedProtocol, FramedIPAddress, AcctStartDelay) values('%{Acct-
Session-Id}', '%{Acct-Unique-Session-Id}', '%{SQL-User-Name}', '%
{NAS-IP-Address}', '%{NAS-Port}', '%{NAS-Port-Type}', DATE_SUB('%S',INTERVAL (%
{Acct-Session-Time:-0} + %{Acct-Delay-Time:-0}) SECOND), '%{Acct-Session-Time}',
'%{Acct-Authentic}', '', '%{Acct-Input-Octets}', '%{Acct-Output-Octets}', '%
{Called-Station-Id}', '%{Calling-Station-Id}', '%{Service-Type}', '%{Framed-
Protocol}', '%{Framed-IP-Address}', '0')"
sql: accounting_start_query = "INSERT into radacct (AcctSessionId,
AcctUniqueId, UserName, Realm, NASIPAddress, NASPortId, NASPortType,

```

```

AcctStartTime, AcctStopTime, AcctSessionTime, AcctAuthentic, ConnectInfo_start,
ConnectInfo_stop, AcctInputOctets, AcctOutputOctets, CalledStationId,
CallingStationId, AcctTerminateCause, ServiceType, FramedProtocol,
FramedIPAddress, AcctStartDelay, AcctStopDelay) values('%{Acct-Session-Id}', '%
{Acct-Unique-Session-Id}', '%{SQL-User-Name}', '%{Realm}', '%{NAS-IP-Address}',
'%{NAS-Port}', '%{NAS-Port-Type}', '%S', '0', '0', '%{Acct-Authentic}', '%
{Connect-Info}', '', '0', '0', '%{Called-Station-Id}', '%{Calling-Station-Id}',
'', '%{Service-Type}', '%{Framed-Protocol}', '%{Framed-IP-Address}', '%{Acct-
Delay-Time}', '0')"
sql: accounting_start_query_alt = "UPDATE radacct SET AcctStartTime = '%S',
AcctStartDelay = '%{Acct-Delay-Time}', ConnectInfo_start = '%{Connect-Info}'
WHERE AcctSessionId = '%{Acct-Session-Id}' AND UserName = '%{SQL-User-Name}' AND
NASIPAddress = '%{NAS-IP-Address}'"
sql: accounting_stop_query = "UPDATE radacct SET AcctStopTime = '%S',
AcctSessionTime = '%{Acct-Session-Time}', AcctInputOctets = '%{Acct-Input-
Octets}', AcctOutputOctets = '%{Acct-Output-Octets}', AcctTerminateCause = '%
{Acct-Terminate-Cause}', AcctStopDelay = '%{Acct-Delay-Time}', ConnectInfo_stop
= '%{Connect-Info}' WHERE AcctSessionId = '%{Acct-Session-Id}' AND UserName = '%
{SQL-User-Name}' AND NASIPAddress = '%{NAS-IP-Address}'"
sql: accounting_stop_query_alt = "INSERT into radacct (AcctSessionId,
AcctUniqueId, UserName, Realm, NASIPAddress, NASPortId, NASPortType,
AcctStartTime, AcctStopTime, AcctSessionTime, AcctAuthentic, ConnectInfo_start,
ConnectInfo_stop, AcctInputOctets, AcctOutputOctets, CalledStationId,
CallingStationId, AcctTerminateCause, ServiceType, FramedProtocol,
FramedIPAddress, AcctStartDelay, AcctStopDelay) values('%{Acct-Session-Id}', '%
{Acct-Unique-Session-Id}', '%{SQL-User-Name}', '%{Realm}', '%{NAS-IP-Address}',
'%{NAS-Port}', '%{NAS-Port-Type}', DATE_SUB('%S', INTERVAL (%{Acct-Session-
Time:-0} + %{Acct-Delay-Time:-0}) SECOND), '%S', '%{Acct-Session-Time}', '%
{Acct-Authentic}', '', '%{Connect-Info}', '%{Acct-Input-Octets}', '%{Acct-
Output-Octets}', '%{Called-Station-Id}', '%{Calling-Station-Id}', '%{Acct-
Terminate-Cause}', '%{Service-Type}', '%{Framed-Protocol}', '%{Framed-IP-
Address}', '0', '%{Acct-Delay-Time}')"
sql: group_membership_query = "SELECT GroupName FROM usergroup WHERE
UserName='%{SQL-User-Name}'"
sql: connect_failure_retry_delay = 60
sql: simul_count_query = ""
sql: simul_verify_query = "SELECT RadAcctId, AcctSessionId, UserName,
NASIPAddress, NASPortId, FramedIPAddress, CallingStationId, FramedProtocol FROM
radacct WHERE UserName='%{SQL-User-Name}' AND AcctStopTime = 0"
sql: postauth_table = "radpostauth"
sql: postauth_query = "INSERT into radpostauth (id, user, pass, reply, date)
values ('', '%{User-Name}', '%{User-Password:-Chap-Password}', '%{reply:Packet-
Type}', NOW())"
sql: safe-characters =
"@abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789.-_: /"
rlm_sql (sql): Driver rlm_sql_mysql (module rlm_sql_mysql) loaded and linked
rlm_sql (sql): Attempting to connect to radiusmysql@localhost:/radius
rlm_sql (sql): starting 0
rlm_sql (sql): Attempting to connect rlm_sql_mysql #0
rlm_sql_mysql: Starting connect to MySQL server for #0
rlm_sql (sql): Connected new DB handle, #0
rlm_sql (sql): starting 1
rlm_sql (sql): Attempting to connect rlm_sql_mysql #1
rlm_sql_mysql: Starting connect to MySQL server for #1
rlm_sql (sql): Connected new DB handle, #1
rlm_sql (sql): starting 2
rlm_sql (sql): Attempting to connect rlm_sql_mysql #2
rlm_sql_mysql: Starting connect to MySQL server for #2
rlm_sql (sql): Connected new DB handle, #2
rlm_sql (sql): starting 3
rlm_sql (sql): Attempting to connect rlm_sql_mysql #3

```

```

rlm_sql_mysql: Starting connect to MySQL server for #3
rlm_sql (sql): Connected new DB handle, #3
rlm_sql (sql): starting 4
rlm_sql (sql): Attempting to connect rlm_sql_mysql #4
rlm_sql_mysql: Starting connect to MySQL server for #4
rlm_sql (sql): Connected new DB handle, #4
Module: Instantiated sql (sql)
Module: Loaded Acct-Unique-Session-Id
  acct_unique: key = "User-Name, Acct-Session-Id, NAS-IP-Address, Client-IP-Address, NAS-Port"
Module: Instantiated acct_unique (acct_unique)
Module: Loaded files
  files: usersfile = "/usr/local/freeradius-1.0.2/etc/raddb/users"
  files: acctusersfile = "/usr/local/freeradius-1.0.2/etc/raddb/acct_users"
  files: preproxy_usersfile = "/usr/local/freeradius-1.0.2/etc/raddb/preproxy_users"
  files: compat = "no"
Module: Instantiated files (files)
Module: Loaded detail
  detail: detailfile = "/usr/local/freeradius-1.0.2/var/log/radius/radacct/{Client-IP-Address}/detail-%Y%m%d"
  detail: detailperm = 384
  detail: dirperm = 493
  detail: locking = no
Module: Instantiated detail (detail)
Module: Loaded radutmp
  radutmp: filename = "/usr/local/freeradius-1.0.2/var/log/radius/radutmp"
  radutmp: username = "%{User-Name}"
  radutmp: case_sensitive = yes
  radutmp: check_with_nas = yes
  radutmp: perm = 384
  radutmp: callerid = yes
Module: Instantiated radutmp (radutmp)
Listening on authentication *:1812
Listening on accounting *:1813
Listening on proxy *:1814
Ready to process requests.

```

l) Interrogation de la base MySQL.

```

[root@ordi ~]# /usr/local/freeradius-1.0.2/bin/radtest testlocal local localhost
1812 secretpartage
Sending Access-Request of id 113 to 127.0.0.1:1812
  User-Name = "testlocal"
  User-Password = "local"
  NAS-IP-Address = ordi.u-bourgogne.fr
  NAS-Port = 1812
rad_recv: Access-Accept packet from host 127.0.0.1:1812, id=113, length=20
[root@ordi ~]#

```

m) Résultat du test.

```

rad_recv: Access-Request packet from host 127.0.0.1:32777, id=113, length=61
  User-Name = "testlocal"
  User-Password = "local"
  NAS-IP-Address = 255.255.255.255
  NAS-Port = 1812

```

```

Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 0
modcall[authorize]: module "preprocess" returns ok for request 0
modcall[authorize]: module "chap" returns noop for request 0
modcall[authorize]: module "mschap" returns noop for request 0
    rlm_realm: No '@' in User-Name = "testlocal", looking up realm NULL
    rlm_realm: No such realm "NULL"
modcall[authorize]: module "suffix" returns noop for request 0
    rlm_eap: No EAP-Message, not doing EAP
modcall[authorize]: module "eap" returns noop for request 0
radius_xlat: 'testlocal'
rlm_sql (sql): sql_set_user escaped user --> 'testlocal'
radius_xlat: 'SELECT id,UserName,Attribute,Value,op FROM radcheck WHERE
Username = 'testlocal' ORDER BY id'
rlm_sql (sql): Reserving sql socket id: 4
radius_xlat: 'SELECT
radgroupcheck.id,radgroupcheck.GroupName,radgroupcheck.Attribute,radgroupcheck.V
alue,radgroupcheck.op FROM radgroupcheck,usergroup WHERE usergroup.Username =
'testlocal' AND usergroup.GroupName = radgroupcheck.GroupName ORDER BY
radgroupcheck.id'
radius_xlat: 'SELECT id,UserName,Attribute,Value,op FROM radreply WHERE
Username = 'testlocal' ORDER BY id'
radius_xlat: 'SELECT
radgroupreply.id,radgroupreply.GroupName,radgroupreply.Attribute,radgroupreply.V
alue,radgroupreply.op FROM radgroupreply,usergroup WHERE usergroup.Username =
'testlocal' AND usergroup.GroupName = radgroupreply.GroupName ORDER BY
radgroupreply.id'
rlm_sql (sql): Released sql socket id: 4
modcall[authorize]: module "sql" returns ok for request 0
modcall: group authorize returns ok for request 0
auth: type Local
auth: user supplied User-Password matches local User-Password
Sending Access-Accept of id 113 to 127.0.0.1:32777
Finished request 0
Going to the next request
--- Walking the entire request list ---
Waking up in 6 seconds...

```

n) Installation de MySQL en tant que service du système.

```

[root@ordi ~]# cd /usr/local/mysql-5.0.4/share/mysql
[root@ordi mysql]# cp mysql.server /etc/rc.d/init.d
[root@ordi mysql]# cd /etc/rc.d/init.d/
[root@ordi init.d]# chkconfig --add mysql.server
[root@ordi init.d]# ll
total 632
-rwxr-xr-x 1 root root 1128 août 9 2004 acpid
-rwxr-xr-x 1 root root 834 sep 28 2004 anacron
-rwxr-xr-x 1 root root 1429 jun 22 2004 apmd
-rwxr-xr-x 1 root root 1176 mar 8 20:45 atd
[...]
-rwxr-xr-x 1 root root 8548 mai 17 17:58 mysql.server
[...]
[root@ordi init.d]# chkconfig --level 345 mysql.server on
[root@ordi init.d]# cd
[root@ordi ~]#

```

o) Démarrage puis arrêt de MySQL (juste pour tester le service).

```
[root@ordi ~]# service mysql.server start
[root@ordi ~]# service mysql.server stop
Killing mysql.server with pid 4142
Wait for mysql.server to exit. done
[root@ordi ~]#
```

Le test de fonctionnement proposé est local avec l'utilisateur « testlocal » qui ne renvoie pas d'attributs. Il suffit de tester avec les autres comptes pour voir le résultat en EAP. L'avantage de MySQL est que la modification de la base des utilisateurs est dynamique, il n'y a pas besoin de relancer le processus « radiusd » à chaque fois.

6.2. OpenLDAP

OpenLDAP est un annuaire LDAP (*Lightweight Directory Access Protocol*) libre, ouvert et gratuit téléchargeable sur Internet à l'adresse « <http://www.openldap.org> ». Il permet de mettre en place un annuaire pour le système d'information d'un établissement contenant de nombreuses informations sur les utilisateurs, les ordinateurs, les bâtiments, les enseignements... Pour simplifier, l'annuaire est une base de données hiérarchique organisée sous la forme d'un arbre. Freeradius va authentifier les utilisateurs à l'aide de requêtes LDAP qui vont permettre à la base d'utilisateurs d'être dynamique, il n'y aura donc pas besoin de relancer « radius » à chaque modification de la base des utilisateurs.

a) Décompression de l'archive compressée « .tar.gz »:

```
[root@ordi tmp]# ll
total 2576
-rw-rw-r-- 1 jeje jeje 2626629 mai 11 18:06 openldap-stable-20050429.tgz
[root@ordi tmp]# tar xvzf openldap-stable-20050429.tgz
openldap-2.2.26/
openldap-2.2.26/doc/
openldap-2.2.26/doc/man/
openldap-2.2.26/doc/man/Makefile.in
openldap-2.2.26/doc/man/man1/
openldap-2.2.26/doc/man/man1/Makefile.in
openldap-2.2.26/doc/man/man1/ldapcompare.1
[...]
openldap-2.2.26/tests/scripts/test018-syncreplication-persist
openldap-2.2.26/tests/scripts/test019-syncreplication-cascade
openldap-2.2.26/tests/scripts/test020-proxycache
openldap-2.2.26/tests/scripts/test021-certificate
[root@ordi tmp]#
```

b) Déplacement dans le répertoire OpenLDAP.

```
[root@ordi tmp]# cd openldap-2.2.26/
[root@ordi openldap-2.2.26]# ll
total 1000
-rw-rw-r-- 1 2000 2000 2570 jan 20 18:00 acconfig.h
-rw-rw-r-- 1 2000 2000 650 jan 20 18:00 acinclude.m4
-rw-rw-r-- 1 2000 2000 129642 jan 20 18:00 aclocal.m4
-rw-rw-r-- 1 2000 2000 3439 jan 20 18:00 ANNOUNCEMENT
drwxrwxr-x 2 2000 2000 4096 avr 28 05:05 build
-rw-rw-r-- 1 2000 2000 14981 avr 28 03:59 CHANGES
```

```

drwxrwxr-x 3 2000 2000 4096 avr 28 05:05 clients
-rwxrwxr-x 1 2000 2000 633977 mar 23 00:20 configure
-rw-rw-r-- 1 2000 2000 75861 mar 14 18:06 configure.in
drwxrwxr-x 6 2000 2000 4096 avr 28 05:05 contrib
-rw-rw-r-- 1 2000 2000 2241 jan 20 17:56 COPYRIGHT
drwxrwxr-x 8 2000 2000 4096 avr 28 05:05 doc
drwxrwxr-x 3 2000 2000 4096 avr 28 05:05 include
-rw-rw-r-- 1 2000 2000 4425 jan 20 18:00 INSTALL
drwxrwxr-x 8 2000 2000 4096 avr 28 05:05 libraries
-rw-rw-r-- 1 2000 2000 2214 déc 1 2003 LICENSE
-rw-rw-r-- 1 2000 2000 993 jan 20 18:00 Makefile.in
-rw-rw-r-- 1 2000 2000 3547 fév 21 19:41 README
drwxrwxr-x 4 2000 2000 4096 avr 28 05:05 servers
drwxrwxr-x 5 2000 2000 4096 avr 28 05:05 tests
[root@ordi openldap-2.2.26]#

```

c) Configuration de l'installation dans « /usr/local/openldap-2.2.6 ».

```

[root@ordi openldap-2.2.26]# ./configure --prefix=/usr/local/openldap-2.2.6 --
enable-debug --enable-crypt --enable-bdb --enable-ldbm --with-ldbm-api=berkeley
--enable-monitor --enable-local --enable-cldap --disable-rlookups --with-tls --
with-cyrus-sasl --enable-passwd --enable-shell --enable-cleartext --enable-
spasswd --enable-meta --enable-ldap --enable-rewrite
Copyright 1998-2005 The OpenLDAP Foundation. All rights reserved.
Restrictions apply, see COPYRIGHT and LICENSE files.
Configuring OpenLDAP 2.2.26-Release ...
checking host system type... i686-pc-linux-gnu
checking target system type... i686-pc-linux-gnu
checking build system type... i686-pc-linux-gnu
checking for a BSD compatible install... /usr/bin/install -c
checking whether build environment is sane... yes
[...]
creating tests/run
creating tests/progs/Makefile
creating include/portable.h
include/portable.h is unchanged
creating include/ldap_features.h
include/ldap_features.h is unchanged
creating include/lber_types.h
include/lber_types.h is unchanged
Please run "make depend" to build dependencies
[root@ordi openldap-2.2.26]#

```

d) Construction des dépendances.

```

[root@ordi openldap-2.2.26]# make depend
Making depend in /tmp/tmp/tmp/openldap-2.2.26
Entering subdirectory include
make[1]: Entering directory `/tmp/tmp/tmp/openldap-2.2.26/include'
make[1]: Rien à faire pour « depend ».
make[1]: Leaving directory `/tmp/tmp/tmp/openldap-2.2.26/include'

Entering subdirectory libraries
make[1]: Entering directory `/tmp/tmp/tmp/openldap-2.2.26/libraries'
Making depend in /tmp/tmp/tmp/openldap-2.2.26/libraries
Entering subdirectory liblutil
make[2]: Entering directory `/tmp/tmp/tmp/openldap-2.2.26/libraries/liblutil'

```



```

../../build/mkdep -d "." -c "cc" -m "-M" -I../../include -I../../include
base64.c csn.c entropy.c sasl.c signal.c hash.c passfile.c md5.c passwd.c sha1.c
getpass.c lockf.c utils.c uuid.c sockpair.c avl.c ldif.c fetch.c testavl.c
setproctitle.c getpeereid.c detach.c
[...]
make[3]: Rien à faire pour « depend ».
make[3]: Leaving directory `/tmp/tmp/tmp/openldap-2.2.26/doc/man/man5'

Entering subdirectory man8
make[3]: Entering directory `/tmp/tmp/tmp/openldap-2.2.26/doc/man/man8'
make[3]: Rien à faire pour « depend ».
make[3]: Leaving directory `/tmp/tmp/tmp/openldap-2.2.26/doc/man/man8'

make[2]: Leaving directory `/tmp/tmp/tmp/openldap-2.2.26/doc/man'

make[1]: Leaving directory `/tmp/tmp/tmp/openldap-2.2.26/doc'

[root@ordi openldap-2.2.26]#

```

e) Lancement de la compilation.

```

[root@ordi openldap-2.2.26]# make
Making all in /tmp/tmp/tmp/openldap-2.2.26
Entering subdirectory include
make[1]: Entering directory `/tmp/tmp/tmp/openldap-2.2.26/include'
make[1]: Rien à faire pour « all ».
make[1]: Leaving directory `/tmp/tmp/tmp/openldap-2.2.26/include'
Entering subdirectory libraries
make[1]: Entering directory `/tmp/tmp/tmp/openldap-2.2.26/libraries'
Making all in /tmp/tmp/tmp/openldap-2.2.26/libraries
Entering subdirectory liblutil
make[2]: Entering directory `/tmp/tmp/tmp/openldap-2.2.26/libraries/liblutil'
rm -f version.c
../../build/mkversion -v "2.2.26" liblutil.a > version.c
cc -g -O2 -I../../include -I../../include -c -o base64.o base64.c
[...]
make[3]: Leaving directory `/tmp/tmp/tmp/openldap-2.2.26/doc/man/man8'

make[2]: Leaving directory `/tmp/tmp/tmp/openldap-2.2.26/doc/man'

make[1]: Leaving directory `/tmp/tmp/tmp/openldap-2.2.26/doc'

[root@ordi openldap-2.2.26]#

```

f) Installation des exécutables.

```

[root@ordi openldap-2.2.26]# make install
Making all in /tmp/tmp/tmp/openldap-2.2.26
Entering subdirectory include
make[1]: Entering directory `/tmp/tmp/tmp/openldap-2.2.26/include'
make[1]: Rien à faire pour « all ».
make[1]: Leaving directory `/tmp/tmp/tmp/openldap-2.2.26/include'
[...]
installing /usr/local/openldap-2.2.6/man/man8/slurpd.8
make[3]: Leaving directory `/tmp/tmp/tmp/openldap-2.2.26/doc/man/man8'

make[2]: Leaving directory `/tmp/tmp/tmp/openldap-2.2.26/doc/man'

```

```
make[1]: Leaving directory `/tmp/tmp/tmp/openldap-2.2.26/doc'
```

```
[root@ordi openldap-2.2.26]#
```

g) Configuration de la base des utilisateurs radius dans « ldap.conf ».

```
[root@ordi openldap-2.2.6]# cd etc/
[root@ordi etc]# ll
total 8
drwxr-xr-x  3 root root 4096 mai 12 15:51 openldap
[root@ordi etc]# cd openldap/
[root@ordi openldap]# ll
total 40
-rw-r--r--  1 root root  246 mai 12 15:51 ldap.conf
-rw-r--r--  1 root root  246 mai 12 15:51 ldap.conf.default
drwxr-xr-x  2 root root 4096 mai 12 15:51 schema
-rw-----  1 root root 2260 mai 12 15:51 slapd.conf
-rw-----  1 root root 2260 mai 12 15:51 slapd.conf.default

[root@ordi openldap]# vi ldap.conf
#
# LDAP Defaults
#

# See ldap.conf(5) for details
# This file should be world readable but not world writable.

#BASE    dc=example, dc=com
#URI      ldap://ldap.example.com ldap://ldap-master.example.com:666
#HOST     localhost
#BASE     dc=iutlecreusot, dc=local

#SIZELIMIT      12
#TIMELIMIT      15
#DEREF          never

[root@ordi openldap]#
```

h) Copie du schéma des attributs freeradius dans openldap.

```
[root@ordi openldap]# cp /usr/local/freeradius-1.0.2/share/doc/freeradius-
1.0.2/RADIUS-LDAPV3.schema /usr/local/openldap-2.2.6/etc/openldap/schema/
[root@ordi openldap]#
```

i) Configuration de la base des utilisateurs radius dans « slapd.conf ».

```
[root@ordi openldap]# vi /usr/local/openldap-2.2.6/etc/openldap/slapd.conf
#
# See slapd.conf(5) for details on configuration options.
# This file should NOT be world readable.
#
include      /usr/local/openldap-2.2.6/etc/openldap/schema/core.schema
include      /usr/local/openldap-2.2.6/etc/openldap/schema/cosine.schema
include      /usr/local/openldap-2.2.6/etc/openldap/schema/inetorgperson.schema
```

```

include      /usr/local/openldap-2.2.6/etc/openldap/schema/RADIUS-LDAPv3.schema

# Define global ACLs to disable default read access.

# Do not enable referrals until AFTER you have a working directory
# service AND an understanding of referrals.
#referral    ldap://root.openldap.org

pidfile      /usr/local/openldap-2.2.6/var/run/slapd.pid
argsfile     /usr/local/openldap-2.2.6/var/run/slapd.args

# Load dynamic backend modules:
# modulepath  /usr/local/openldap-2.2.6/libexec/openldap
# moduleload  back_bdb.la
# moduleload  back_ldap.la
# moduleload  back_ldbm.la
# moduleload  back_passwd.la
# moduleload  back_shell.la

# Sample security restrictions
#   Require integrity protection (prevent hijacking)
#   Require 112-bit (3DES or better) encryption for updates
#   Require 63-bit encryption for simple bind
# security    ssf=1 update_ssf=112 simple_bind=64

# Sample access control policy:
#   Root DSE: allow anyone to read it
#   Subschema (sub)entry DSE: allow anyone to read it
#   Other DSEs:
#       Allow self write access
#       Allow authenticated users read access
#       Allow anonymous users to authenticate
#   Directives needed to implement policy:
# access to dn.base="" by * read
# access to dn.base="cn=Subschema" by * read
# access to *
#   by self write
#   by users read
#   by anonymous auth
#
# if no access controls are present, the default policy
# allows anyone and everyone to read anything but restricts
# updates to rootdn. (e.g., "access to * by * read")
#
# rootdn can always read and write EVERYTHING!

#####
# BDB database definitions
#####

database     bdb
#suffix      "dc=my-domain,dc=com"
#rootdn      "cn=Manager,dc=my-domain,dc=com"
suffix       "dc=iutlecreusot,dc=local"
rootdn       "cn=root,dc=iutlecreusot,dc=local"
# Cleartext passwords, especially for the rootdn, should
# be avoided. See slapd(8) and slapd.conf(5) for details.
# Use of strong authentication encouraged.
rootpw       mdpjeromelandre
# The database directory MUST exist prior to running slapd AND
# should only be accessible by the slapd and slap tools.

```

```
# Mode 700 recommended.
directory /usr/local/openldap-2.2.6/var/openldap-data
# Indices to maintain
index objectClass eq
```

j) Définition d'un script de création d'un utilisateur.

```
[root@ordi ~]# cd /usr/local/freeradius-1.0.2/etc/raddb/
[root@ordi raddb]# mkdir ldap
[root@ordi raddb]# cd ldap
[root@ordi ldap]# vi creation

echo ""
echo "ajoute un utilisateur a la base LDAP RADIUS"
echo "utilisation: ./creation nom password groupe vlan"
echo ""

echo "dn: cn=\"$1\",dc=iutlecreusot,dc=local" > mdp.ldif
echo "objectclass: radiusprofile" >> mdp.ldif
echo "cn: \"$1" >> mdp.ldif
echo "radiusGroupName: \"$3" >> mdp.ldif
echo "radiusTunnelPassword: \"$2" >> mdp.ldif
echo "radiusTunnelType: 13" >> mdp.ldif
echo "radiusTunnelMediumType: 6" >> mdp.ldif
echo "radiusTunnelPrivateGroupId: \"$4" >> mdp.ldif
echo "radiusAuthType: EAP" >> mdp.ldif

/usr/local/openldap-2.2.6/bin/ldapmodify -a -x -D
"cn=root,dc=iutlecreusot,dc=local" -f mdp.ldif -w mdpjeromelandre

[root@ordi ldap]#
```

k) lancement de « slapd ».

```
[root@ordi openldap]# /usr/local/openldap-2.2.6/libexec/slapd -d 1
@(#) $OpenLDAP: slapd 2.2.26 (May 16 2005 18:11:16) $
root@ordi.u-bourgogne.fr:/tmp/tmp/tmp/openldap-2.2.26/servers/slapd
daemon_init: listen on ldap:///
daemon_init: 1 listeners to open...
ldap_url_parse_ext(ldap:///)
daemon: initialized ldap:///
daemon_init: 2 listeners opened
slapd init: initiated server.
slap_sasl_init: initialized!
bdb_back_initialize: initialize BDB backend
bdb_back_initialize: Sleepycat Software: Berkeley DB 4.2.52: (September 21,
2004)
>>> dnNormalize: <cn=Subschema>
=> ldap_bv2dn(cn=Subschema,0)
ldap_err2string
<= ldap_bv2dn(cn=Subschema)=0 Success
=> ldap_dn2bv(272)
ldap_err2string
<= ldap_dn2bv(cn=subschema)=0 Success
<<< dnNormalize: <cn=subschema>
bdb_db_init: Initializing BDB database
>>> dnPrettyNormal: <dc=iutlecreusot,dc=local>
```

```

=> ldap_bv2dn(dc=iutlecreusot,dc=local,0)
ldap_err2string
<= ldap_bv2dn(dc=iutlecreusot,dc=local)=0 Success
=> ldap_dn2bv(272)
ldap_err2string
<= ldap_dn2bv(dc=iutlecreusot,dc=local)=0 Success
=> ldap_dn2bv(272)
ldap_err2string
<= ldap_dn2bv(dc=iutlecreusot,dc=local)=0 Success
<<< dnPrettyNormal: <dc=iutlecreusot,dc=local>, <dc=iutlecreusot,dc=local>
>>> dnPrettyNormal: <cn=root,dc=iutlecreusot,dc=local>
=> ldap_bv2dn(cn=root,dc=iutlecreusot,dc=local,0)
ldap_err2string
<= ldap_bv2dn(cn=root,dc=iutlecreusot,dc=local)=0 Success
=> ldap_dn2bv(272)
ldap_err2string
<= ldap_dn2bv(cn=root,dc=iutlecreusot,dc=local)=0 Success
=> ldap_dn2bv(272)
ldap_err2string
<= ldap_dn2bv(cn=root,dc=iutlecreusot,dc=local)=0 Success
<<< dnPrettyNormal: <cn=root,dc=iutlecreusot,dc=local>,
<cn=root,dc=iutlecreusot,dc=local>
matching_rule_use_init
1.2.840.113556.1.4.804 (integerBitOrMatch): matchingRuleUse:
( 1.2.840.113556.1.4.804 NAME 'integerBitOrMatch' APPLIES
( radiusSimultaneousUse $ supportedLDAPVersion ) )
1.2.840.113556.1.4.803 (integerBitAndMatch): matchingRuleUse:
( 1.2.840.113556.1.4.803 NAME 'integerBitAndMatch' APPLIES
( radiusSimultaneousUse $ supportedLDAPVersion ) )
1.3.6.1.4.1.1466.109.114.2 (caseIgnoreIA5Match): matchingRuleUse:
( 1.3.6.1.4.1.1466.109.114.2 NAME 'caseIgnoreIA5Match' APPLIES ( radiusReplyItem
$ radiusCheckItem $ radiusExpiration $ dialupAccess $ radiusUserCategory $
radiusLoginTime $ radiusTunnelClientEndpoint $ radiusVSA $ radiusTunnelType $
radiusTunnelServerEndpoint $ radiusTunnelPrivateGroupId $ radiusTunnelPreference
$ radiusTunnelPassword $ radiusTunnelMediumType $ radiusTunnelAssignmentId $
radiusTerminationAction $ radiusSessionTimeout $ radiusServiceType $ radiusRealm
$ radiusReplicateToRealm $ radiusProxyToRealm $ radiusPrompt $ radiusPortLimit $
radiusPasswordRetry $ radiusLoginTCPPort $ radiusLoginService $
radiusLoginLATService $ radiusLoginLATPort $ radiusLoginLATNode $
radiusLoginLATGroup $ radiusLoginIPHost $ radiusIdleTimeout $
radiusHuntgroupName $ radiusHint $ radiusGroupName $ radiusFramedRouting $
radiusFramedRoute $ radiusFramedProtocol $ radiusFramedMTU $
radiusFramedIPXNetwork $ radiusFramedIPNetmask $ radiusFramedIPAddress $
radiusFramedCompression $ radiusFramedAppleTalkZone $
radiusFramedAppleTalkNetwork $ radiusFramedAppleTalkLink $ radiusFilterId $
radiusClientIPAddress $ radiusClass $ radiusCallingStationId $
radiusCalledStationId $ radiusCallbackNumber $ radiusCallbackId $ radiusAuthType
$ radiusArapZoneAccess $ radiusArapSecurity $ radiusArapFeatures $ email $
associatedDomain $ dc $ mail $ altServer ) )
1.3.6.1.4.1.1466.109.114.1 (caseExactIA5Match): matchingRuleUse:
( 1.3.6.1.4.1.1466.109.114.1 NAME 'caseExactIA5Match' APPLIES ( radiusReplyItem
$ radiusCheckItem $ radiusExpiration $ dialupAccess $ radiusUserCategory $
radiusLoginTime $ radiusTunnelClientEndpoint $ radiusVSA $ radiusTunnelType $
radiusTunnelServerEndpoint $ radiusTunnelPrivateGroupId $ radiusTunnelPreference
$ radiusTunnelPassword $ radiusTunnelMediumType $ radiusTunnelAssignmentId $
radiusTerminationAction $ radiusSessionTimeout $ radiusServiceType $ radiusRealm
$ radiusReplicateToRealm $ radiusProxyToRealm $ radiusPrompt $ radiusPortLimit $
radiusPasswordRetry $ radiusLoginTCPPort $ radiusLoginService $
radiusLoginLATService $ radiusLoginLATPort $ radiusLoginLATNode $
radiusLoginLATGroup $ radiusLoginIPHost $ radiusIdleTimeout $
radiusHuntgroupName $ radiusHint $ radiusGroupName $ radiusFramedRouting $

```

```

radiusFramedRoute      $      radiusFramedProtocol      $      radiusFramedMTU      $
radiusFramedIPXNetwork $      radiusFramedIPNetmask    $      radiusFramedIPAddress $
radiusFramedCompression      $      radiusFramedAppleTalkZone      $
radiusFramedAppleTalkNetwork $      radiusFramedAppleTalkLink $      radiusFilterId $
radiusClientIPAddress      $      radiusClass      $      radiusCallingStationId $
radiusCalledStationId $      radiusCallbackNumber $      radiusCallbackId $      radiusAuthType
$      radiusArapZoneAccess $      radiusArapSecurity $      radiusArapFeatures $      email $
associatedDomain $ dc $ mail $ altServer ) )
    2.5.13.35 (certificateMatch): matchingRuleUse: ( 2.5.13.35 NAME
'certificateMatch' APPLIES ( cACertificate $ userCertificate ) )
    2.5.13.34 (certificateExactMatch): matchingRuleUse: ( 2.5.13.34 NAME
'certificateExactMatch' APPLIES ( cACertificate $ userCertificate ) )
    2.5.13.30 (objectIdentifierFirstComponentMatch): matchingRuleUse:
( 2.5.13.30 NAME 'objectIdentifierFirstComponentMatch' APPLIES
( supportedApplicationContext $ ldapSyntaxes $ supportedFeatures $
supportedExtension $ supportedControl ) )
    2.5.13.29 (integerFirstComponentMatch): matchingRuleUse: ( 2.5.13.29 NAME
'integerFirstComponentMatch' APPLIES ( radiusSimultaneousUse $
supportedLDAPVersion ) )
    2.5.13.27 (generalizedTimeMatch): matchingRuleUse: ( 2.5.13.27 NAME
'generalizedTimeMatch' APPLIES ( modifyTimestamp $ createTimeStamp ) )
    2.5.13.24 (protocolInformationMatch): matchingRuleUse: ( 2.5.13.24 NAME
'protocolInformationMatch' APPLIES protocolInformation )
    2.5.13.23 (uniqueMemberMatch): matchingRuleUse: ( 2.5.13.23 NAME
'uniqueMemberMatch' APPLIES uniqueMember )
    2.5.13.22 (presentationAddressMatch): matchingRuleUse: ( 2.5.13.22 NAME
'presentationAddressMatch' APPLIES presentationAddress )
    2.5.13.20 (telephoneNumberMatch): matchingRuleUse: ( 2.5.13.20 NAME
'telephoneNumberMatch' APPLIES telephoneNumber )
    2.5.13.17 (octetStringMatch): matchingRuleUse: ( 2.5.13.17 NAME
'octetStringMatch' APPLIES userPassword )
    2.5.13.16 (bitStringMatch): matchingRuleUse: ( 2.5.13.16 NAME
'bitStringMatch' APPLIES x500UniqueIdentifier )
    2.5.13.14 (integerMatch): matchingRuleUse: ( 2.5.13.14 NAME 'integerMatch'
APPLIES ( radiusSimultaneousUse $ supportedLDAPVersion ) )
    2.5.13.13 (booleanMatch): matchingRuleUse: ( 2.5.13.13 NAME 'booleanMatch'
APPLIES ( radiusStripUserName $ hasSubordinates ) )
    2.5.13.11 (caseIgnoreListMatch): matchingRuleUse: ( 2.5.13.11 NAME
'caseIgnoreListMatch' APPLIES ( registeredAddress $ postalAddress ) )
    2.5.13.8 (numericStringMatch): matchingRuleUse: ( 2.5.13.8 NAME
'numericStringMatch' APPLIES ( internationalISDNNumber $ x121Address ) )
    2.5.13.7 (caseExactSubstringsMatch): matchingRuleUse: ( 2.5.13.7 NAME
'caseExactSubstringsMatch' APPLIES ( dnQualifier $ destinationIndicator $
serialNumber ) )
    2.5.13.6 (caseExactOrderingMatch): matchingRuleUse: ( 2.5.13.6 NAME
'caseExactOrderingMatch' APPLIES ( dnQualifier $ destinationIndicator $
serialNumber ) )
    2.5.13.5 (caseExactMatch): matchingRuleUse: ( 2.5.13.5 NAME 'caseExactMatch'
APPLIES ( uid $ dmdName $ houseIdentifier $ dnQualifier $ generationQualifier $
initials $ givenName $ destinationIndicator $ physicalDeliveryOfficeName $
postOfficeBox $ postalCode $ businessCategory $ description $ title $ ou $ o $
street $ st $ l $ c $ serialNumber $ sn $ knowledgeInformation $ labeledURI $ cn
$ name $ ref $ vendorVersion $ vendorName $ supportedSASLMechanisms ) )
    2.5.13.4 (caseIgnoreSubstringsMatch): matchingRuleUse: ( 2.5.13.4 NAME
'caseIgnoreSubstringsMatch' APPLIES ( dnQualifier $ destinationIndicator $
serialNumber ) )
    2.5.13.3 (caseIgnoreOrderingMatch): matchingRuleUse: ( 2.5.13.3 NAME
'caseIgnoreOrderingMatch' APPLIES ( dnQualifier $ destinationIndicator $
serialNumber ) )
    2.5.13.2 (caseIgnoreMatch): matchingRuleUse: ( 2.5.13.2 NAME
'caseIgnoreMatch' APPLIES ( uid $ dmdName $ houseIdentifier $ dnQualifier $

```

```

generationQualifier $ initials $ givenName $ destinationIndicator $
physicalDeliveryOfficeName $ postOfficeBox $ postalCode $ businessCategory $
description $ title $ ou $ o $ street $ st $ l $ c $ serialNumber $ sn $
knowledgeInformation $ labeledURI $ cn $ name $ ref $ vendorVersion $ vendorName
$ supportedSASLMechanisms ) )
    2.5.13.1 (distinguishedNameMatch): matchingRuleUse: ( 2.5.13.1 NAME
'distinguishedNameMatch' APPLIES ( radiusProfileDn $ seeAlso $ roleOccupant $
owner $ member $ distinguishedName $ aliasedObjectName $ namingContexts $
subschemaSubentry $ modifiersName $ creatorsName ) )
    2.5.13.0 (objectIdentifierMatch): matchingRuleUse: ( 2.5.13.0 NAME
'objectIdentifierMatch' APPLIES ( supportedApplicationContext $
supportedFeatures $ supportedExtension $ supportedControl ) )
slapd startup: initiated.
backend_startup: starting "dc=iutlecreusot,dc=local"
bdb_db_open: dbenv_open(/usr/local/openldap-2.2.6/var/openldap-data)
slapd starting

```

l) Lancement d'une requête de test.

```

[root@ordi ldap]# /usr/local/openldap-2.2.6/bin/ldapsearch -x -b "" -s base *
# extended LDIF
#
# LDAPv3
# base <> with scope base
# filter: (objectclass=*)
# requesting: creation mdp.ldif
#
#
dn:

# search result
search: 2
result: 0 Success

# numResponses: 2
# numEntries: 1

```

m) Résultat de la requête côté serveur « openldap ».

```

connection_get(10): got connid=0
connection_read(10): checking for input on id=0
ber_get_next
ber_get_next: tag 0x30 len 12 contents:
ber_get_next
ber_get_next on fd 10 failed errno=11 (Resource temporarily unavailable)
do_bind
ber_scanf fmt ({imt) ber:
ber_scanf fmt (m) ber:
>>> dnPrettyNormal: <>
<<< dnPrettyNormal: <>, <>
do_bind: version=3 dn="" method=128
send_ldap_result: conn=0 op=0 p=3
send_ldap_response: msgid=1 tag=97 err=0
ber_flush: 14 bytes to sd 10
do_bind: v3 anonymous bind
connection_get(10): got connid=0

```

```

connection_read(10): checking for input on id=0
ber_get_next
ber_get_next: tag 0x30 len 57 contents:
ber_get_next
ber_get_next on fd 10 failed errno=11 (Resource temporarily unavailable)
do_search
ber_scanf fmt ({miiib} ber:
>>> dnPrettyNormal: <>
<<< dnPrettyNormal: <>, <>
ber_scanf fmt (m) ber:
ber_scanf fmt ({M}) ber:
=> send_search_entry: dn=""
ber_flush: 11 bytes to sd 10
<= send_search_entry
send_ldap_result: conn=0 op=1 p=3
send_ldap_response: msgid=2 tag=101 err=0
ber_flush: 14 bytes to sd 10
connection_get(10): got connid=0
connection_read(10): checking for input on id=0
ber_get_next
ber_get_next: tag 0x30 len 5 contents:
ber_get_next
ber_get_next on fd 10 failed errno=11 (Resource temporarily unavailable)
connection_get(10): got connid=0
connection_read(10): checking for input on id=0
ber_get_next
ber_get_next on fd 10 failed errno=0 (Success)
connection_read(10): input error=-2 id=0, closing.
connection_closing: readying conn=0 sd=10 for close
connection_close: deferring conn=0 sd=10
do_unbind
connection_resched: attempting closing conn=0 sd=10
connection_close: conn=0 sd=10

```

n) Modification de AUXILIARY en STRUCTURAL dans le fichier « RADIUS-LDAPv3.schema ».

```

[root@ordi ldap]# vi /usr/local/openldap-2.2.6/etc/openldap/schema/RADIUS-
LDAPv3.schema
[...]
```

by jerome landre - SUP top AUXILIARY

```

objectclass
( 1.3.6.1.4.1.3317.4.3.2.1
  NAME 'radiusprofile'
  SUP top STRUCTURAL
  DESC ''
  MUST cn
  MAY ( radiusArapFeatures $ radiusArapSecurity $ radiusArapZoneAccess $
        radiusAuthType $ radiusCallbackId $ radiusCallbackNumber $
[...]
```

o) Création de la base de l'annuaire.

```

[root@ordi ldap]# vi init.ldif
dn: dc=iutlecreusot,dc=local
objectclass: dcObject

```



```
objectclass: organization
o: iutlecreusot
dc: iutlecreusot
```

```
dn: cn=root,dc=iutlecreusot,dc=local
objectclass: organizationalRole
cn: root
```

```
[root@ordi ldap]# /usr/local/openldap-2.2.6/bin/ldapadd -x -D
'cn=root,dc=iutlecreusot,dc=local' -W -f init.ldif
Enter LDAP Password:
adding new entry "dc=iutlecreusot,dc=local"

adding new entry "cn=root,dc=iutlecreusot,dc=local"

[root@ordi ldap]#
```

p) Ajout d'un utilisateur.

```
[root@ordi ldap]# ./creation j.landre mdpjerome personnel 100
```

```
ajoute un utilisateur a la base LDAP RADIUS
utilisation: ./creation nom password groupe vlan
```

```
Enter LDAP Password:
adding new entry "cn=j.landre,dc=iutlecreusot,dc=local"
```

```
[root@ordi ldap]# /usr/local/openldap-2.2.6/bin/ldapsearch -x -b
'dc=iutlecreusot,dc=local' '(objectclass=*)'
# extended LDIF
#
# LDAPv3
# base <dc=iutlecreusot,dc=local> with scope sub
# filter: (objectclass=*)
# requesting: ALL
#
# iutlecreusot.local
dn: dc=iutlecreusot,dc=local
objectClass: dcObject
objectClass: organization
o: iutlecreusot
dc: iutlecreusot

# root, iutlecreusot.local
dn: cn=root,dc=iutlecreusot,dc=local
objectClass: organizationalRole
cn: root

# j.landre, iutlecreusot.local
dn: cn=j.landre,dc=iutlecreusot,dc=local
objectClass: radiusprofile
cn: j.landre
radiusGroupName: personnel
radiusTunnelPassword: mdpjerome
radiusTunnelType: 13
radiusTunnelMediumType: 6
radiusTunnelPrivateGroupId: 100
radiusAuthType: EAP
```

```
# search result
search: 2
result: 0 Success
```

```
# numResponses: 4
# numEntries: 3
[root@ordi ldap]#
```

p) Configuration de freeradius.

```
[root@ordi ldap]# cd /usr/local/freeradius-1.0.2/etc/raddb/
[root@ordi raddb]# vi radiusd.conf
[...]
# Lightweight Directory Access Protocol (LDAP)
#
# This module definition allows you to use LDAP for
# authorization and authentication (Auth-Type := LDAP)
#
# See doc/rlm_ldap for description of configuration options
# and sample authorize{} and authenticate{} blocks
ldap {
    server = "localhost"
    identity = "cn=root,dc=iutlecreusot,dc=local"
    password = mdpjeromelandre
    basedn = "dc=iutlecreusot,dc=local"
    filter = "(cn=%{Stripped-User-Name:-%{User-Name}})"
    # base_filter = "(objectclass=radiusprofile)"

    # set this to 'yes' to use TLS encrypted connections
    # to the LDAP database by using the StartTLS extended
    # operation.
    # The StartTLS operation is supposed to be used with normal
    # ldap connections instead of using ldaps (port 689) connections
    start_tls = no

    # tls_cacertfile = /path/to/cacert.pem
    # tls_cacertdir = /path/to/ca/dir/
    # tls_certfile = /path/to/radius.crt
    # tls_keyfile = /path/to/radius.key
    # tls_randfile = /path/to/rnd
    # tls_require_cert = "demand"

    # default_profile = "cn=radprofile,ou=dialup,o=My Org,c=UA"
    # profile_attribute = "radiusProfileDn"
    access_attr = "dialupAccess"

    # Mapping of RADIUS dictionary attributes to LDAP
    # directory attributes.
    dictionary_mapping = ${raddbdir}/ldap.attrmap

    ldap_connections_number = 5
}
[...]
authorize {
[...]
#
# Look in an SQL database. The schema of the database
# is meant to mirror the "users" file.
#
# See "Authorization Queries" in sql.conf
#
sql
```

```

#
# If you are using /etc/smbpasswd, and are also doing
# mschap authentication, the un-comment this line, and
# configure the 'etc_smbpasswd' module, above.
# etc_smbpasswd

#
# The ldap module will set Auth-Type to LDAP if it has not
# already been set
ldap
[...]
authenticate {
[...]
    # Uncomment it if you want to use ldap for authentication
    #
    # Note that this means "check plain-text password against
    # the ldap database", which means that EAP won't work,
    # as it does not supply a plain-text password.
    Auth-Type LDAP {
        ldap
    }

    #
    # Allow EAP authentication.
    eap
}
[...]
```

p) Exemple de trace (log) freeradius.

```

[root@ordi ldap]# /usr/local/freeradius-1.0.2/sbin/radiusd -X
Starting - reading configuration files ...
reread_config: reading radiusd.conf
Config: including file: /usr/local/freeradius-1.0.2/etc/raddb/proxy.conf
Config: including file: /usr/local/freeradius-1.0.2/etc/raddb/clients.conf
Config: including file: /usr/local/freeradius-1.0.2/etc/raddb/snmp.conf
Config: including file: /usr/local/freeradius-1.0.2/etc/raddb/eap.conf
Config: including file: /usr/local/freeradius-1.0.2/etc/raddb/sql.conf
main: prefix = "/usr/local/freeradius-1.0.2"
main: localstatedir = "/usr/local/freeradius-1.0.2/var"
main: logdir = "/usr/local/freeradius-1.0.2/var/log/radius"
main: libdir = "/usr/local/freeradius-1.0.2/lib"
main: radacctdir = "/usr/local/freeradius-1.0.2/var/log/radius/radacct"
main: hostname_lookups = no
main: max_request_time = 30
main: cleanup_delay = 5
main: max_requests = 1024
main: delete_blocked_requests = 0
main: port = 0
main: allow_core_dumps = no
main: log_stripped_names = no
main: log_file = "/usr/local/freeradius-1.0.2/var/log/radius/radius.log"
main: log_auth = no
main: log_auth_badpass = no
main: log_auth_goodpass = no
main: pidfile = "/usr/local/freeradius-1.0.2/var/run/radiusd/radiusd.pid"
main: user = "(null)"
main: group = "(null)"
main: usercollide = no
```

```

main: lower_user = "no"
main: lower_pass = "no"
main: nospace_user = "no"
main: nospace_pass = "no"
main: checkrad = "/usr/local/freeradius-1.0.2/sbin/checkrad"
main: proxy_requests = yes
proxy: retry_delay = 5
proxy: retry_count = 3
proxy: synchronous = no
proxy: default_fallback = yes
proxy: dead_time = 120
proxy: post_proxy_authorize = yes
proxy: wake_all_if_all_dead = no
security: max_attributes = 200
security: reject_delay = 1
security: status_server = no
main: debug_level = 0
read_config_files: reading dictionary
read_config_files: reading naslist
Using deprecated naslist file. Support for this will go away soon.
read_config_files: reading clients
read_config_files: reading realms
radiusd: entering modules setup
Module: Library search path is /usr/local/freeradius-1.0.2/lib
Module: Loaded exec
exec: wait = yes
exec: program = "(null)"
exec: input_pairs = "request"
exec: output_pairs = "(null)"
exec: packet_type = "(null)"
rlm_exec: Wait=yes but no output defined. Did you mean output=none?
Module: Instantiated exec (exec)
Module: Loaded expr
Module: Instantiated expr (expr)
Module: Loaded PAP
pap: encryption_scheme = "crypt"
Module: Instantiated pap (pap)
Module: Loaded CHAP
Module: Instantiated chap (chap)
Module: Loaded MS-CHAP
mschap: use_mppe = yes
mschap: require_encryption = no
mschap: require_strong = no
mschap: with_ntdomain_hack = no
mschap: passwd = "(null)"
mschap: authtype = "MS-CHAP"
mschap: ntlm_auth = "(null)"
Module: Instantiated mschap (mschap)
Module: Loaded System
unix: cache = no
unix: passwd = "(null)"
unix: shadow = "(null)"
unix: group = "(null)"
unix: radwtmp = "/usr/local/freeradius-1.0.2/var/log/radius/radwtmp"
unix: usegroup = no
unix: cache_reload = 600
Module: Instantiated unix (unix)
Module: Loaded LDAP
ldap: server = "localhost"
ldap: port = 389
ldap: net_timeout = 1

```

```

ldap: timeout = 4
ldap: timelimit = 3
ldap: identity = "cn=root,dc=iutlecreusot,dc=local"
ldap: tls_mode = no
ldap: start_tls = no
ldap: tls_cacertfile = "(null)"
ldap: tls_cacertdir = "(null)"
ldap: tls_certfile = "(null)"
ldap: tls_keyfile = "(null)"
ldap: tls_randfile = "(null)"
ldap: tls_require_cert = "allow"
ldap: password = "toto"
ldap: basedn = "dc=iutlecreusot,dc=local"
ldap: filter = "(cn=%{Stripped-User-Name}-%{User-Name})"
ldap: base_filter = "(objectclass=radiusprofile)"
ldap: default_profile = "(null)"
ldap: profile_attribute = "(null)"
ldap: password_header = "(null)"
ldap: password_attribute = "(null)"
ldap: access_attr = "dialupAccess"
ldap: groupname_attribute = "cn"
ldap: groupmembership_filter = "(&(&(objectClass=GroupOfNames)(member=%{Ldap-UserDn}))(&(objectClass=GroupOfUniqueNames)(uniquemember=%{Ldap-UserDn})))"
ldap: groupmembership_attribute = "(null)"
ldap: dictionary_mapping = "/usr/local/freeradius-1.0.2/etc/raddb/ldap.attrmap"
ldap: ldap_debug = 0
ldap: ldap_connections_number = 5
ldap: compare_check_items = no
ldap: access_attr_used_for_allow = yes
ldap: do_xlat = yes
rlm_ldap: Registering ldap_groupcmp for Ldap-Group
rlm_ldap: Registering ldap_xlat with xlat_name ldap
rlm_ldap: reading ldap<->radius mappings from file /usr/local/freeradius-1.0.2/etc/raddb/ldap.attrmap
rlm_ldap: LDAP radiusCheckItem mapped to RADIUS $GENERIC$
rlm_ldap: LDAP radiusReplyItem mapped to RADIUS $GENERIC$
rlm_ldap: LDAP radiusAuthType mapped to RADIUS Auth-Type
rlm_ldap: LDAP radiusSimultaneousUse mapped to RADIUS Simultaneous-Use
rlm_ldap: LDAP radiusCalledStationId mapped to RADIUS Called-Station-Id
rlm_ldap: LDAP radiusCallingStationId mapped to RADIUS Calling-Station-Id
rlm_ldap: LDAP lmPassword mapped to RADIUS LM-Password
rlm_ldap: LDAP ntPassword mapped to RADIUS NT-Password
rlm_ldap: LDAP acctFlags mapped to RADIUS SMB-Account-CTRL-TEXT
rlm_ldap: LDAP radiusExpiration mapped to RADIUS Expiration
rlm_ldap: LDAP radiusServiceType mapped to RADIUS Service-Type
rlm_ldap: LDAP radiusFramedProtocol mapped to RADIUS Framed-Protocol
rlm_ldap: LDAP radiusFramedIPAddress mapped to RADIUS Framed-IP-Address
rlm_ldap: LDAP radiusFramedIPNetmask mapped to RADIUS Framed-IP-Netmask
rlm_ldap: LDAP radiusFramedRoute mapped to RADIUS Framed-Route
rlm_ldap: LDAP radiusFramedRouting mapped to RADIUS Framed-Routing
rlm_ldap: LDAP radiusFilterId mapped to RADIUS Filter-Id
rlm_ldap: LDAP radiusFramedMTU mapped to RADIUS Framed-MTU
rlm_ldap: LDAP radiusFramedCompression mapped to RADIUS Framed-Compression
rlm_ldap: LDAP radiusLoginIPHost mapped to RADIUS Login-IP-Host
rlm_ldap: LDAP radiusLoginService mapped to RADIUS Login-Service
rlm_ldap: LDAP radiusLoginTCPPort mapped to RADIUS Login-TCP-Port
rlm_ldap: LDAP radiusCallbackNumber mapped to RADIUS Callback-Number
rlm_ldap: LDAP radiusCallbackId mapped to RADIUS Callback-Id
rlm_ldap: LDAP radiusFramedIPXNetwork mapped to RADIUS Framed-IPX-Network
rlm_ldap: LDAP radiusClass mapped to RADIUS Class
rlm_ldap: LDAP radiusSessionTimeout mapped to RADIUS Session-Timeout

```

```

rlm_ldap: LDAP radiusIdleTimeout mapped to RADIUS Idle-Timeout
rlm_ldap: LDAP radiusTerminationAction mapped to RADIUS Termination-Action
rlm_ldap: LDAP radiusLoginLATService mapped to RADIUS Login-LAT-Service
rlm_ldap: LDAP radiusLoginLATNode mapped to RADIUS Login-LAT-Node
rlm_ldap: LDAP radiusLoginLATGroup mapped to RADIUS Login-LAT-Group
rlm_ldap: LDAP radiusFramedAppleTalkLink mapped to RADIUS Framed-AppleTalk-Link
rlm_ldap: LDAP radiusFramedAppleTalkNetwork mapped to RADIUS Framed-AppleTalk-
Network
rlm_ldap: LDAP radiusFramedAppleTalkZone mapped to RADIUS Framed-AppleTalk-Zone
rlm_ldap: LDAP radiusPortLimit mapped to RADIUS Port-Limit
rlm_ldap: LDAP radiusLoginLATPort mapped to RADIUS Login-LAT-Port
conns: 0x8a8ce60
Module: Instantiated ldap (ldap)
Module: Loaded eap
  eap: default_eap_type = "md5"
  eap: timer_expire = 60
  eap: ignore_unknown_eap_types = no
  eap: cisco_accounting_username_bug = no
rlm_eap: Loaded and initialized type md5
rlm_eap: Loaded and initialized type leap
  gtc: challenge = "Password: "
  gtc: auth_type = "PAP"
rlm_eap: Loaded and initialized type gtc
  mschapv2: with_ntdomain_hack = no
rlm_eap: Loaded and initialized type mschapv2
Module: Instantiated eap (eap)
Module: Loaded preprocess
  preprocess: huntgroups = "/usr/local/freeradius-1.0.2/etc/raddb/huntgroups"
  preprocess: hints = "/usr/local/freeradius-1.0.2/etc/raddb/hints"
  preprocess: with_ascend_hack = no
  preprocess: ascend_channels_per_line = 23
  preprocess: with_ntdomain_hack = no
  preprocess: with_specialix_jetstream_hack = no
  preprocess: with_cisco_vsa_hack = no
Module: Instantiated preprocess (preprocess)
Module: Loaded realm
  realm: format = "suffix"
  realm: delimiter = "@"
  realm: ignore_default = no
  realm: ignore_null = no
Module: Instantiated realm (suffix)
Module: Loaded Acct-Unique-Session-Id
  acct_unique: key = "User-Name, Acct-Session-Id, NAS-IP-Address, Client-IP-
Address, NAS-Port"
Module: Instantiated acct_unique (acct_unique)
Module: Loaded files
  files: usersfile = "/usr/local/freeradius-1.0.2/etc/raddb/users"
  files: acctusersfile = "/usr/local/freeradius-1.0.2/etc/raddb/acct_users"
  files:      preproxy_usersfile      =      "/usr/local/freeradius-
1.0.2/etc/raddb/preproxy_users"
  files: compat = "no"
Module: Instantiated files (files)
Module: Loaded detail
  detail: detailfile = "/usr/local/freeradius-1.0.2/var/log/radius/radacct/%
{Client-IP-Address}/detail-%Y%m%d"
  detail: detailperm = 384
  detail: dirperm = 493
  detail: locking = no
Module: Instantiated detail (detail)
Module: Loaded radutmp
  radutmp: filename = "/usr/local/freeradius-1.0.2/var/log/radius/radutmp"

```

```

radutmp: username = "%{User-Name}"
radutmp: case_sensitive = yes
radutmp: check_with_nas = yes
radutmp: perm = 384
radutmp: callerid = yes
Module: Instantiated radutmp (radutmp)
Listening on authentication *:1812
Listening on accounting *:1813
Listening on proxy *:1814
Ready to process requests.

rad_recv: Access-Request packet from host 127.0.0.1:32775, id=54, length=61
  User-Name = "j.landre"
  User-Password = "mdpjerome"
  NAS-IP-Address = 255.255.255.255
  NAS-Port = 1812
  Processing the authorize section of radiusd.conf
modcall: entering group authorize for request 0
  modcall[authorize]: module "preprocess" returns ok for request 0
  modcall[authorize]: module "chap" returns noop for request 0
  modcall[authorize]: module "mschap" returns noop for request 0
    rlm_realm: No '@' in User-Name = "j.landre", looking up realm NULL
    rlm_realm: No such realm "NULL"
  modcall[authorize]: module "suffix" returns noop for request 0
  rlm_eap: No EAP-Message, not doing EAP
  modcall[authorize]: module "eap" returns noop for request 0
rlm_ldap: - authorize
rlm_ldap: performing user authorization for j.landre
radius_xlat: '(cn=j.landre)'
radius_xlat: 'dc=iutlecreusot,dc=local'
rlm_ldap: ldap_get_conn: Checking Id: 0
rlm_ldap: ldap_get_conn: Got Id: 0
rlm_ldap: attempting LDAP reconnection
rlm_ldap: (re)connect to localhost:389, authentication 0
rlm_ldap: bind as cn=root,dc=iutlecreusot,dc=local/toto to localhost:389
rlm_ldap: waiting for bind result ...
rlm_ldap: Bind was successful
rlm_ldap: performing search in dc=iutlecreusot,dc=local, with filter
(cn=j.landre)
rlm_ldap: checking if remote access for j.landre is allowed by dialupAccess
rlm_ldap: looking for check items in directory...
rlm_ldap: Adding radiusAuthType as Auth-Type, value EAP & op=21
rlm_ldap: looking for reply items in directory...
rlm_ldap: user j.landre authorized to use remote access
rlm_ldap: ldap_release_conn: Release Id: 0
  modcall[authorize]: module "ldap" returns ok for request 0
modcall: group authorize returns ok for request 0
  rad_check_password: Found Auth-Type EAP
auth: type "EAP"
  Processing the authenticate section of radiusd.conf
modcall: entering group authenticate for request 0
[...]
Sending Access-Accept of id 59 to 10.0.0.189:21657
  Tunnel-Type:0 = VLAN
  Tunnel-Medium-Type:0 = IEEE-802
  Tunnel-Private-Group-Id:0 = "100"
  Message-Authenticator = 0x00000000000000000000000000000000
  User-Name = "j.landre"
Finished request 0
Going to the next request
--- Walking the entire request list ---

```

```

Waking up in 1 seconds...
--- Walking the entire request list ---
Waking up in 1 seconds...
--- Walking the entire request list ---
Sending Access-Reject of id 54 to 127.0.0.1:32775
Waking up in 4 seconds...

```

OpenLDAP est un annuaire libre, Microsoft a préféré développer une version d'annuaire avec plus de services, Active Directory.

6.3. Active Directory

L'I.U.T. du Creusot dispose d'une base d'utilisateurs gérée par un annuaire Microsoft Active Directory. Afin de mettre en place une politique de signature unique (*SSO: Single Sign On*), l'idéal serait de faire en sorte que le serveur RADIUS authentifie les utilisateurs directement sur le serveur Active Directory de l'établissement. C'est ce qui a été réalisé grâce à l'aide de nos collègues du C.R.I. de Dijon: Fabien Bole, Christine Browaeys, Jean-Claude Joly et Olivier Perrot que nous remercions et que nous associons à cette partie.

Afin de permettre la communication entre le serveur RADIUS et le serveur Windows 2003 server, nous allons intégrer le serveur freeradius au domaine Windows de l'établissement grâce à SAMBA. L'identification d'une machine sur un domaine Windows utilise le protocole Kerberos 5. Il faut donc installer SAMBA 3 et Kerberos 5 sur le serveur avant de l'intégrer au domaine.

Ici, on ne va pas partir des sources, mais nous allons installer SAMBA 3 et Kerberos 5 à partir de paquetages complets « .rpm ».

a) Installation de Kerberos 5.

```

[root@ordi ~]# rpm -ivh krb5-libs-1.3.6-2.i386.rpm
Préparation... ##### [100%]
1: krb5-libs ##### [100%]
[root@ordi ~]# rpm -ivh krb5-server-1.3.6-2.i386.rpm
Préparation... ##### [100%]
1: krb5-server ##### [100%]
[root@ordi ~]# rpm -ivh krb5-devel-1.3.6-2.i386.rpm
Préparation... ##### [100%]
1: krb5-devel ##### [100%]
[root@ordi ~]# rpm -ivh krb5-workstation-1.3.6-2.i386.rpm
Préparation... ##### [100%]
1: krb5-workstation ##### [100%]
[root@ordi ~]#

```

b) Installation de Samba 3.

```

[root@ordi ~]# rpm -ivh samba-3.0.10-1.fc3.i386.rpm
Préparation... ##### [100%]
1: samba ##### [100%]
[root@ordi ~]# rpm -ivh samba-client-3.0.10-1.fc3.i386.rpm
Préparation... ##### [100%]
1: samba-client ##### [100%]
[root@ordi ~]# rpm -ivh samba-common-3.0.10-1.fc3.i386.rpm

```



```

Préparation... ##### [100%]
1: samba-common ##### [100%]
[root@ordi ~]#

```

c) Configuration de Kerberos 5.

```

[root@ordi ~]# vi /etc/krb5.conf
[logging]
default = FILE:/var/log/krb5libs.log
kdc = FILE:/var/log/krb5kdc.log
admin_server = FILE:/var/log/kadmind.log

[libdefaults]
default_realm = sitecreusot.LOCAL
dns_lookup_realm = false
dns_lookup_kdc = false

[realms]
sitecreusot.LOCAL = {
    kdc = iutlecreusot.sitecreusot.local:88
    admin_server = iutlecreusot.sitecreusot.local:749
    default_domain = sitecreusot.local
    kdc = *
}

[domain_realm]
.sitecreusot.local = sitecreusot.LOCAL
sitecreusot.local = sitecreusot.LOCAL

[kdc]
profile = /var/kerberos/krb5kdc/kdc.conf

[appdefaults]
pam = {
    debug = false
    ticket_lifetime = 36000
    renew_lifetime = 36000
    forwardable = true
    krb4_convert = false
}

```

d) Configuration de Samba 3.

```

[root@ordi ~]# vi /etc/samba/smb.conf

# This is the main Samba configuration file. You should read the
# smb.conf(5) manual page in order to understand the options listed
# here. Samba has a huge number of configurable options (perhaps too
# many!) most of which are not shown in this example
#
# Any line which starts with a ; (semi-colon) or a # (hash)
# is a comment and is ignored. In this example we will use a #
# for commentry and a ; for parts of the config file that you
# may wish to enable
#
# NOTE: Whenever you modify this file you should run the command "testparm"
# to check that you have not made any basic syntactic errors.

```

```

#
#===== Global Settings =====
[global]

# workgroup = NT-Domain-Name or Workgroup-Name
#   workgroup = sitecreusot

# server string is the equivalent of the NT Description field
#   server string = iutlecreusot

# This option is important for security. It allows you to restrict
# connections to machines which are on your local network. The
# following example restricts access to two C class networks and
# the "loopback" interface. For more examples of the syntax see
# the smb.conf man page
;   hosts allow = 192.168.1. 192.168.2. 127.

# if you want to automatically load your printer list rather
# than setting them up individually then you'll need this
#   printcap name = /etc/printcap
#   load printers = yes

# It should not be necessary to spell out the print system type unless
# yours is non-standard. Currently supported print systems include:
# bsd, sysv, plp, lprng, aix, hpux, qnx
;   printing = cups

# This option tells cups that the data has already been rasterized
cups options = raw

# Uncomment this if you want a guest account, you must add this to /etc/passwd
# otherwise the user "nobody" is used
;   guest account = pcguest

# this tells Samba to use a separate log file for each machine
# that connects
#   log file = /var/log/samba/%m.log
# all log information in one file
#   log file = /var/log/samba/smbd.log

# Put a capping on the size of the log files (in Kb).
#   max log size = 50

# Security mode. Most people will want user level security. See
# security_level.txt for details.
#   security = ads
# Use password server option only with security = server
;   password server = <NT-Server-Name>

# Password Level allows matching of _n_ characters of the password for
# all combinations of upper and lower case.
;   password level = 8
;   username level = 8

# You may wish to use password encryption. Please read
# ENCRYPTION.txt, Win95.txt and WinNT.txt in the Samba documentation.
# Do not enable this option unless you have read those documents
;   encrypt passwords = yes
;   smb passwd file = /etc/samba/smbpasswd

# The following are needed to allow password changing from Windows to

```

```

# update the Linux system password also.
# NOTE: Use these with 'encrypt passwords' and 'smb passwd file' above.
# NOTE2: You do NOT need these to allow workstations to change only
#         the encrypted SMB passwords. They allow the Unix password
#         to be kept in sync with the SMB password.
; unix password sync = Yes
; passwd program = /usr/bin/passwd %u
; passwd chat = *New*UNIX*password* %n\n *ReType*new*UNIX*password* %n\n
*passwd:*all*authentication*tokens*updated*successfully*

# Unix users can map to different SMB User names
; username map = /etc/samba/smbusers

# Using the following line enables you to customise your configuration
# on a per machine basis. The %m gets replaced with the netbios name
# of the machine that is connecting
; include = /etc/samba/smb.conf.%m

# Most people will find that this option gives better performance.
# See speed.txt and the manual pages for details
; socket options = TCP_NODELAY SO_RCVBUF=8192 SO_SNDBUF=8192

# Configure Samba to use multiple interfaces
# If you have multiple network interfaces then you must list them
# here. See the man page for details.
; interfaces = 192.168.12.2/24 192.168.13.2/24

# Configure remote browse list synchronisation here
# request announcement to, or browse list sync from:
#     a specific host or from / to a whole subnet (see below)
; remote browse sync = 192.168.3.25 192.168.5.255
# Cause this host to announce itself to local subnets here
; remote announce = 192.168.1.255 192.168.2.44

# Browser Control Options:
# set local master to no if you don't want Samba to become a master
# browser on your network. Otherwise the normal election rules apply
; local master = no

# OS Level determines the precedence of this server in master browser
# elections. The default value should be reasonable
; os level = 33

# Domain Master specifies Samba to be the Domain Master Browser. This
# allows Samba to collate browse lists between subnets. Don't use this
# if you already have a Windows NT domain controller doing this job
; domain master = yes

# Preferred Master causes Samba to force a local browser election on startup
# and gives it a slightly higher chance of winning the election
; preferred master = yes

# Enable this if you want Samba to be a domain logon server for
# Windows95 workstations.
; domain logons = yes

# if you enable domain logons then you may want a per-machine or
# per user logon script
# run a specific logon batch file per workstation (machine)
; logon script = %m.bat
# run a specific logon batch file per username

```

```

;   logon script = %U.bat

# Where to store roving profiles (only for Win95 and WinNT)
#       %L substitutes for this servers netbios name, %U is username
#       You must uncomment the [Profiles] share below
;   logon path = \\%L\Profiles\%U

# All NetBIOS names must be resolved to IP Addresses
# 'Name Resolve Order' allows the named resolution mechanism to be specified
# the default order is "host lmhosts wins bcast". "host" means use the unix
# system gethostbyname() function call that will use either /etc/hosts OR
# DNS or NIS depending on the settings of /etc/host.config, /etc/nsswitch.conf
# and the /etc/resolv.conf file. "host" therefore is system configuration
# dependant. This parameter is most often of use to prevent DNS lookups
# in order to resolve NetBIOS names to IP Addresses. Use with care!
# The example below excludes use of name resolution for machines that are NOT
# on the local network segment
# - OR - are not deliberately to be known via lmhosts or via WINS.
; name resolve order = wins lmhosts bcast

# Windows Internet Name Serving Support Section:
# WINS Support - Tells the NMBD component of Samba to enable it's WINS Server
;   wins support = yes

# WINS Server - Tells the NMBD components of Samba to be a WINS Client
#       Note: Samba can be either a WINS Server, or a WINS Client, but NOT
#       both
;   wins server = w.x.y.z

# WINS Proxy - Tells Samba to answer name resolution queries on
# behalf of a non WINS capable client, for this to work there must be
# at least one WINS Server on the network. The default is NO.
;   wins proxy = yes

# DNS Proxy - tells Samba whether or not to try to resolve NetBIOS names
# via DNS nslookups. The built-in default for versions 1.9.17 is yes,
# this has been changed in version 1.9.18 to no.
;   dns proxy = no

# Case Preservation can be handy - system default is _no_
# NOTE: These can be set on a per share basis
;   preserve case = no
;   short preserve case = no
# Default case is normally upper case for all DOS files
;   default case = lower
# Be very careful with case sensitivity - it can break things!
;   case sensitive = no

#===== Share Definitions =====
realm = sitecreusot.LOCAL
#encrypt passwords = yes
encrypt passwords = no
idmap uid = 16777216-33554431
idmap gid = 16777216-33554431
template shell = /bin/false
winbind use default domain = yes
password server = 193.52.240.254
[homes]
comment = Home Directories
browseable = no
writable = yes

```

```

# Un-comment the following and create the netlogon directory for Domain Logons
; [netlogon]
;   comment = Network Logon Service
;   path = /home/netlogon
;   guest ok = yes
;   writable = no
;   share modes = no

# Un-comment the following to provide a specific roving profile share
# the default is to use the user's home directory
;[Profiles]
;   path = /home/profiles
;   browseable = no
;   guest ok = yes

# NOTE: If you have a BSD-style print system there is no need to
# specifically define each individual printer
[printers]
    comment = All Printers
    path = /var/spool/samba
    browseable = no
# Set public = yes to allow user 'guest account' to print
    guest ok = no
    writable = no
    printable = yes

# This one is useful for people to share files
;[tmp]
;   comment = Temporary file space
;   path = /tmp
;   read only = no
;   public = yes

# A publicly accessible directory, but read only, except for people in
# the "staff" group
;[public]
;   comment = Public Stuff
;   path = /home/samba
;   public = yes
;   read only = yes
;   write list = @staff

# Other examples.
#
# A private printer, usable only by fred. Spool data will be placed in fred's
# home directory. Note that fred must have write access to the spool
# directory,
# wherever it is.
;[fredsprn]
;   comment = Fred's Printer
;   valid users = fred
;   path = /homes/fred
;   printer = fredsprn
;   public = no
;   writable = no
;   printable = yes

# A private directory, usable only by fred. Note that fred requires write

```

```

# access to the directory.
;[fredsdir]
; comment = Fred's Service
; path = /usr/somewhere/private
; valid users = fred
; public = no
; writable = yes
; printable = no

# a service which has a different directory for each machine that connects
# this allows you to tailor configurations to incoming machines. You could
# also use the %u option to tailor it by user name.
# The %m gets replaced with the machine name that is connecting.
;[pchome]
; comment = PC Directories
; path = /usr/pc/%m
; public = no
; writable = yes

# A publicly accessible directory, read/write to all users. Note that all
# files
# created in the directory by users will be owned by the default user, so
# any user with access can delete any other user's files. Obviously this
# directory must be writable by the default user. Another user could of course
# be specified, in which case all files would be owned by that user instead.
;[public]
; path = /usr/somewhere/else/public
; public = yes
; only guest = yes
; writable = yes
; printable = no

# The following two entries demonstrate how to share a directory so that two
# users can place files there that will be owned by the specific users. In
# this
# setup, the directory should be writable by both users and should have the
# sticky bit set on it to prevent abuse. Obviously this could be extended to
# as many users as required.
;[myshare]
; comment = Mary's and Fred's stuff
; path = /usr/somewhere/shared
; valid users = mary fred
; public = no
; writable = yes
; printable = no
; create mask = 0765

```

e) Intégration du serveur au domaine Windows « sitecreusot.LOCAL ».

```

[root@ordi ~]# net ads join -U jerome@sitecreusot.LOCAL
jerome@sitecreusot.LOCAL's password:
Using short domain name -- sitecreusot
Joined 'ordi' to realm 'sitecreusot.LOCAL'
[root@ordi ~]#

```

f) Test d'authentification d'un utilisateur.

```
[root@ordi ~]# kinit jerome@sitecreusot.LOCAL
Password for jerome@sitecreusot.LOCAL:
[root@ordi ~]#
```

g) Liste des tickets Kerberos obtenus.

```
[root@ordi ~]# klist
Ticket cache: FILE:/tmp/krb5cc_0
Default principal: jerome@sitecreusot.LOCAL

Valid starting      Expires            Service principal
05/13/05 11:11:53   05/13/05 21:11:57  krbtgt/sitecreusot.LOCAL@sitecreusot.LOCAL
        renew until 05/14/05 11:11:53

Kerberos 4 ticket cache: /tmp/tkt0
klist: You have no tickets cached
[root@ordi ~]#
```

h) Récupération de la liste des utilisateurs et des groupes depuis le serveur Active Directory.

```
[root@ordi raddb]# wbinform -u
[...]
jerome
alain
DROOPY$
DALTON$
PLUTO$
AMPHI-1$
SERVINFO$
[...]

[root@ordi raddb]# wbinform -g
[...]
etudiants-dueti
etudiants-dueti
etudiants-cfao
etudiants- duchine
[...]
```

i) Configuration de freeradius.

```
[root@ordi ~]# vi /etc/raddb/radiusd.conf
[...]
    mschap {
        #
        # As of 0.9, the mschap module does NOT support
        # reading from /etc/smbpasswd.
        #
        # If you are using /etc/smbpasswd, see the 'passwd'
        # module for an example of how to use /etc/smbpasswd
```

```

# authtype value, if present, will be used
# to overwrite (or add) Auth-Type during
# authorization. Normally should be MS-CHAP
authtype = MS-CHAP

# if use_mppe is not set to no mschap will
# add MS-CHAP-MPPE-Keys for MS-CHAPv1 and
# MS-MPPE-Recv-Key/MS-MPPE-Send-Key for MS-CHAPv2
#
use_mppe = no

# if mppe is enabled require_encryption makes
# encryption moderate
#
#require_encryption = yes

# require_strong always requires 128 bit key
# encryption

#require_strong = yes

# Windows sends us a username in the form of
# DOMAIN\user, but sends the challenge response
# based on only the user portion. This hack
# corrects for that incorrect behavior.
#
with_ntdomain_hack = no

# The module can perform authentication itself, OR
# use a Windows Domain Controller. This configuration
# directive tells the module to call the ntlm_auth
# program, which will do the authentication, and return
# the NT-Key. Note that you MUST have "winbindd" and
# "nmbd" running on the local machine for ntlm_auth
# to work. See the ntlm_auth program documentation
# for details.
#
# Be VERY careful when editing the following line!
#
ntlm_auth = "/usr/bin/ntlm_auth --request-nt-key
--username=%{Stripped-User-Name:-%{User-Name:-None}}
--challenge=%{mschap:Challenge:-00} --nt-response=%{mschap:NT-Response:-00}"
}
[...]
ldap {
    server = "iutlecreusot.sitecreusot.LOCAL"
    identity = "jerome@sitecreusot.local"
    password = "motdepassedejeromelandre"
    basedn = "OU=Personnels,DC=sitecreusot,DC=LOCAL"
    filter = "(SAMAccountName=%{Stripped-User-Name:-%{User-Name}})"

    # set this to 'yes' to use TLS encrypted connections
    # to the LDAP database by using the StartTLS extended
    # operation.
    # The StartTLS operation is supposed to be used with normal
    # ldap connections instead of using ldaps (port 689) connections
start_tls = no

    # tls_cacertfile          = /path/to/cacert.pem
    # tls_cacertdir           = /path/to/ca/dir/

```



```

# tls_certfile           = /path/to/radius.crt
# tls_keyfile            = /path/to/radius.key
# tls_randfile           = /path/to/rnd
# tls_require_cert       = "demand"

# default_profile = "cn=radprofile,ou=dialup,o=My Org,c=UA"
# profile_attribute = "radiusProfileDn"
#access_attr = "dialupAccess"

# Mapping of RADIUS dictionary attributes to LDAP
# directory attributes.
dictionary_mapping = ${raddbdir}/ldap.attrmap
ldap_connections_number = 5

#
# NOTICE: The password_header directive is NOT case
# insensitive
#
# password_header = "{clear}"
#
# Set:
#     password_attribute = nspmPassword
#
# to get the user's password from a Novell eDirectory
# backend. This will work *only if* freeRADIUS is
# configured to build with --with-edir option.
#
#
# The server can usually figure this out on its own, and pull
# the correct User-Password or NT-Password from the database.
#
# Note that NT-Passwords MUST be stored as a 32-digit hex
# string, and MUST start off with "0x", such as:
#
#     0x000102030405060708090a0b0c0d0e0f
#
# Without the leading "0x", NT-Passwords will not work.
# This goes for NT-Passwords stored in SQL, too.
#
password_attribute = userPassword
#
# Un-comment the following to disable Novell eDirectory
# account
# policy check and intruder detection. This will work *only
# if*
# FreeRADIUS is configured to build with --with-edir option.
#
# edir_account_policy_check=no
#
# groupname_attribute = cn
# groupmembership_filter =
#     # "(&(&(objectClass=GroupOfNames)(member=%{Ldap-UserDn}))(&
(objectClass=GroupOfUniqueNames)(uniquemember=%{Ldap-UserDn}))"
# groupmembership_attribute = radiusGroupName
timeout = 4
timelimit = 3
net_timeout = 1
# compare_check_items = yes
# do_xlat = yes
# access_attr_used_for_allow = yes
}

```

```
[...]
authorize{
[...]
    ldap
[...]
authenticate {
[...]
    # Uncomment it if you want to use ldap for authentication
    #
    # Note that this means "check plain-text password against
    # the ldap database", which means that EAP won't work,
    # as it does not supply a plain-text password.
    Auth-Type LDAP {
        ldap
    }
[...]
[root@ordi ~]#
```

Le serveur freeradius authentifie les utilisateurs sur l'Active Directory seulement avec les EAP qui utilisent MSCHAPv2 (le protocole *handshake* de Microsoft). C'est-à-dire EAP-PEAP et EAP-TTLS. Pour le protocole LEAP de l'I.U.T., cela ne marche pas car les échanges ne sont pas effectués en MSCHAPv2 (dommage...).

7. Le Wi-Fi distant

Dans le cadre d'un accord entre l'I.U.T. et la plateforme technologique 3D <http://www.plateform3d.com> du Creusot, l'équipe informatique de l'I.U.T. a déployé un réseau Wi-Fi étendu entre les trois sites de la plateforme. Cette expérience a montré qu'il était possible de déployer un réseau Wi-Fi sécurisé distant entre plusieurs sites en s'appuyant sur la sécurisation mise en place dans l'établissement.

Pour des raisons juridiques (réglementation de l'ART), les sites distants ne sont pas autorisés à sortir sur Internet (sinon l'I.U.T. devient fournisseur d'accès Internet), le Wi-Fi distant ne sert qu'à créer un groupe de travail plateforme 3d local (bien que techniquement, l'accès Internet sécurisé soit possible). Il ne transite donc que des fichiers de données cryptés sur cette liaison.

La première contrainte à respecter est que les points d'accès diffusant l'information doivent se « voir ». Pour le lycée Jaurès, pas de problème, il se situe à 200 m de l'I.U.T. avec une vue dégagée. Pour le lycée Lavoisier, le bâtiment à relier était situé en haut d'une colline à 800 m, derrière un autre bâtiment à vue. Il a été nécessaire d'utiliser une antenne relais sur ce premier bâtiment à vue dont le rôle est simplement de relayer l'information du bâtiment 1 au bâtiment 2.

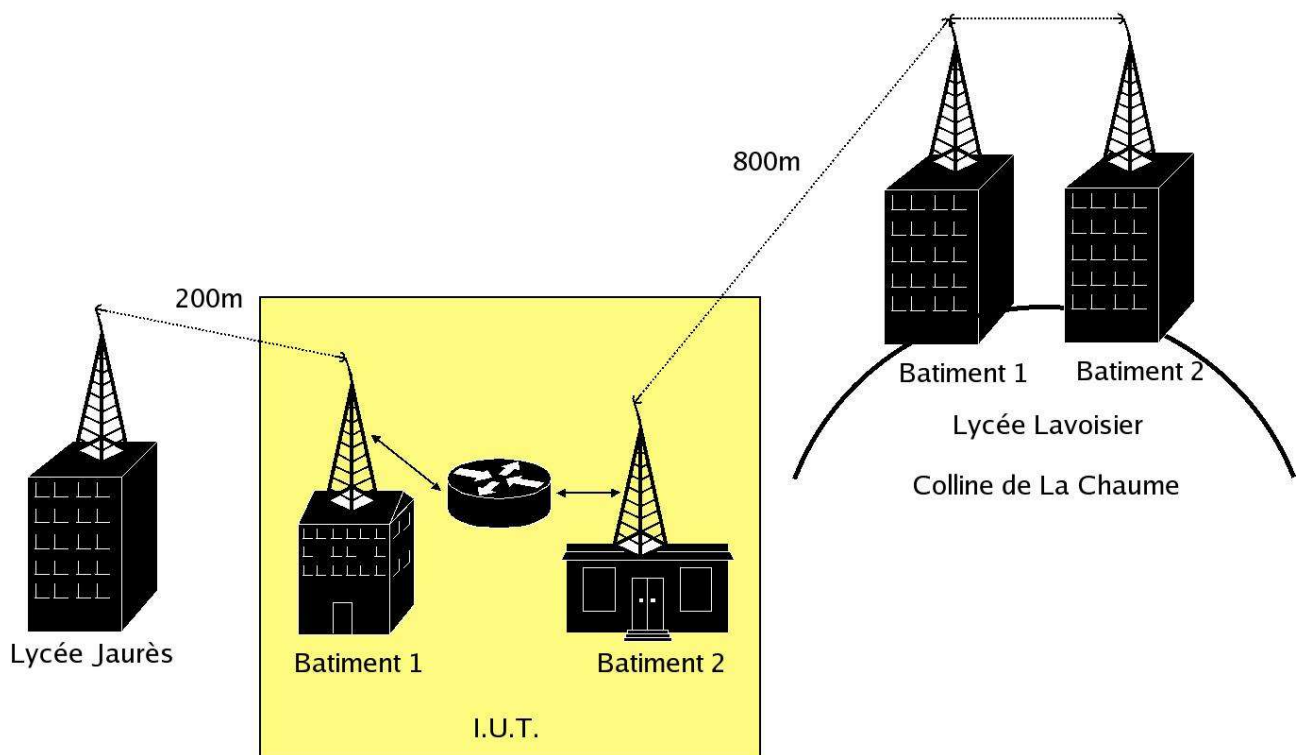


Figure 22: Schéma de la liaison Wi-Fi distante entre l'I.U.T. et les deux autres sites de la plateforme technologique 3d

La portée annoncée des antennes Wi-Fi est de sept kilomètres, nous n'avons pas vérifié cette portée, mais sur une distance de 800 m, le Wi-Fi fonctionne très bien. Mais si cette distance de 7 km est vérifiée, cela signifie qu'on peut relier les sites distants sur un même campus ou bien à l'autre bout de la ville (à condition d'avoir une vue dégagée !!!) avec cette technologie.

Les liaisons utilisent des antennes directionnelles qui se comportent comme des points d'accès qui émettent seulement dans une certaine direction.

Les antennes utilisées sont:

- IUT Bâtiment 1-Jaurès: Cisco 350-Cisco 350,
- IUT Bâtiment 2-Lavoisier: Cisco 350-Cisco 350,
- Lavoisier Bâtiment 1-Lavoisier Bâtiment 2: Cisco 1200-Cisco WorkBridge 350.

La sécurisation retenue est l'EAP LEAP comme dans le cas du réseau recherche de l'établissement avec un WVLAN et un VLAN dédiés à la plateforme 3D. Ainsi, seuls les ordinateurs de la plateforme peuvent communiquer entre eux. Ils ont leur propre classe d'adresse et leur propre ESSID, définis sur chaque point d'accès de l'établissement et sur chaque point d'accès distant.

Après des essais de positionnement visuels des antennes directionnelles, l'alignement des antennes a été réalisé au laser pour avoir une bonne précision. Il faut savoir que les canaux doivent être choisis avec précaution car l'une des antennes à l'I.U.T. perturbait la réception d'un autre point d'accès de l'établissement situé lui aussi sur le canal 13. Les antennes sont directionnelles, mais le faisceau obtenu est conique et inonde l'établissement d'ondes qui peuvent interférer avec les points d'accès locaux.



Figure 23: L'antenne Wi-Fi dirigée vers le lycée Lavoisier, 800 m à vol d'oiseau.

Après de nombreux essais, de nombreux passages sur les toits des différents bâtiments, de nombreux ajustement des antennes, la liaison fonctionne parfaitement. Il faut savoir que le réglage de la puissance d'émission a influencé grandement les performances du réseau. Ainsi par exemple, pour la liaison IUT Bâtiment 2-Lavoisier Bâtiment 1, le fait de passer de 20 mW à 10 mW a amélioré la bande passante de 250 Ko/s à plus de 600 Ko/s (sans rien changer d'autre sur l'installation !).

8. Les outils des pirates

Il existe sur Internet de nombreux outils pour espionner un réseau Wi-Fi, récupérer la clé WEP, se connecter au réseau et récupérer des données. Nous avons testé quelques-uns de ces outils afin d'évaluer la sécurité de nos solutions. Le PC utilisé était un Céléron 500 avec 256 Mo de RAM sous Linux RedHat 9 avec une carte Cisco PCMCIA 802.11abg.

Contrairement à freeradius, MySQL, OpenLDAP et Samba, vous ne trouverez pas de procédure d'installation de ces outils. La raison en est simple, on ne peut pas d'un côté protéger un réseau Wi-Fi et de l'autre publier ouvertement un mode d'emploi pour les pirater.

8.1. airtsnort

Airtsnort est l'un des plus célèbres logiciels d'espionnage de réseau Wi-Fi. Son fonctionnement est assez simple et ne nécessite pas de connaissance particulière. Il se débrouille tout seul pour déchiffrer la clé WEP d'un système.

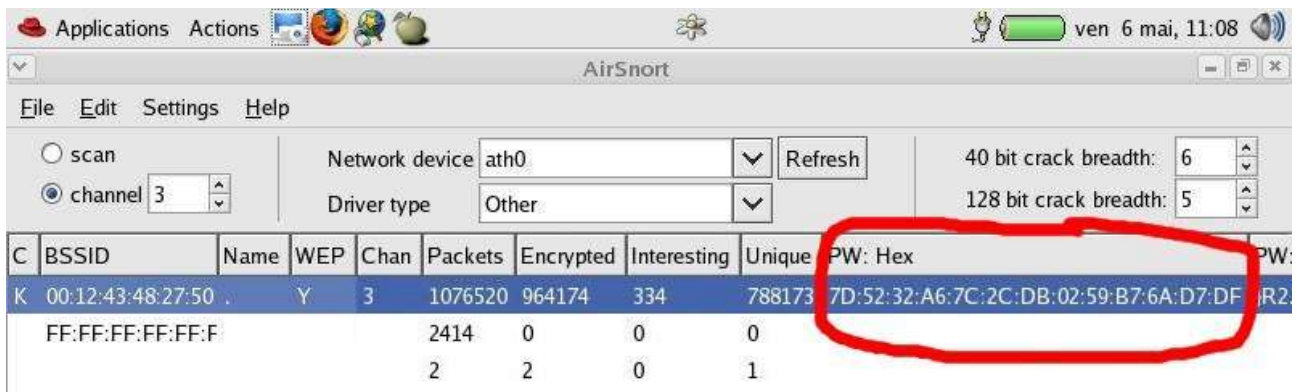


Figure 24: airsnort découvre la clé WEP d'un réseau Wi-Fi.

8.2. airodump et aircrack

Le couple airodump et aircrack est particulièrement efficace. Airodump est chargé de récupérer et stocker des paquets Wi-Fi par l'écoute du réseau. A tout moment, aircrack est capable de déchiffrer la clé WEP à partir des fichiers générés par airodump. La figure 25 illustre un exemple de récupération de clé WEP.

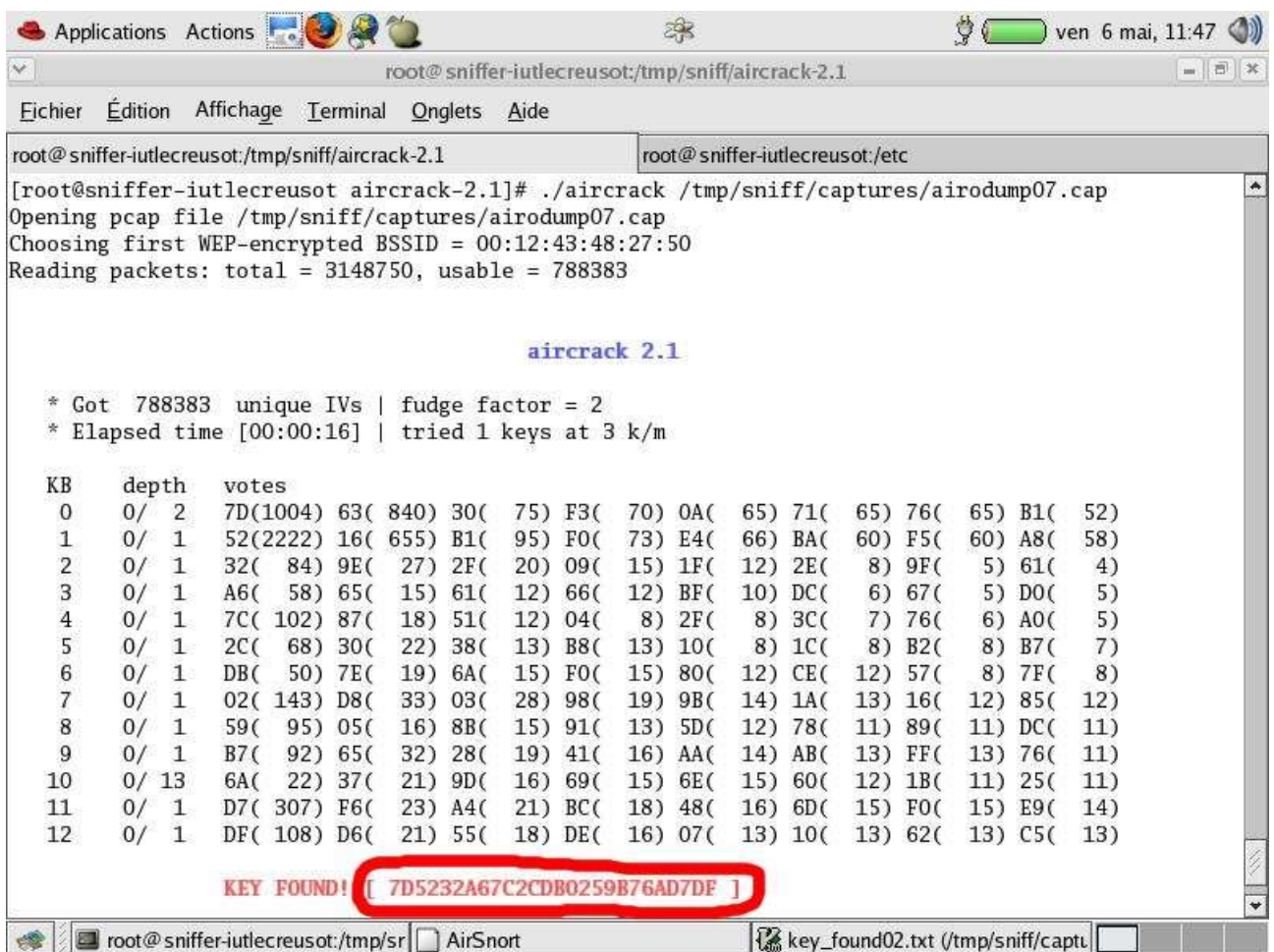


Figure 25: aircrack lancé sur un fichier créé par airodump.

8.3. NetStumbler et MiniStumbler

NetStumbler (figure 26) est un logiciel de repérage de réseaux Wi-Fi (*war driving*). Il scanne sans cesse les bandes de fréquences Wi-Fi, tente de localiser un point d'accès, récupère son adresse MAC et l'ESSID associé. MiniStumbler (figure 27) est la version pocketPC de NetStumbler avec moins d'options, mais tout aussi efficace.

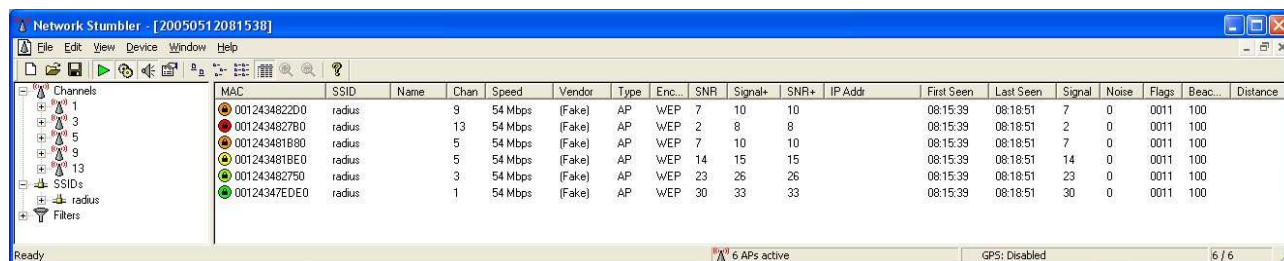


Figure 26: netStumbler, le découvreur de réseaux (war-driving).



Figure 27: miniStumbler, le logiciel d'espionnage sur PocketPC.

8.4. Kismet

Kismet est une boîte à outil de repérage, de découverte d'ESSID (même si celui-ci n'est pas diffusé) et de points d'accès. C'est un outil complet, difficile à maîtriser parfois, mais très efficace.

Tous ces outils sont malheureusement disponibles très facilement sur Internet, ce qui implique de renforcer au maximum sa sécurité avec WPA et si possible WPA2.

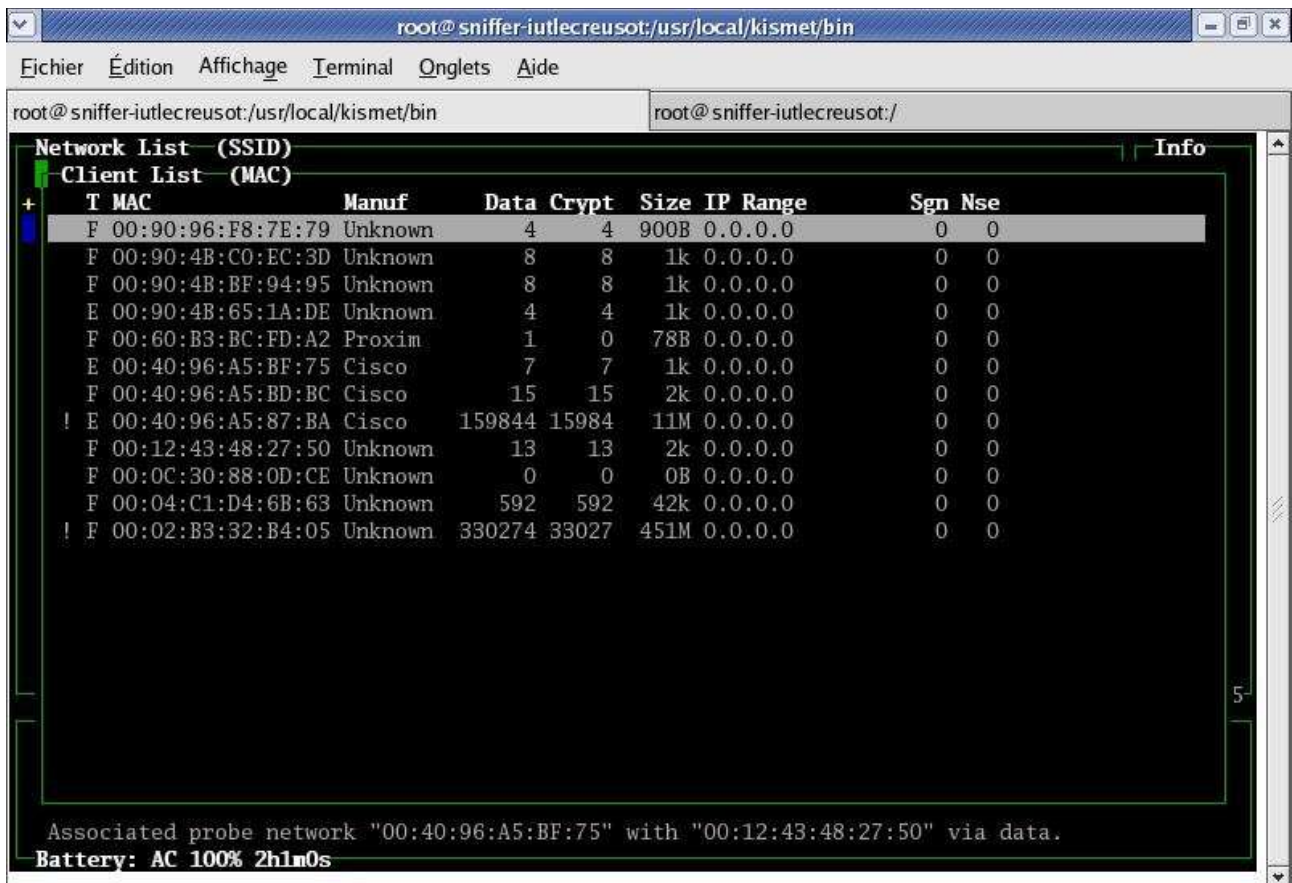


Figure 28: Kismet et la liste des découvertes Wi-Fi.

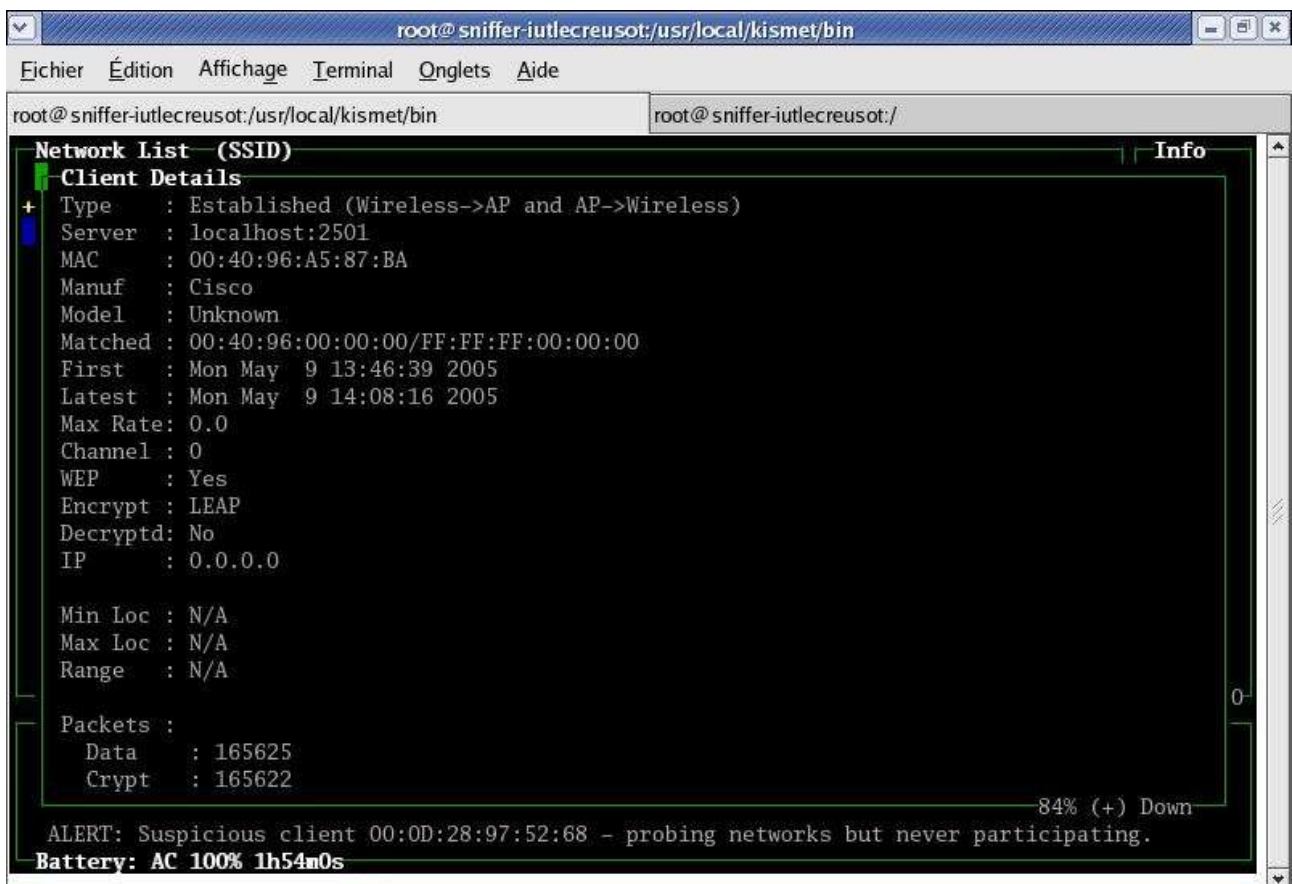


Figure 29: Kismet et le détail d'un client en LEAP.

9. Conclusion

Ce mémoire présente les travaux Wi-Fi réalisés par la modeste équipe informatique de l'I.U.T. du Creusot (quatre permanents) lors de la mise en place du réseau Wi-Fi de l'établissement (septembre 2003) et de son ouverture pour les étudiants (septembre 2004).

D'une part le portail captif fonctionne très bien et ne nécessite qu'une faible intervention sur l'ordinateur client (configuration DHCP et entrée de l'ESSID). Ce minimum d'installation est très important pour nous car nous ne pouvions pas nous permettre de passer la journée à installer du Wi-Fi sur les postes des étudiants en raison de la gestion des interventions quotidiennes très prenante. Les étudiants et les invités ont donc des droits limités par le firewall.

Ce service offert aux étudiants compte, à ce jour, une quarantaine ordinateurs portables (pour environ 800 étudiants sur le site). Les étudiants respectent la charte d'utilisation et les principales applications utilisées sont l'Internet, la mise à jour Windows, le mail (POP et SMTP), la mise à jour de l'antivirus et de l'anti-spyware (obligatoires pour obtenir l'accès au Wi-Fi).

D'autre part la solution EAP-LEAP retenue permet une configuration simple des postes clients de la recherche (environ 20 ordinateurs) avec les mêmes droits que sur le réseau filaire. La sécurité de WPA ajoutée à celle de LEAP rend le système très sûr et très fiable sans avoir à gérer une infrastructure de certificats très complexe. Les utilisateurs sont basculés automatiquement dans le WVLAN de leur groupe et utilisent le réseau de façon transparente où qu'ils soient dans l'établissement. Cette solution est provisoire en attendant d'avoir des clients WPA2 (AES).

Les choix que nous avons retenus sont issus de notre propre expérience et ne constituent pas des modèles mais le résultat d'expérimentations menées sur le terrain. Chaque établissement doit faire ses propres choix en fonction de ses ressources et des services qu'il veut offrir.

Si ce mémoire peut vous aider en quoi que ce soit à propos du Wi-Fi, nous en serions très heureux. C'est dans ce but qu'il a été rédigé, afin de permettre à tous d'avoir une meilleure idée sur la sécurisation d'un réseau Wi-Fi.

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